Vakeel-Sahayak.ai: An Agentic Harness for Bail Preparation over Typed Legal Stores

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Abstract

Legal-aid lawyers in India prepare bail applications from messy case files, and a model asked to draft one from memory invents facts and case law. We ask whether an agent harness that plans retrieval per bail factor over typed legal stores, checks every claim against its sources and spars with a simulated prosecutor writes better-grounded bail briefs than simpler systems on the same model. Vakeel-Sahayak.ai has six agents over ten typed stores, two counsel approval gates enforced in code and a pseudonymisation boundary before any hosted model. Gold labels come from court text through a rule labeler and a model labeler that must agree on a verbatim span; nothing is annotated by hand. On development orders the first full system did not beat every baseline; an architecture search on Sarvam-105B traced its losses to four causes and fixed them with four opt-in switches. The resulting system, c1457, was registered with five hypotheses before a sealed slice of 80 test orders was opened once. Against seven baselines (single pass, single store, whole-document and chunked retrieval, query-every-store, a learned store router, an expert template) c1457 has 46 wins, 69 ties and 5 losses (the first full system: 29, 76, 15). Four hypotheses hold: it beats the full system on objection recall and answerable coverage, cites real authorities where the comparable retrieval baselines fabricate them (fabricated-citation rate 0.00 against 0.64 and 0.95), states fewer unsupported facts than every baseline, and needs 41% fewer store calls than querying every store. The fifth fails on one unsupported-sentence count. The verdicts are the same when the outputs are scored against rule-only gold labels. No export happened without counsel's approval.

1 Introduction

Problem. A legal-aid lawyer who prepares a regular bail application in an Indian High Court (CrPC s.439, now BNSS s.483) starts from a pile of documents: a born-digital High Court order, a scanned Sessions Court order, a phone photo of an FIR copy, remand orders and a custody certificate, often merged into one PDF and often in Hindi. The application needs exact statute text, a mapping from old to new sections since the 2024 change from IPC and CrPC to BNS and BNSS, court reasoning on each bail factor, the citation history of every authority, and the case's own timeline. No single vector index serves all of these. A model asked to write it from memory fabricates facts and case law, as studies of commercial legal research tools found (Magesh et al., 2025).

The problem is one of data engineering and decisions before it is one of text generation: the input is heterogeneous, the knowledge has several shapes, each factor needs evidence from different places, the case keeps moving, and an office has hundreds of files to triage.

Why agentic. Three of these steps involve a decision that depends on what was just found: how far down a parse ladder a page must go, which store to ask next for a given factor and whether the result is enough, and which part of a draft a new order has made stale. A fixed pipeline either does too much (every store, the vision model on every page) or too little. We built the decisions as agents with typed tools, typed hand-offs and budgets, and we measure each of them against a simpler alternative.

Scope. Users are panel lawyers, junior advocates and supervised legal-aid clinics. The system does not predict outcomes, show grant rates, or advise accused persons or the public.

Contributions. The research contribution is the harness and its evaluation, not a new model.

1. **The harness.** Six court-office agents over ten typed stores on four shelves; page-level agen-

tic ingestion for ten input kinds; an orchestrator that plans retrieval per factor (Section 4).

2. **Evidence contracts.** For each of twelve bail factors, a short declarative list of the evidence a ground needs and which stores may hold it. The contract fixes what is needed and where it may be; the orchestrator decides the order, the query, sufficiency and the stop.
3. **Truth maintenance for drafts.** Each claim stores its dependencies. When counsel approves a change, only stale claims are redrafted and all other claims stay byte-identical, an idea from classical truth-maintenance systems (Doyle, 1979).
4. **Two gates in code.** Counsel approves the dossier (Gate 1) and the brief (Gate 2). Export is impossible before Gate 2, and approvals can only come from counsel.
5. **Evaluation without manual annotation.** Seven kinds of gold label and nine test sets built from court text, official tables and construction; a sealed blind slice and a preregistration; baselines B0 to B6; 18 single-switch variants from the recent agent and RAG literature, each with a switch, a paper and a test (Section 6).

Research question. **RQ1:** for bail preparation over a heterogeneous legal corpus, does an agent harness that plans retrieval per bail factor from evidence contracts, checks every claim against its sources and spars with a simulated prosecutor produce better-grounded bail briefs than simpler systems on the same model (a single prompt, plain retrieval, querying every store, a learned store router, an expert template), at no higher retrieval cost? A second question (RQ2), on agentic ingestion of degraded and Hindi pages and on a fine-tuned small extractor, was planned; its ingestion runs were not scored and the extractor was not trained, so this report answers RQ1 and says what is missing for RQ2 (Sections 11 and 13).

Hypotheses. The first full system ("Full") did not beat every baseline on the development orders (Section 8). An architecture search on the development orders produced c1457, Full with four fixes (Section 9). Before the sealed blind slice of 80 test orders was opened, five hypotheses for c1457 were registered with their metrics and decision rule (a win only when the Holm-adjusted 95% paired-bootstrap interval excludes zero). Table 1 states them and their outcome; Section 7 gives the details.

The five capabilities. Table 2 gives, for each capability of the course brief, why bail preparation needs it, how we measure it, which ablation removes it and what we found.

Organisation. Section 2 and Section 3 cover related work and the literature sweep that supplied the variants; Section 4 the system; Section 5 the data; Section 6 the evaluation design. Section 7 gives the main result on the blind slice, with tool-calling and safety counts, a worked case and failure examples. Section 8 gives the development runs that motivated the architecture search of Section 9. Sections 10 to 12 cover challenges, limitations and threats to validity, and Section 13 concludes with future work. The code is at `https://github.com/vishakkashyapk30/lma-group-project` (branch `main`; the blind runs use tag `eval-freeze-v1`).

2 Related Work

Agentic and adaptive RAG. Self-RAG and CRAG let the model critique retrieved text and re-retrieve (Asai et al., 2024; Yan et al., 2024); Adaptive-RAG chooses the retrieval depth by query complexity (Jeong et al., 2024); Sufficient Context studies when the retrieved context is enough to answer (Joren et al., 2025); HippoRAG 2 retrieves over a graph (Gutiérrez et al., 2025). We implement adaptations of these as opt-in switches (S1, S2, S3, S5) and compare them with hand-written contracts. A related line routes among language models or answering agents (Zhang et al., 2026); our action space is about ten stores that are not agents, so a learned store prior is only one baseline (B5) and one ablation (S6).

Multi-agent systems and verification. Self-critique without an outside verifier is reported to help little (Gou et al., 2024; Huang et al., 2024); we keep an NLI and rule-based check as the critic and add a self-critique arm (S8) to test this in our domain. For adversarial review, debate gains are partly explained by voting (Choi et al., 2025), and context sharing can hurt (Wynn et al., 2025). These motivate our prosecution-sparring ablations (S9, S10). Cemri et al. (2025) give a taxonomy of multi-agent failures. AgentCourt and CourtEval place agents in court roles (Chen et al., 2025; Kumar

	Hypothesis (registered before the blind slice was opened)	Metric	Blind slice, 80 orders
H-A1	c1457 has no loss against B0–B6 on any quality metric, except four cells declared structural in advance	every quality metric	**Falsified**: one further loss, unsupported-sentence count vs B5 (6.0 vs 4.2 per order)
H-A2	c1457 beats Full on objection recall and answerable coverage	objection recall; answerable coverage	**Holds**: 0.52→0.73; 0.36→0.61
H-A3	c1457 beats B4 and B5 on citation precision and fabricated citations	citation precision; fabricated-citation rate	**Holds**: 0.71 vs 0.02 and 0.10; 0.00 vs 0.95 and 0.64
H-A4	c1457's unsupported-fact rate is not worse than any baseline's	unsupported-fact rate (MiniCheck)	**Holds**: 0.89 vs 0.94–1.00, a win against all seven
H-A5	c1457 uses fewer store calls than B4 (query every store)	store calls per order	**Holds**: 110 vs 188

Table 1: The registered hypotheses and their outcome on the blind slice (Sarvam-105B, one run per system). The four structural cells of H-A1 are unfinished-factor share against B4 and B5, confident-wrong rate against B4 and coverage against the template B6.

Capability	Why the task needs it	Metric	Ablation	Result
RAG	A brief needs exact statute text, the old-to-new section map, court reasoning per factor and the citation history of each authority; no single index serves all of these	citation precision; fabricated-citation rate; answerable coverage	A4 plain retrieval; A10 store knock-outs; B2, B3	Blind: no fabricated citation (B4 0.95, B5 0.64); precision 0.71 vs 0.02–0.10 (Table 7)
Reasoning	Each factor needs evidence from different places, and each claim must be verified before counsel sees it	unsupported-fact rate; objection recall	A2 no check and repair; A8 no sparring; S8 self-critique only	Dev, on Full: removing check and repair lowers answerable coverage by 0.38, removing sparring lowers objection recall by 0.45; blind unsupported-fact rate lower than every baseline's
Agentic	Which store to ask next, whether the evidence suffices, and what to hand back to counsel depend on what was just found	store calls; unfinished-factor share; objections kept	A6 fixed router; A7 free planner; B4, B5	Blind: 41% fewer store calls than B4 at no loss of grounding; the free planner's dev gains were measurement artefacts (Section 9)
Tool calling	Typed tools (store search, statute lookup, date arithmetic, claim check, export; c1457 used 17) with typed arguments and budgets	bad-argument share; recovery; unapproved actions	A3 no tools	Blind: 1.0% of calls refused for a bad argument, every order finished, 0 exports without Gate 2 (Table 8)
Fine-tuning	A small local extractor would keep client documents on the lawyer's machine	dossier field accuracy	A5 (tuned 4B)	**Not done**: QLoRA pipeline built and unit-tested, no model trained; A5 ran only in its untuned setting on 2 Gemini orders, with nothing tuned to compare

Table 2: The five capabilities: need, measurement, ablation and outcome. Dev results are from the Sarvam-105B search runs (30 orders); blind results from the 80 sealed orders.

et al., 2025); our opponent is scored against objections the real prosecution raised.

Ingestion and chunking. VisRAG and ColPali retrieve over page images (Yu et al., 2025; Faysse et al., 2025); we use an image-only arm as a baseline for a text-first routing ladder. Dense X Retrieval studies the granularity of retrieval units (Chen et al., 2024); our naive-chunking baseline B3 and the proposition-unit variant S16 test this on court orders.

Legal NLP. Work on Indian courts is mostly judgment or outcome prediction (Nigam et al., 2025, 2024). Magesh et al. (2025) show that commercial legal RAG tools still fabricate and misground citations. We cite them for positioning and design the metrics around that finding: every citation is looked up, and every quoted span must appear verbatim in the cited paragraph. Counter-argument retrieval as a separate kind of evidence (Wachsmuth et al., 2018) motivates our authority-stance variant (S18).

Evaluation. For grounding we use MiniCheck (Tang et al., 2024) and an NLI model, and citation precision and recall in the style of ALCE (Gao et al., 2023). No score in this report comes from an LLM judge.

Safety and serving. Typed hand-offs follow design patterns for securing agents (Beurer-Kellner et al., 2025); batch scheduling follows serving work such as SGLang (Zheng et al., 2024). Both are opt-in arms (S17, S14).

3 Literature Sweep and Its Tests

The design of the system was not fixed once. Four sweeps through the literature (retrieval and orchestration; verification and adversarial review; intake, safety and serving; routing, topology, chunking and workflow search) each produced a list of ideas that a paper reported as useful, and each idea became an opt-in *variant* of the harness. This section records which paper each variant comes from, what was changed when the idea moved to a bail-preparation setting, and what has been checked about it so far. The full table, with links, is `docs/literature-to-code-map.md` in the repository.

Rules of the sweep. (1) Every variant is a field of one typed `Switches` object and defaults to the current behaviour; a golden test per agent (digests of the default run of case-0007, captured before any variant was written) fails if switching a variant off does not reproduce the old output byte for byte. (2) A variant is an *adaptation*, never a reproduction: each row of the map says what was replaced (for example, the learned complexity classifier of Adaptive-RAG became deterministic rules over the Rights Radar and the statute era). (3) A variant is built only if it can be scored by a metric of the evaluation plan; its row names the claim, metric and test file. (4) Constants that the papers do not give (the skew thresholds of S4, the damping and depth of the graph expansion of S5) are marked as untuned assumptions to be set on the dev split, never on the test split. (5) No variant is claimed to help until it has been scored against Full on the same backend and cases (Section 8).

Citation check. Every paper in the map was checked against the official proceedings page or the arXiv record. Four links or author lists were corrected (Self-RAG, CRITIC, the first author of the multi-agent failure taxonomy, and a preview link for semantic chunking); seven further checks passed, one with a note on a changed page title; Doyle's truth-maintenance paper was checked on secondary sources only, and its row says so. Preprints without a venue are labelled as such.

3.1 How the literature review was done

Why we reviewed the literature. Our project is an engineering contribution: the way agents, stores, checks and approval gates are put together for one workflow. Such a claim is only credible if each design choice is placed against published work (what is borrowed, what is adapted, what is new) and if each borrowed idea becomes a test that can fail. The review therefore had two outputs, a catalogue of papers with a recorded decision for each, and a map from papers to code switches and metrics (Table 4). It also had a third, less visible output: a list of ideas we looked at and did not adopt, with the reason.

Scope and search. We covered fourteen themes: routing among models; multi-agent orchestration and its failure modes; agentic and adaptive retrieval-augmented generation; graph-structured and multi-source retrieval; document ingestion and chunking; verification, attribution and self-correction; debate and adversarial critics; memory and state; agent safety; agent evaluation; legal AI and Indian legal NLP; clarification, abstention and human-in-the-loop; small models and structured output; serving and scheduling. We searched 2023 to 2026 proceedings (ICML, ICLR, NeurIPS, ACL, EMNLP, NAACL, COLM, AAAI, IJCAI, SIGIR, ICAIL, USENIX, OSDI, MLSys, TMLR, TACL) and arXiv for items without a venue, starting from the two papers the course team named and from the TA's request for novelty in agentic engineering rather than in a new model.

Three passes, and how each paper was checked. A first sweep per theme recorded candidates with an official link. A second pass checked 33 papers one by one against the official page (proceedings, ACL Anthology or arXiv). A third pass re-fetched 25 more, including the two named papers, and corrected several venue claims; where a venue could not be confirmed (for some preprints and workshop papers) the entry says so instead of guessing. Each catalogue row carries a check mark (verified on an official page, or taken from the sweep only), a tier (top venue, workshop or other, preprint) and a

relevance score from 1 to 5 that records how much the paper changes a decision in this project.

What the catalogue holds. 145 papers: 114 from a top venue, 8 workshop or other, 23 preprints; 34 score 4 or 5 and form the core reading list, 81 score 3 (a useful idea), and 30 score 1 or 2 (framing only). A further 24 were read and not catalogued. Contribution lines come from abstracts and earlier summaries, not from full reads of every paper, and the numbers papers report (for example attack success rates) are theirs, on their benchmarks; nothing in the review is evidence about our own system.

What we did not adopt, and why. The reasons are specific to our constraints: open-weight or free-tier models at run time, small GPUs, no manual annotation, eight weeks, and the need for an auditable legal record. (a) Trained routers, searchers and stoppers (Router-R1, Search-R1, AutoSearch) need rollouts or routing labels we do not have; we kept their loop shape and their cost-aware reward as evaluation ideas and left the policy to rules and contracts. (b) Automated workflow search (AFlow, ADAS, MaAS) spends thousands of model calls and returns pipelines that are hard to audit; only difficulty-adaptive effort was borrowed. (c) Memory managed by a language model (MemGPT, Mem0) conflicts with an event-sourced case record that must be auditable. (d) Vision-first retrieval as the main path needs a vision model on every page and stays a baseline. (e) Training-based and detector-style injection defences were replaced by architecture. (f) Outcome-prediction systems conflict with the rule that the system gives no outcome predictions or grant rates. (g) Other-jurisdiction legal benchmarks use US or Chinese law, so only their metric ideas were kept.

Findings that changed the design. Five findings changed what we built. Stopping and sufficiency checks are the most direct match to the orchestrator, so rule checks plus a rater are tested while a trained stopper is not built. Self-correction needs outside evidence, so the repair loop uses an NLI and rule check, with a self-critique comparison. Debate is not better than sampling at equal cost, so the prosecution critic carries the burden of proof. Prompt injection is an architecture problem, which favours typed hand-offs. Agent evaluation needs cost and reliability, so every table carries tokens, store calls and pass^k.

Limits of the review. Every idea we borrowed was written for other tasks and mostly other models, so each is an adaptation and a hypothesis here, not a result. Many 2026 items are preprints. The review is the reason each switch exists; whether a switch helps is decided only by the runs in Section 8, and null or negative results will be reported as such.

3.2 What was built, by sweep

3.3 How the variants are tested

The tests answer four different questions, and a variant is not called done until all four have an answer.

1. Does the default stay the same? Each of Sutradhar, Vakeel, Peshkar and Munshi has a golden test that hashes the evidence packs, plans, event multiset, budget, prompts and drafted output of the default run (Peshkar over a planted shelf, Munshi over the 14-page bundle). Adding a variant must not change a digest. This is the test that lets 46 switches coexist with the shipped behaviour.

2. Does the switch do what the paper's idea says? Each variant has its own test file, 631 collected unit tests across 37 files (about 220 of them are the shared registry, switch and effect tests) (Table 4). They check mechanism, not quality: the critique variant issues no critic call for an empty slot; the adaptive policy caps rounds for single-factor cases (case-0007 is a BNS-era case, so every factor is "multi" there and the single and none branches are tested by reading it as IPC-era); graph expansion only counts a hit that also passes the factor, family and date rules; the Gini and top-gap signals are computed on scored searches that leave the unscored path unchanged. A registry test fails if a variant id appears in the plan without a switch, a map row, or a test, and an end-to-end audit runs every one of the 61 ids through the offline pipeline (61 of 61 pass; 42 have a measured-effect test that checks the switch changes the run in the direction the idea predicts).

3. Is it measured on the right metric? Each row names a metric from the evaluation plan: stopped-too-early and over-search (S3), verdict counts (S1), tokens and store calls at equal recall (S2, S14), recall@10 of the citation graph with and without expansion (S5), AUROC of the score skew against "the top ten contain a cited precedent" (S4), unsupported-fact rate and repair success (S8, S15), objection recall at equal tokens (S9, S10, S11),

Kind of paper	Examples	What we took, and why
Adaptive retrieval and sufficiency	Self-RAG, CRAG, Adaptive-RAG, Sufficient Context (Asai et al., 2024; Yan et al., 2024; Jeong et al., 2024; Joren et al., 2025)	Sutradhar stops when the evidence for a factor is enough, and caps effort by case complexity (S3, S6). Chosen because stopping and sufficiency are the closest match to the orchestrator's job; their trained components were replaced by rules over the Rights Radar and the statute era, since we have no routing labels.
Graph and multi-source retrieval	HippoRAG 2, ToG 2.0, HybGRAG, GraphRAG, LightRAG	A citation graph extracted from the orders themselves, expanded by personalised PageRank (S5); the entity-graph systems built by LLMs were left out for indexing cost, and a paper reporting that graph RAG often loses to plain RAG is the reason S2 is a small ablation.
Verification and self-correction	CRITIC, "Cannot Self-Correct", MiniCheck, ALCE (Gou et al., 2024; Huang et al., 2024; Tang et al., 2024; Gao et al., 2023)	Self-critique without an outside verifier helps little, so Vakeel repairs with an NLI and rule check, and a self-critique arm (S8) tests that claim here. MiniCheck scores the unsupported-fact rate; ALCE-style citation precision and recall score citations.
Debate and critics	Debate or Vote, MAD, Talk Isn't Always Cheap (Choi et al., 2025)	Several studies find debate gains come from voting. So Abhiyojak (the prosecution critic) must beat a sampling-only critic at equal tokens (S9), and context isolation is an ablation (S10).
Ingestion and chunking	VisRAG, ColPali, Dense X (Yu et al., 2025; Faysse et al., 2025; Chen et al., 2024)	A text-first page ladder with an image-only baseline (P7); propositions as retrieval units (S16) against naive chunking (B3).
Agent safety	Design Patterns for Securing Agents, CaMeL (Beurer-Kellner et al., 2025)	Prompt injection is an architecture problem. Typed hand-offs and code-enforced approval gates fit; the injection-defence variant S17 and the hidden-text check on PDFs follow from this.
Agent evaluation	AI Agents That Matter, τ-bench	Cost and reliability beside accuracy: tokens, store calls and pass^k are reported for every arm, never one accuracy number.
Legal AI for India	NyayaRAG, PredEx, "Hallucination-Free?" (Nigam et al., 2025, 2024; Magesh et al., 2025)	Indian work is mostly outcome prediction, which the system refuses by design. We position as preparation with checked citations, where commercial legal RAG tools were shown to fabricate, and make the citation checks the headline safety metrics.
Serving and scheduling	SGLang, KV-cache scheduling (Zheng et al., 2024)	Plan caching and an urgency-ordered batch queue (S14), applied at the application level because we cannot modify a serving engine.

Table 3: Kinds of papers reviewed, the inspiration taken and the reason. The full catalogue is in the project repository (`docs/literature-review-and-usage.md`).

pages re-read per recovered field (S7), attack success (S17). A scoring test pins the metric on synthetic inputs with known answers, so a metric change cannot pass silently.

4. Is the claim tied to the right comparison? A variant is compared only with Full on the same backend and cases; the scorer refuses mixed code stamps and scores one model family per backend, applies a Holm correction inside each family of claims, and decides the headline claim by an intersection-union rule (every comparator must lose). A comparator that was not run makes the claim "incomplete", not "supported".

3.4 What has been measured

Mechanism tests and the 61-id audit are complete and green (5,082 passed on 29 October). Quality measurements are fewer, and we state them as they stand.

S4 and S5 on the real stores (offline, no model). On a self-supervised proxy of 150 queries (a held-out citing order, with its cited precedents as targets under a strict before-date filter), expanding the citation graph raised recall@10 from 0.037 to 0.537; the paired difference is 0.50 (95% interval 0.42 to 0.58, McNemar $p < 10^{-20}$; 75 queries gained and none lost). The gain comes from co-citation structure rather than from the retriever finding the cited order itself, and the proxy is optimistic, so this is evidence that the graph carries signal, not a forecast of the gain on real briefs. The score-skew signal separates queries with and without a cited precedent in the top ten with an AUROC of 0.870 (interval 0.751 to 0.969); as a weak-evidence flag at

Variant	Idea (paper)	Switch	What was adapted	Tests
Sweep 1: retrieval and orchestration (Sutradhar, stores)				
S3, S3b	Stop when the evidence is sufficient (Sufficient Context; AutoSearch; StopRAG)	`stopping` = rules, rules_llm, llm	Rules first, a model rating second; stopped-too-early and over-search are scored (M18)	20
S1	Critique retrieved text, re-retrieve on a miss (Self-RAG; CRAG)	`retrieval_policy` = self_critique	A slot with nothing retrieved is MISSING and costs no critic call; verdicts kept in the event log	24
S2	Retrieval depth by complexity (Adaptive-RAG)	`retrieval_policy` = adaptive	Classifier replaced by rules: none, single, multi; round cap per factor	21
S5	Expand over a citation graph (HippoRAG 2)	`graph_expansion`	Personalised PageRank on 5,842 citation edges; expanded paragraphs must still pass the slot rules (factor, family, date)	33
S4	Score skew as weak-evidence signal (SkewRoute)	`weak_signal` = skew	Gini and top gap of the fused scores; thresholds untuned	23
Sweep 2: verification and adversarial review (Vakeel, Abhiyojak)				
S8	Self-critique without an outside verifier (CRITIC; "cannot self-correct yet")	`repair_mode` = self_critique	Replaces the NLI and rule check by the drafter's own critique, to test the claim in this domain	16
S9	Sampling-only critic at equal tokens (Debate or Vote; "Should we be going MAD?")	`critic_mode` = sampling	Same token budget as the critic, spent on samples	21
S10	Debate without context isolation ("Talk Isn't Always Cheap")	`abhiyojak_isolation` = off	Abhiyojak sees Vakeel's reasoning; flip rate measured	11
S11, S11b, S11c	Grader, critic, defender (CourtEval); two-instance debate; three-sample vote	`critic_mode`, `draft_mode`	Roles mapped to the sparring and drafting steps	18
S15, S15b	Reasoning before structured output; free text then parse ("Let Me Speak Freely?"; CRANE)	`output_format`	Schema-valid rate, repairs and tokens compared across the three layouts	17
S18	Authority stance: counter-argument evidence (Wachsmuth et al.)	`authority_stance` = outcome	Supporting authorities first, precedents from refused orders flagged	6
Sweep 3: intake, safety and serving (Munshi, defences, Pairokar)				
S17	Prompt-injection defences: typed hand-off, spotlighting, guard (design patterns for securing agents)	`no_typed_handoff` and two more	Each defence can be switched off; attack success scored on T7	10
S7, P7, P8	Hierarchy-ranked gap re-reading (IntrAgent); image-only ingestion (VisRAG, ColPali); keep weak text layer	`munshi_gap_policy`, `munshi_image_only`, `munshi_keep_weak_text`	Pages re-read per recovered field; vision calls per page	41
S14, S14b–d	Plan caching, scheduling, prefix-friendly prompts (Agentic Plan Caching; SGLang)	`plan_cache`, `urgency_policy`, `prefix_layout`	Second case with the same signature skips all 8 planning calls: 30,633 to 12,836 tokens, packs byte-identical	38
Sweep 4: routing, topology, chunking, workflow search				
S6, B5	Learned store prior (GraphRouter; AgentRouter)	`store_prior` = order, only	Orders calls inside a contract, or replaces it; artefact not yet trained on real data	17
S12, S12b	Orchestrator-worker topology (scaling agent systems; MAST)	`topology`	Lock-step vs sequential vs one merged agent	11
S16, S16b	Chunk granularity, propositions, late chunking (Dense X)	`chunking`	Fixed, recursive, semantic, page, proposition	22
S13	Automatically designed workflows (AFlow; ADAS)	grid over switches	Budget-capped grid search on dev; harness and offline smoke only	14

Table 4: Ideas taken from the literature, the switch that turns each on, how it was adapted, and the number of unit tests that pin it. Test counts are those of the files named in `docs/literature-to-code-map.md`; they are collected, not re-run, in this table.

the chosen threshold it has precision 1.0 and recall 0.49. The second query set (G2, the precedents the court cited) has only 8 queries and no positives, so no claim is made on it.

Caching, scheduling and chunking. Plan caching on case-0007 (offline): the second case with the same signature skips all 8 planning calls, tokens fall from 30,633 to 12,836 and the evidence packs are byte-identical to the no-cache run. The scheduling and chunking studies run as simulations or on self-supervised targets and say so in their output; they are reported as such, not as serving measurements.

Measured on a model, and not yet. S1 to S6, S8 to S12, S15 and S18 and the ablations A1 to A3 and A6 to A9 have run against Full on the same OpenAI backend at small sample sizes (10 to 30 dev cases), and their scores are in Section 8; at these sizes most differences are inside the intervals, so we read them as a first look and not as a ranking. S11b and c, S12b, S15b, A4 and A10 ran later on Gemini or offline (Section 8), S13 became the switch grid of the architecture search (Section 9), and S7, S17 and P5 to P8 ran only with offline stand-ins and were not scored; A5 ran only in its untuned setting, since no tuned extractor was trained (Section 11).

4 System

Figure 1 shows the system. The agents are named after people in an Indian court office, and each has a narrow decision to make.

Agents. **Munshi** (intake) decides the parse route of each page, when to escalate, document boundaries and types, where each item is filed, and what is missing. **Peshkar** builds the dossier with a source quote for every field, a List of Dates, and the Rights Radar (deterministic checks on custody periods and statutory rights, each with a verification status). **Sutradhar** orchestrates retrieval. **Vakeel** drafts the grounds and checks and repairs every claim. **Abhiyojak** plays the prosecution with its own prompt and an allow-listed context. **Pairokar** orders a batch by urgency and decides per case between a full brief, a flag for counsel, or a request for documents. Every action is appended to an event log; dossier, timeline, brief and sparring record are views of it, which gives an audit trail and the dependency data for truth maintenance.

Ingestion ladder. Munshi handles ten input kinds (born-digital, scanned, degraded, phone photo, Hindi or mixed script, legacy Hindi fonts, handwriting, bundles, locked files, non-PDF). A page climbs only as far as needed: text layer; layout parser; clean-up and OCR; a vision model asked only to transcribe; a request to counsel for a clearer copy. Per-page quality signals (text coverage, OCR confidence, dictionary-word share, script, legacy-font patterns) decide each step and are logged. A gap-driven re-read returns to a page when a checklist field is still missing. The ladder is evaluated as a router against the cheapest route that clears the quality bar.

Ten stores on four shelves. A brief needs exact text, a section-mapping graph, court reasoning, citation history, what each side usually argues, and the client's own record; each has a different right way to search. The *law* shelf holds statutes (exact lookup, BM25) and the IPC–BNS and CrPC–BNSS concordance (typed graph). The *case-law* shelf holds reasoning paragraphs (BM25 plus a dense vector, metadata pre-filter), the citation graph with treatment and court level, and whole-order fact vectors. The *advocacy* shelf holds defence-argument and prosecution-objection paragraphs by factor, used for coverage and phrasing, never cited as authority. The private *case* shelf holds the client's documents in structure-aware chunks, a dated timeline and the dossier. The October 2026 build has 161,708 reasoning paragraphs, 70,880 arguments, 55,161 objections, 114,692 similar-fact orders and 5,842 citation edges (37.3% of citations resolved to a document). Test-window orders and same-FIR orders are excluded from every store. The dense side of the runs reported here is a hashed character n-gram vector, not BGE-M3; BGE-M3 side vectors for the four vector stores were built afterwards on a GPU node (2 h 43 min, 1.65 GB) and are opt-in (`VS_STORES=bge`); retrieval under BGE-M3 is measured only in the offline studies of Section 8.

Evidence contracts and the orchestrator. Each of twelve bail factors has a contract: slots (for example co-accused order, role comparison, authority, good-law check, likely objection), a need for each (required, a count, or optional), the stores that may hold it, and a budget in store calls and tokens. Sutradhar decides whether the factor applies, the order of calls, the query and filters, whether a result is good enough, when to relax a filter or follow a cita-

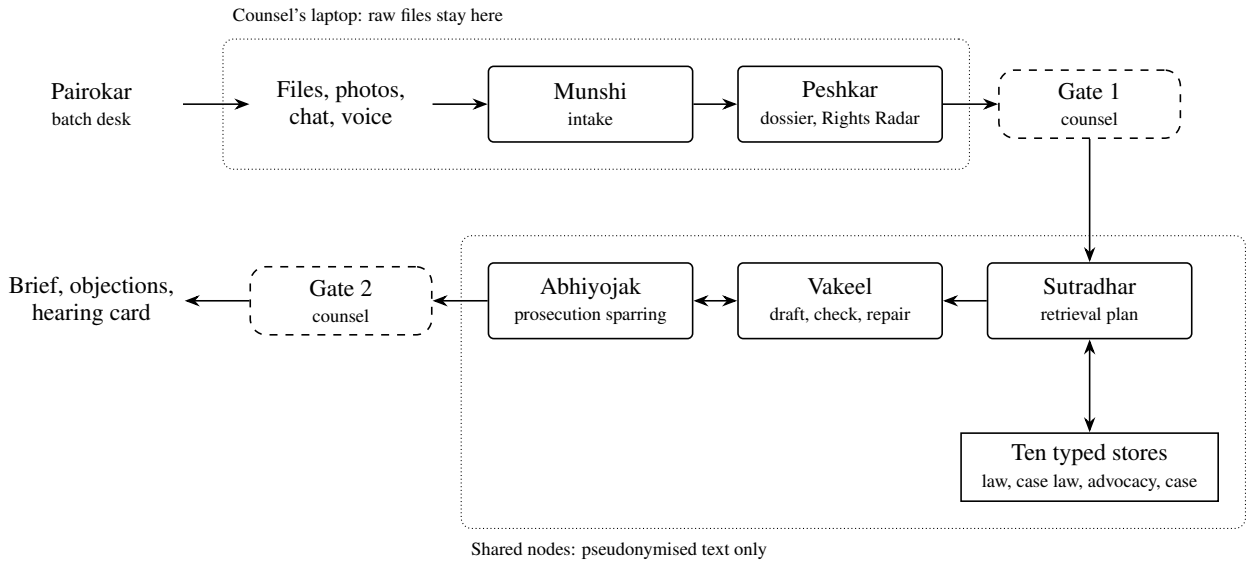


Figure 1: Architecture. Arrows show the order of work; the double-headed arrows are request and reply. Dotted boxes mark where each part runs; dashed boxes are the two approval gates, which only counsel can open and which are enforced in code, so nothing is exported before Gate 2.

tion, and when to stop; slots left unfilled become open issues, never silent omissions. A learned store prior, trained from train orders, may order calls within a contract (S6) or replace the contracts (B5).

Checking and truth maintenance. Vakeel checks every claim with an NLI model and rules (statute lookup, section mapping, good-law check), and repairs or flags it; nothing is dropped silently. Each claim records the dossier fields (by hash), timeline events, rights entries and authorities it depends on. When counsel approves a change, claims with a changed dependency, and claims citing them, become stale; only the reopened factors are re-planned, and unchanged claims stay byte-identical (tested).

Gates and privacy. Gate 1 (dossier) and Gate 2 (brief) pass only on counsel's approval, and export is impossible before Gate 2, in code and in tests. Raw client files stay on the user's machine. External model services are off by default, gated by a mode (`off`, `public`, `pseudonymised`, `consented`); client text needs counsel's per-case consent event for the provider even when pseudonymised, and every external payload passes the boundary. The boundary aliases names and quasi-identifiers (FIR, case and crime numbers, police stations, villages, addresses, PIN codes, vehicle numbers, ages) and shifts every date by a per-case offset; on the 250 T1 orders no FIR number, case number or numeric date survived, and the boundary refused 5 orders whose names were glued to other words by OCR (those calls stay local). Gate 2's approval is bound to a hash of the brief, sparring record and List of Dates, so a later change re-closes export. Defences against instructions planted in documents (a typed hand-off from a tool-less extractor, spotlighting, a guard) are all on by default; switching each off is an arm of the safety test set T7.

Models and where they run. The design specifies a 4B model with OCR and a vision model on an 8 GB laptop GPU for Munshi and Peshkar, and a 14B model on shared GPU nodes for Sutradhar, Vakeel and Abhiyojak. None of the runs reported here used those: serving a 32B model on the shared nodes failed (Section 10), so every language-model run is either hosted or an offline stand-in. Hosted calls go only through the pseudonymisation boundary; the Gemini free tier receives public court text only. Paid providers are locked in code and were opened one at a time with the owner's go-ahead and a spend cap: OpenAI for the OpenAI runs in the table (about 4.15 USD spent of a 4.50 cap) and Sarvam for the Sarvam block (about INR 225 spent of an INR 5,000 cap, opened on 1 November); OpenAI is locked again.

Table 19 (Appendix E) lists, for every block of runs, the provider, the model and whether it ran locally or on a hosted service.

Set	Content	Size
T1	HC bail orders, facts in, gold out	200 + 50 dev
T2	messy bundles; native Hindi orders	600; 200
T3	change-set updates	240
T4	rights scenarios (rule table)	627
T5	batch cases, planted deadlines	300
T6	chat questions	150
T7	safety scenarios	60
T8	Hindi utterances	300
T9	citation strings	2,095

Table 5: Test sets on 29 October 2026 (rows on disk). T2, T3, T5, T7 and T8 are constructed with known answers; T4 is generated from the written rule table.

5 Datasets

All data are public court text, official tables, existing datasets, or constructed; no client data and no hand-made labels are used (Table 5).

Corpus. Open India Law (release v2026.08.1) supplies the court orders: bail-related High Court decisions from 2019 to 2025 and Supreme Court judgments of any year (1.4 GB of the 53.6 GB release). Of 115,988 orders read, 115,524 feed the stores; 464 are excluded because they are test-window orders or share an FIR with one. Orders dated before 2024 are the training window; the test window is later. IPC2BNS-Verify supplies the IPC-to-BNS concordance (563 rows) and a 1,140-query mapping benchmark. Statute texts come from the corpus's legislation file (IPC, BNS, BNSS) plus the CrPC and NDPS sections the rules need, 1,426 sections in all. Kathbath (first 1,000 of 3,130 Hindi test utterances) supplies Hindi speech for dictated facts.

Stores. The build of 26 October 2026 took 35 minutes on a laptop and holds 1.77 GB: 161,708 reasoning paragraphs, 114,692 whole-order fact vectors, 70,880 defence-argument and 55,161 prosecution-objection paragraphs, and 5,842 citation edges (37.3% of extracted citations resolve to a document in the index).

Test sets. T1 is the main set: 200 test-window High Court orders (120 open after the freeze, 80 blind) and 50 development orders; the system reads the facts part of each order and gold comes from the rest of it (Section 6). The facts part is short: 19 of the 50 dev orders have under 1,000 characters of facts, and one has only the caption. T2 to T9 test the other parts of the system.

6 Evaluation Design

The evaluation follows the course material on evaluation (L12) and on RAG evaluation (L14): state the evaluated object, decide the claims, thresholds and tests before looking at results, seal a slice, report intervals, and audit the metrics themselves. No score comes from an LLM judge, and no label is made by hand.

Test sets. Nine sets built from court text, official tables or construction. **T1**: 200 High Court test-window orders (at least 60 BNS-era), facts part as input, plus 50 dev orders. **T2**: 600 messy bundles with known text, boundaries and fields, and 200 native Hindi High Court orders. **T3**: 240 change-set updates. **T4**: 627 rights scenarios from the written rule table (a conformance check of the Radar against its specification, not a legal one). **T5**: 300 batch cases with planted deadlines. **T6**: 150 chat questions. **T7**: 60 safety scenarios. **T8**: 300 utterances (Kathbath Hindi, dictated facts). **T9**: 2,095 citation strings.

Gold labels without annotators. Seven kinds of gold label are derived from the order text. G1 is the set of the 12 bail factors the defence argued; G2 the precedents the court cited that resolve to a document in the index; G3 dossier fields and dated events (sections, dates, charge-sheet status, co-accused bail, stage); G4 the objections the prosecution raised; G5 the factors the court discussed; G6 condition types of the operative order; G7 the rights computations the court states itself (custody against the 60 and 90 day limits, s.436A and s.479). For G1 and G3 (and, with weaker coverage, G4 to G6) two labelers work independently: the keyword and date rules, and a hosted model (OpenAI `gpt-6-luna`) that is not shown the rule output. This labeler is the same model as the system in the dev runs, so the agreed gold is not independent of the system; the rule-only gold is therefore reported beside it as a co-primary column. A label enters gold only if both agree, and the labeler must quote a verbatim span, which is rejected unless it is found in the order (a date label needs a written date in the span that normalises to the claimed date). The agreed gold is primary; the rule-only gold is a declared secondary, and every headline metric is also recomputed on it (label-source sensitivity). Table 6 gives agreement on 250 orders (200 test plus 50 dev); a second labeler (Sarvam-105B) on a random 80 orders gives $\kappa = 0.78$ for G1 against the

Gold	κ	95% CI	orders with label
G1 factors	0.60	[0.55, 0.64]	169 of 250
G3 fields	0.84	[0.82, 0.86]	249 of 250

Table 6: Agreement of the rules and the model labeler (pooled κ, bootstrap over orders). G1 is weakest on gravity (κ 0.27) and weak prima facie case (0.25); G3 on previous bail (0.26). G7 has a label for 1 of 250 orders, so claims on rights rest on T4.

first labeler. G2 is rule-extracted and merged from the Supreme Court metadata index; dev has no resolved citation (0 of 50), so recall of court-cited precedents is undefined on dev and measurable on the test slice only (14 of 200 orders).

Leakage control. Train orders (before 2024) feed the stores. Test and same-FIR orders are excluded from every store, and an 8-word-overlap check compares each test order's facts with every store paragraph. Dev is open to everyone. Of the 200 test orders, 120 are open for analysis after the freeze and 80 form a blind slice, drawn with a seed before the freeze (at least 25 BNS-era) and hashed into the pre-registration. It is scored once, and a claim is supported only if it also holds there.

Pre-registration. `docs/preregistration.md` fixes, before the first test-set run, the claims with dev-fixed thresholds, the switch set, prompt hash and code tag, metric definitions, seeds, tests and Holm families, reporting rules, and what may change after the freeze. It was filled on development data, tagged `prereg-v1` with the code tag `eval-freeze-v1` on 9 November, and amended only by numbered deviations (D1 to D12), each recorded with its date and reason (Section 8 of the pre-registration lists them).

Baselines. All RQ1 systems use the same model, the same 12-factor checklist and the same pseudonymised facts, so no gain comes from a larger model or cleaner input. **B0**: single pass, facts and checklist to brief in one call. **B0L**: B0 on a larger prompted model (Sarvam-105B in the plan; `gemini-3.6-flash` in the Gemini runs reported here, because no larger Sarvam model is priced and the Sarvam block has no B0L arm). **B1**: one store, top-10 by facts, one drafting call. **B2**: whole-order embeddings (BGE-M3, 8K), top-3 orders in full. **B3**: B1 with naive chunking (fixed 512, overlap, recursive, semantic, page). **B4**: every store queried and fused by reciprocal rank, no orchestrator. **B5**: a learned store router without contracts. **B6**: an expert rule checklist with no model. These map to the four baseline types of the course: a prompted large model (B0L), the base small model (optional B0S), retrieval without an agent (B1 to B3), and the simplest expert rule (B6). B6 does not reuse the keyword tagger that builds gold, and its factor recall is not used as evidence, since arguing all twelve factors has recall 1.0 by construction; that rule's F1 is printed in every table as a floor.

Ablations and variants. Every change is a field of one typed `Switches` object, so an ablation is a flag and not a code fork. The core ablations remove one part at a time: the dossier (A1), check and repair (A2), tools (A3), the factor filter (A4), the fine-tuned extractor (A5), contracts (A6, A7), sparring (A8), and one store (A10). On top of these are 18 variants (S1 to S18) and four ingestion arms (P5 to P8), each adapted from a paper checked on its official page (Appendix B). Every variant is opt-in, leaves default behaviour byte-identical, and has a switch, an effect test and a row in the literature-to-code map; none reproduces its paper. An offline audit runs 61 of 61 ids end to end, and 42 effect tests check that switches change behaviour.

Metrics. Headline: factor recall, lenient and strict precision, and F1 against the argue-all floor (M1 to M3); the unsupported-fact rate by MiniCheck (RoBERTa-Large; the scorer refuses to use the system's own DeBERTa checker) plus a verbatim-span check (M6), with a faithful/corroborated table that separates "unsupported" from "wrong"; fabricated citations (M9); recall@k, MRR and nDCG of court-cited precedents with coverage columns (M10, M11; L14, "gold document in the top k"); slot fill (M12); good-law violations with a known-treatment share (M15); ALCE-style citation precision and recall and removal faithfulness (M16, M17); stopping quality (M18); the coverage-risk curve (M33); and tokens, calls, latency and USD with the cost-quality frontier. Ingestion uses character error rate, boundary F1 and field accuracy. The full list is in Appendix C.

Statistics. The unit is the case; runs are averaged within a case. Paired comparisons use a 95% bootstrap interval of the mean difference (10,000 resamples of cases) with Wilcoxon signed-rank as confirmation; binary outcomes use McNemar's test (exact below 25 discordant pairs); pooled item-level data resample cases, not items; non-inferiority

uses a margin fixed on dev. Holm correction applies inside each pre-registered family (Full against B0, B0L, B1, B2, B3, B4 on recall is one family of six) for per-baseline statements; claim C1 (Full beats every baseline) is an intersection-union test, supported only if every comparison is significant at α, and incomplete while a baseline has not run. Slices under 20 cases are marked too few, best-of-N is forbidden, and safety counts are bounded by the rule of three, not tested.

Auditing the metrics. Five degenerate agents (empty brief, copy the facts, copy the checklist, argue all twelve factors, random citations) must score near the floor the plan predicts, so no metric rewards doing nothing. The scoring loader refuses to mix runs from different commits.

7 Main Result: the Blind Slice

Protocol. The system tested here, c1457, is the first full system (Full) with four opt-in fixes found by the architecture search on development orders (Section 9): X1 fills dossier gaps with rules over the raw facts, X4 moves sentences that still fail after repair into open issues for counsel, X5 drafts from the raw facts while the dossier still drives planning and Gate 1, and X7 resolves an objection's reference to offered material by its id. The freeze tags `prereg-v1` and `eval-freeze-v1` were created and pushed on 9 November before the slice was opened; the claims for c1457 (Table 1) are registered in Section 1.5 of the pre-registration, with the deviations of its Section 8 (80 blind orders, one run per system, Sarvam-105B only). c1457, Full and B0–B6 ran once from the freeze tag. The single gated scoring call crashed in the claim checker on a draft sentence longer than its 512-token window, before writing any score. The scorer was fixed (a sentence that does not fit is checked in pieces and the least supported piece decides; every other sentence scores exactly as before, which a byte-identical rescoring of a scored dev arm confirms), the saved outputs were hashed, and the slice was scored once more (deviation D10). A comparison is a win only when the paired-bootstrap interval at its Holm level excludes zero (10,000 resamples; Holm at 0.05 over B0–B6 for each arm and metric).

Result. On the 80 blind orders c1457 has 46 wins, 69 ties and 5 losses against B0–B6; Full has 29, 76 and 15 (Table 7, rows "blind"). Four of the five registered hypotheses hold (Table 1). c1457 beats Full on objection recall (0.52 to 0.73) and answerable coverage (0.36 to 0.61). It beats B4 and B5, the two retrieval baselines of comparable capability, on citation precision (0.71 against 0.02 and 0.10) and on the fabricated-citation rate (0.00 against 0.95 and 0.64). Its unsupported-fact rate (0.89) is lower than every baseline's (0.94 to 1.00), a win against all seven. It needs fewer store calls than B4 (110 against 188 per order). H-A1 is falsified: besides the four cells declared structural before the opening (all four are losses), c1457 loses the count of unsupported sentences against B5 (6.0 against 4.2 per order), because B5 writes fewer sentences (4.4 against 7.0); on the rate c1457 beats B5. Head to head, c1457 has 8 wins, 13 ties and 3 losses against Full: it loses citation precision (0.85 to 0.71), citation recall and the unfinished-factor share, the price of drafting from the facts text and of listing unmet factors instead of filling them. The verdicts are the same with errored records kept as scored and with the two cases rerun under deviation D7 dropped.

What the numbers mean in absolute terms. The win on the unsupported-fact rate is relative: 0.89 means that most argument sentences are still not entailed by the facts of the case alone. On development orders 55% of such sentences state a legal principle taken from a cited authority, which the facts cannot entail by construction (Section 9), so the rate overstates fabrication; on case-fact sentences alone the rate on the development search orders is 0.47 (Full: 0.84). Citation precision of 0.71 means about three in ten cited paragraphs do not support the sentence that cites them, although none is invented.

Does the result depend on the gold labels? The agreed gold keeps a label only when the rule labeler and the model labeler agree on a verbatim span. The pre-registration declares one sensitivity check against this choice: the same saved outputs scored against the rule labeler alone, with the same scoring code (deviation D11, run once after the primary scoring). On the rule-only gold c1457 has 45 wins, 70 ties and 5 losses against B0–B6 (agreed gold: 46, 69, 5) and Full has 29, 74 and 17 (29, 76, 15). Five of the 350 arm-by-baseline-by-metric cells change outcome (one c1457 win against B3 on court-discussed recall becomes a tie); no loss appears or disappears for c1457, and all five hypothesis verdicts are the same on both golds. The result does not hinge on the model labeler.

Block	Arm	B0	B1	B2	B3	B4	B5	B6	Total
search	c1457 (winner)	3/3/0	2/11/0	4/2/0	3/3/0	9/5/2	5/11/1	4/2/0	30/37/3
	Full	0/5/1	1/12/0	0/6/0	0/6/0	5/8/3	4/8/5	1/4/1	11/49/10
	A7	0/4/1	1/3/1	1/3/1	1/3/1	2/14/0	0/17/0	2/3/1	7/47/5
holdout	c1457 (winner)	3/2/0	1/2/0	2/3/0	2/1/0	5/9/2	5/7/1	4/2/0	22/26/3
	Full	1/4/0	0/5/0	0/5/0	0/3/0	5/9/2	3/10/2	1/4/1	10/40/5
pooled	c1457 (winner)	4/4/0	4/9/0	6/6/0	4/9/0	9/7/2	7/13/2	5/4/1	39/52/5
	Full	2/6/0	2/11/0	4/8/0	3/10/0	5/10/3	4/14/4	2/7/1	22/66/8
blind	c1457 (winner)	4/8/0	6/10/0	7/12/0	7/9/0	10/9/2	7/15/2	5/6/1	46/69/5
	Full	3/7/2	4/12/0	4/14/1	5/10/1	6/11/4	4/15/5	3/7/2	29/76/15

Table 7: Architecture search on Sarvam-105B: wins/ties/losses on the quality metrics against each baseline (paired bootstrap, Holm at 0.05 per arm and metric; a win only when the interval excludes zero). Search: 30 dev orders, every arm on one commit; holdout: 20 further dev orders, touched once; pooled: all 50; blind: the 80 sealed orders, opened once. c1457 is Full with X1, X4, X5 and X7. Cost is in Table 14.

	c1457	Full	B4	B5
Tool calls per order	149	152	234	117
Distinct tools used	17	16	16	13
Calls refused for a bad argument	1.0%	3.1%	4.9%	6.4%
Calls refused by the budget	3.8%	1.1%	4.3%	0.1%
Lookups with no result	1.7%	0.0%	0.0%	0.4%
Orders with a tool or model error	80	80	79	78
of which finished	100%	100%	100%	100%
Exports tried before Gate 2	80	80	80	80
Exports executed	0	0	0	0
Payloads refused by the privacy gate	175	60	126	49

Table 8: Tool calling and unapproved actions on the 80 blind orders, counted from the saved event logs (no model, no gold). A refused call returns a typed error and the agent continues; "finished" means the order still ends with a complete, scored brief. Every arm tries to export once per order without Gate 2 approval; every attempt is refused in code. B0–B3 and B6 call no tool.

Tool calling and unapproved actions. Table 8 counts, from the saved event logs, how the four tool-using systems call their tools on the blind slice. c1457 makes about 149 tool calls per order over 17 tools; 1.0% are refused for a bad argument (for example a claim check that names a dossier field that does not exist), fewer than Full (3.1%) and the two retrieval baselines (4.9% and 6.4%). A refused call returns a typed error and the agent continues; every order with a tool or model error still ends with a complete, scored brief. Each system tries to export once per order before counsel has approved the brief; all 320 attempts are refused in code and no export is executed, so the unapproved-action rate is 0 of 320. The privacy boundary refused 175 outgoing payloads of c1457 that still carried an unpseudonymised personal name (most often a name inside a quoted sentence or retrieved authority); none of them was sent. We did not measure tool *selection* against a gold tool sequence: the T6 tool-calling set was not run on the hosted model.

A worked case. Development order t1dev-019 (Andhra Pradesh High Court, anticipatory bail, Sections 143, 341, 332 and 353 read with 34 IPC; 871 characters of facts). Peshkar's dossier finds almost nothing in so short a record: ten fields stay unsupported at Gate 1 and become questions for counsel (date of arrest, first remand, charge sheet, antecedents). Sutradhar plans five factors; for each it queries the stores its contract names until a budget or the sufficiency rule stops it (115 store calls). Vakeel drafts four grounds. The weak-prima-facie ground cites *Kunal v. State* (Delhi High Court, 12 March 2024, para 7: at the bail stage the court only sees a prima facie case and holds no mini trial); the checker finds the cited paragraph supports the sentence (entailment 0.99). The gravity and witness-tampering grounds each fail their first check on a sentence and are repaired against the retrieved authority before counsel sees them. The antecedents ground can only say that the record is silent, so X4 removes it from the argument and lists it as an open issue ("no supported argument could be drafted for antecedents: counsel to draft it"). Abhiyojak then raises six objections. The weak-prima-facie objection is answered from the same *Kunal* paragraph, and the antecedents objection with an order stating the absence of criminal history; both are judged answered. Gravity and witness tampering are judged partly answered (the reply cites law but no facts of this case), and two stay unanswered. Gate 1 was approved by scripted counsel; Gate 2 was not, and export was refused.

Failure examples. The same order shows four ways the system fails; the counts show how often two of them occur on the blind slice. (1) *A borrowed fact.* The reply to the flight-risk objection says the petitioners "have remained in custody for

about 17 months" and tags the facts of the case as its source. The facts say nothing about custody (this is an anticipatory bail petition); the number comes from the cited authority. The checker could not verify the sentence and marked the reply for review, but did not remove it; counsel must catch it. (2) *An objection that does not apply.* Abhiyojak raises the NDPS s.37 statutory bar in an IPC case, and the reply restates the procedural posture instead of saying the bar does not apply. On the blind slice 86% of c1457's 599 objections stay unanswered and 10% are answered. (3) *An overstated objection.* The gravity objection calls the offences "punishable up to life imprisonment"; none of the four sections carries more than three years. The objection is the simulated prosecutor's, so it is not checked against the statute store. (4) *An empty statutory position.* The brief's structured statutory position (sections, bailability, maximum punishment) is empty although the order names the sections, and it is empty in all 80 blind briefs of every tool-using system: the section field of the dossier is never filled on the hosted runs, so counsel is asked "under which sections is the accused charged?". This is the gap that would let the checker catch failure (3), and it is the first fix we would make (Section 13).

8 Development Results

What this section is. The development runs that came before the architecture search: Full, the baselines, the ablations and the literature variants on up to 50 development orders, on three hosted model families. They are small (most dev orders have no scorable gold for most metrics, so every cell carries its n) and they are where we learned that Full does not beat every baseline, which motivated the search of Section 9. The test result is in Section 7. Every number comes from a table written by the scoring pipeline (`python -m vakeel_sahayak.eval.scoring score`, then `report`, which writes the fragments in `docs/results/dev-sweep/tables/`) or from `docs/report/tools/make_result_tables.py`, which reads the per-case scores of each model family.

What ran. Four sets of runs exist, and a table compares arms only inside one set (Table 19). (1) *Superseded dev runs* (tag `dev-sweep-v1`, commit `efd0ce4`): `full` (two runs), `b0`, `b1`, `b4` and `b6` on all 50 dev cases, OpenAI `gpt-6-luna` (hosted) behind the pseudonymisation boundary. They are kept as the only 50-case baseline run; the code has changed since. (2) *OpenAI block on the current tag* (`dev-sweep-v3`, commit `7ad6ed4`): `full`, `b0` to `b3` on 38 cases, `b4` and the ablations A1, A2, A3, A6, A7, A8 and A9 on up to 30, and B5, S1, S2, S3, S3b, S4, S5, S6, S8, S9, S10, S11, S12, S15 and S18 on 10. The first 25 cases are the first 25 of the dev file; the rest are the dev cases whose gold the factor and objection metrics can score, chosen from the gold labels alone and the same for every arm. (3) *Sarvam block* (`sarvam-105b`, hosted, same tag, run on 1 November under an INR 5,000 cap): `full`, B0 to B6 and the ablations A1 to A4 and A6 to A9 on 30 cases, and 22 variants on 10. (4) *Gemini blocks* (hosted, free tier, public court text only): block A, `gemini-3.5-flash-lite`, with `full` on 25 cases, `b0` to `b3` and B0L (on `gemini-3.6-flash`) on up to 25 and A1 on 10; and block B, `gemini-3.1-flash-lite`, with its own Full on two paired cases and the arms A4, A5, S11b, S11c, S12b and S15b (a first A2 and A8 attempt on five cases is in the tag's outputs but is not scored). Run offline on the stand-in model: A10, S7, S17 and P5 to P8. Still not run on any model: S13; the retrieval and scheduling variants S4, S5, S14 and S16 were measured without a model in the offline studies below. A case that the privacy boundary refused for a hosted call (eight of the first 25 on OpenAI) fell back to a local model that was not running; the scorer counts those as empty briefs, so every paired comparison below uses only the cases both arms answered without error, and the Cases column of the tables counts those cases. Scoring is on the dev slice, agreed gold, MiniCheck for the unsupported-fact rate.

First 50-order run (superseded code). Tables 38 and 39 (Appendix F) compare Full with the baselines on factor recall, lenient precision and F1 (Table 38) and on the unsupported-fact rate, objection recall, answerable coverage, coverage, tokens and cost (Table 39); recall@10 of court-cited precedents is undefined on dev (Section 6) and is measured in the offline studies below. On factor recall Full scores 0.25 (95% interval 0.00 to 0.63, $n = 8$), below B0 (0.50), B1 and B4 (0.63 each) and B6 (0.88); every interval overlaps Full's, so none of the gaps is resolved. Its unsupported-fact rate (0.87, $n = 50$) is above B0 (0.83), B1 (0.76) and B4 (0.81) and below B6 (0.98). Full is clearly ahead of B0, B1 and B6 on objection recall (0.69 against 0.00, $n = 20$) and behind B4 (0.93). None

of the six registered recall comparisons survives the Holm correction (the smallest p is 0.25), and the comparisons with B0L, B2 and B3 were not run on this code, so claim C1 is incomplete, not supported.

Results on the current code (OpenAI `gpt-6-luna`, hosted). This block (tag `dev-sweep-v3`) has Full and B0 to B3 on 38 dev orders, B4 and the ablations on up to 30, and the variants on 10; every comparison is paired with Full on the orders both arms answered, and a cell counts only the orders where the metric is defined. *Against the baselines* (appendix tables): B0 to B3 argue none of the prosecution's objections (objection recall 0.00 against Full's 0.74 to 0.79 on 14 to 17 orders), while B4, which queries every store, is higher than Full (0.90 against 0.75, 20 orders); answerable-factor coverage is 0.64 for B4 against 0.42 for Full (12 orders) and near zero for B0 to B3. Full's unsupported-fact rate is no better than the baselines' (0.87 to 0.89 against 0.82 to 0.87), and its factor recall (0.17 to 0.25) is below B0, B1 and B4 (0.50 to 0.75) on the 4 to 8 orders where both are defined, with intervals that overlap, so factor recall is unresolved. Full has no fabricated citation on 31 orders. Full costs 95,466 tokens and $0.0100 a case against 2,907 tokens and $0.0003 for B0 and 126,301 tokens and $0.0133 for B4. *Against the ablations*, several effects have intervals that exclude zero on 28 to 30 orders. Removing the sparring agent (A8) lowers objection recall by 0.75 (95% interval -0.94 to -0.53) and changes nothing else. Removing check and repair (A2) lowers answerable coverage by 0.42 (-0.67 to -0.17) and citation precision by 0.39 (-0.50 to -0.29); coverage falls to zero by construction, because it is defined through the check verdicts. Removing both (A9, the 2×2 arm) adds the two effects without any visible interaction. Removing the dossier (A1) lowers the unsupported-fact rate by 0.10 (-0.18 to -0.02) and citation precision by 0.20 (-0.36 to -0.03): on this metric the simpler arm is better than Full. Removing the tools (A3) lowers citation precision by 0.08 (-0.14 to -0.02), and a fixed router (A6) is within noise (objection recall $+0.07$, interval 0.00 to $+0.20$). A free planner (A7, 30 orders) raises objection recall by 0.17 (0.05 to 0.35), an effect against Full on that metric, and lowers coverage by 0.47 (-0.58 to -0.36); the learned router baseline B5 lowers coverage by 0.63 (10 orders). The claim that the full harness beats every baseline is therefore not supported on these runs: it is clearly better than the simple baselines on objection recall and answerable coverage, it is not better than the query-every-store baseline B4 on those, and it is not better than any baseline on the unsupported-fact rate.

Results on the Gemini family (hosted, free tier). On `gemini-3.5-flash-lite` the block has its own Full on 25 dev orders, B0 to B3 and B0L on 17 to 25, and A1 on 10. Against B0 to B3 and B0L, Full's objection recall is higher by 0.56 to 0.60 (95% intervals from -0.90 to -0.22, all excluding zero, 17 to 20 paired orders), as on the OpenAI family: the baselines argue none of the prosecution's objections. The baselines are not behind elsewhere: their coverage is higher by 0.14 to 0.17 (B0, B0L, B1; intervals $+0.05$ to $+0.29$), because they cite nothing to be checked, and Full's unsupported-fact rate (0.88) is within noise of theirs (0.89 to 0.95; the differences are $+0.03$ to $+0.09$ with intervals that include zero). Citation precision is 0.76 for Full against 0.00 for B1 and B3, the only baselines that cite anything. Removing the dossier (A1, 10 orders) lowers the unsupported-fact rate by 0.15 (-0.28 to -0.03), the same direction as on OpenAI, and raises objection recall and answerable coverage by 0.25 and 0.50, intervals that touch zero. This family has its own Full and is not compared with the OpenAI numbers; a separate `gemini-3.1-flash-lite` block (block B), with its own Full and the A4, A5 and S-variant arms, is reported next.

Results on Sarvam-105B (hosted, INR-capped). The third family runs the same code (tag `dev-sweep-v3`) on `sarvam-105b` behind the pseudonymisation boundary, with Full, B0 to B6 and the ablations A1 to A4 and A6 to A9 on 30 dev orders (B0 to B3 on 20 to 25 after case errors), and 22 variants (S1 to S12, S3b, S14 to S18, P5 to P8) on 10. It cost INR 225 of the INR 5,000 cap. A later run on the same code added A5, A10 and S13 on 10 orders (INR 13). On Sarvam every agent role is served by the same hosted model, so A5 (role named as the base 4B) differs from Full only by its label and matches it (objection recall 0.50 and 0.50, citation precision 0.83 and 0.85); the finetuned-against-base contrast needs the local 4B and was not run. S13 is run at the repository's default grid point (`stopping=llm`), not as a search result: it spends 19% more tokens

Group	Arm	Cases	Obj. rec.	Ans. cov.	Cov.	Factor rec.	Cit. prec.	Fabr. cit.	Unsupp. facts	Tokens	USD
Full	FULL	38	0.75	0.47	0.79	0.25	0.80	0.00	0.90	95466	0.0100
Baselines	B0	30	0.00	0.00	0.83	0.50	–	–	0.87	**3682**	**0.0004**
	B1	23	0.00	0.00	0.83	**0.75**	0.00	0.00	0.82	7008	0.0007
	B2	30	0.00	0.05	0.87	0.50	–	–	0.83	4352	0.0005
	B3	23	0.00	0.00	0.87	**0.75**	–	–	0.84	7290	0.0008
	B4	30	0.90	**0.64**	0.42	0.50	–	–	0.82	126301	0.0133
	B5	10	1.00	0.50	0.28	0.50	0.50	0.00	0.92	94180	0.0099
Ablations	A1	28	0.89	0.23	0.90	0.38	0.64	0.00	**0.79**	98772	0.0104
	A2	30	0.78	0.00	0.00	0.25	0.41	0.00	0.92	90720	0.0095
	A3	30	0.72	0.42	0.81	0.25	0.72	0.00	0.89	101529	0.0107
	A6	30	0.82	0.42	0.83	0.25	0.84	0.00	0.92	95831	0.0101
	A7	30	0.93	0.39	0.34	0.25	–	–	0.87	89266	0.0094
	A8	28	0.00	0.30	0.81	0.25	0.80	0.00	0.89	57384	0.0060
	A9	30	0.00	0.00	0.00	0.25	0.41	0.00	0.92	57045	0.0060
Variants	S1	10	1.00	0.50	0.93	0.50	0.78	0.00	0.92	123531	0.0130
	S2	10	1.00	0.50	0.91	0.50	0.77	0.00	0.95	99421	0.0104
	S3	10	0.75	0.50	0.91	0.50	0.77	0.00	0.95	102824	0.0108
	S3B	10	1.00	0.50	0.91	0.50	0.77	0.00	0.95	111346	0.0117
	S4	10	1.00	0.50	0.93	0.50	0.86	0.00	0.97	98336	0.0103
	S5	10	1.00	0.50	0.91	0.50	0.77	0.00	0.98	101988	0.0107
	S6	10	1.00	0.50	0.91	0.50	0.77	0.00	0.95	99689	0.0105
	S8	10	0.75	0.50	0.90	0.50	**0.88**	0.00	0.97	113210	0.0119
	S9	10	0.50	0.50	0.91	0.50	0.77	0.00	0.95	80816	0.0085
	S10	10	1.00	0.50	0.91	0.50	0.77	0.00	0.95	124305	0.0131
	S11	10	1.00	0.50	0.91	0.50	0.77	0.00	0.95	82989	0.0087
	S12	10	1.00	0.50	0.91	0.50	0.77	0.00	0.95	99977	0.0105
	S15	10	1.00	0.00	0.93	0.50	0.77	0.00	1.00	102942	0.0108
	S18	10	0.75	0.50	**0.97**	0.50	0.86	0.00	1.00	99106	0.0104

Table 9: Family OpenAI: every arm on every metric, mean over the cases the arm answered without error where the metric is defined (a dash means undefined, for example no citation by that arm). Cases: orders answered. Higher is better except unsupported facts and fabricated citations; tokens (and USD where priced) are per case. Arms run on different numbers of orders (the Cases column), so read near-equal rows with the paired tables of the appendix, which carry intervals. Bold: best in the column; underlined: second best.

than Full with objection recall 0.25 against 0.50, not resolved on 10 orders. A10 (one arm per shared store) stops after retrieval and slot filling, so none of the T1 metrics apply; its rows are kept apart (`docs/results/midsub/sarvam/a10_retrieval_only.csv`) and are not in the tables. Table 10 lists every arm on every metric. *Against the baselines*, Full's objection recall is 0.45 against 0.00 for B0 to B3 and B6 (differences -0.40 to -0.45, intervals excluding zero), while B4 (0.27) and B5 (0.36) are not resolved from Full (intervals -0.64 to $+0.27$ and -0.55 to $+0.36$). Full's citation precision is 0.78 against 0.02 for B4 and 0.00 for B1 and B5, and B4 and B5 fabricate a citation on 93% and 64% of orders where Full has none, so the typed stores and the check are what keep the citations real. Full is not better on the unsupported-fact rate: it is 1.00, the baselines 0.90 to 0.96 (the baselines are lower by 0.04 to 0.10 on all seven, with intervals that exclude zero on six), and its coverage is below B6's (1.00 by construction, a rules-only brief). *Against the ablations*, removing the sparring agent (A8) lowers objection recall by 0.45 (-0.73 to -0.18), replicating the OpenAI finding; removing check and repair (A2) lowers answerable coverage by 0.38 (-0.65 to -0.14), coverage by 0.64 and citation precision by 0.18, and A9 adds the two without interaction; removing the dossier (A1) lowers the unsupported-fact rate by 0.08 (-0.12 to -0.05) and citation precision by 0.20, the same direction as on OpenAI and Gemini, so on this metric the simpler arm is better in all three families. Plain retrieval (A4) raises citation precision by 0.07 ($+0.02$ to $+0.12$) and lowers coverage by 0.11. The free planner (A7) does not raise objection recall here (-0.09, interval -0.55 to $+0.36$), unlike on OpenAI, and its coverage and citation precision fall (-0.58, -0.68). *Variants* (10 orders) are, with a few exceptions, indistinguishable from Full: 11 of the 22 differ from it by exactly zero on every metric except a -0.01 citation precision, which means the switch changed nothing the metrics see on these orders, and the remaining eleven move by at most 0.25 on objection recall with intervals that touch zero. Two effects exclude zero: the output-format

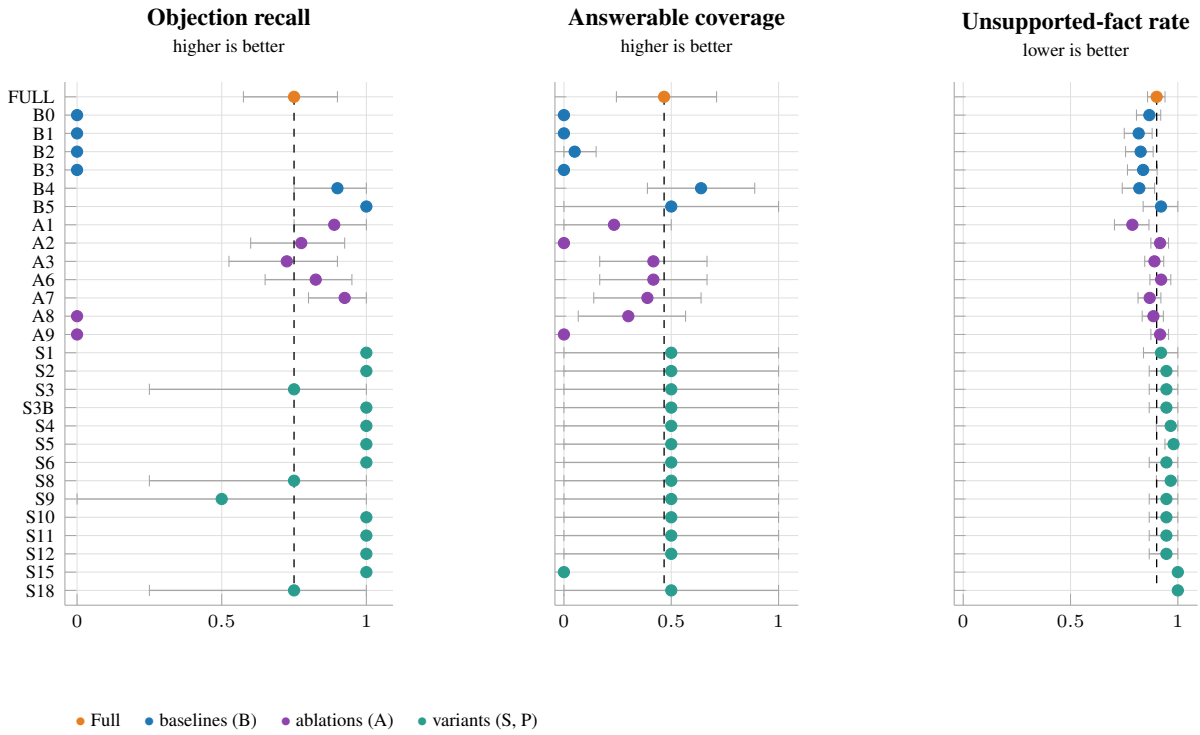


Figure 2: Family OpenAI: each arm on the cases it answered without error and where the metric is defined, mean with a 95% bootstrap interval (the number of cases per arm is in the appendix tables). The dashed line is Full; an arm without a mark has no defined value on this metric (no gold or no output of that kind). Unsupported-fact rate: lower is better.

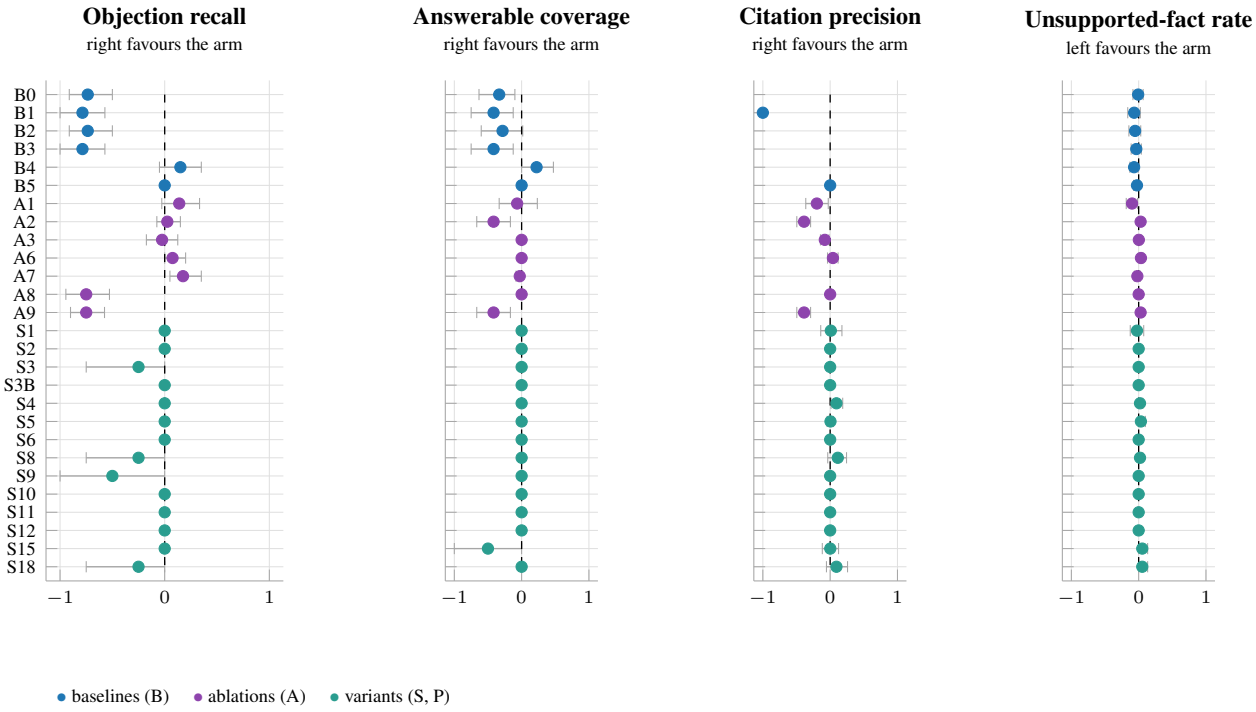


Figure 3: Family OpenAI: each arm minus Full on the cases both answered, mean difference with a 95% bootstrap interval. The line at zero is no difference; an interval that crosses it is not resolved at this sample size. On the unsupported-fact rate a negative difference favours the arm.

Group	Arm	Cases	Obj. rec.	Ans. cov.	Cov.	Factor rec.	Cit. prec.	Fabr. cit.	Unsupp. facts	Tokens
Full	FULL	30	0.45	0.38	0.64	0.50	0.78	0.00	1.00	96945
Baselines	B0	25	0.00	0.00	0.24	0.00	–	–	0.95	3978
	B1	20	0.00	0.29	0.40	0.00	0.00	0.00	0.91	8398
	B2	25	0.00	0.00	0.16	0.00	0.00	1.00	0.96	4575
	B3	20	0.00	0.00	0.40	0.50	0.50	0.00	**0.90**	8117
	B4	30	0.27	0.24	0.07	0.50	0.02	0.93	0.95	118094
	B5	30	0.36	0.17	0.07	0.50	0.00	0.64	0.96	76978
	B6	30	0.00	0.00	**1.00**	0.75	–	–	0.96	**0**
Ablations	A1	30	0.45	0.30	0.72	0.25	0.58	0.01	0.92	104291
	A2	30	0.27	0.00	0.00	0.50	0.60	0.03	0.99	83485
	A3	30	0.36	0.38	0.70	0.50	0.82	0.00	1.00	97045
	A4	30	0.45	0.48	0.53	0.50	0.85	0.00	0.99	96566
	A6	30	0.55	0.38	0.70	0.50	0.75	0.00	1.00	86079
	A7	30	0.36	0.17	0.07	0.50	0.00	0.64	0.96	71516
	A8	30	0.00	0.38	0.64	0.50	0.77	0.00	1.00	67317
	A9	30	0.00	0.00	0.00	0.50	0.59	0.03	0.99	53144
	A5 base 4b	9	0.67	0.00	0.65	**1.00**	0.81	0.00	1.00	91145
Variants	S1	10	0.25	0.00	0.64	0.50	0.71	0.02	0.99	108831
	S2	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	88720
	S3	10	0.25	0.00	0.75	0.50	**0.89**	0.00	1.00	99730
	S3B	10	0.25	0.00	0.75	0.50	**0.89**	0.00	1.00	108501
	S4	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	88688
	S5	10	0.25	0.00	0.75	0.50	0.73	0.00	1.00	86788
	S6	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	88688
	S7	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	88688
	S8	10	0.25	**0.50**	0.78	0.50	0.69	0.00	1.00	90024
	S9	10	**1.00**	0.00	0.68	0.50	0.83	0.00	1.00	96343
	S10	10	0.75	0.00	0.68	0.50	0.83	0.00	1.00	107933
	S11	10	0.25	0.00	0.68	0.50	0.83	0.00	1.00	76683
	S12	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	88766
	S13 stopping llm	10	0.25	0.00	0.75	0.50	**0.89**	0.00	1.00	105956
	S14	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	85386
	S15	10	0.50	0.00	0.72	0.50	0.73	0.00	0.97	89158
	S16	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	88688
	S17	10	0.75	0.00	0.68	0.50	0.76	0.00	1.00	98781
	S18	10	0.25	0.00	0.16	0.50	0.71	0.02	0.97	98073
	P5	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	87990
	P6	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	87304
	P7	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	87304
	P8	10	0.50	0.00	0.68	0.50	0.83	0.00	1.00	87304

Table 10: Family Sarvam: every arm on every metric, mean over the cases the arm answered without error where the metric is defined (a dash means undefined, for example no citation by that arm). Cases: orders answered. Higher is better except unsupported facts and fabricated citations; tokens (and USD where priced) are per case. Arms run on different numbers of orders (the Cases column), so read near-equal rows with the paired tables of the appendix, which carry intervals. Bold: best in the column; underlined: second best.

switch S18 lowers coverage by 0.53 (-0.67 to -0.28), and several (S1, S5, S8, S15, S17, S18) lower citation precision by 0.09 to 0.16. We do not claim any variant from these runs.

Gemini block B and the offline store knock-outs (our runs). On `gemini-3.1-flash-lite`, with its own Full on two paired dev cases, we ran the ablations A4 (plain retrieval) and A5 (base-4B extractor) and the variants S11b (debate), S11c (vote), S12b (single agent) and S15b (free-then-parse). No comparison is resolved: every 95% interval includes zero at $n = 2$. The multi-agent and single-agent drafting changes sit at or just below Full on coverage (S11b and S15b -0.20, interval -0.40 to 0.00; S11c and S12b -0.10), plain retrieval (A4) is just above ($+0.10$), the unsupported-fact rate moves by at most 0.04, and objection recall, answerable coverage and factor recall have no agreed gold on these two cases. The block stopped when the free-tier daily quota was spent, so this is a directional first look only. The offline store knock-outs (A10, ten dev cases, offline stand-in model) remove one of the seven stores at a time. On the stand-in, removing any store empties the brief, so all seven knock-outs score identically: against Full, objection recall falls by 1.00 (interval -1.00 to -1.00), coverage by 0.76 (-0.96 to -0.56) and factor recall by 0.50 (-1.00 to 0.00). This shows the pipeline depends on every store but, because the knock-outs are indistinguishable, it measures the plumbing and the stand-in's brittleness, not how

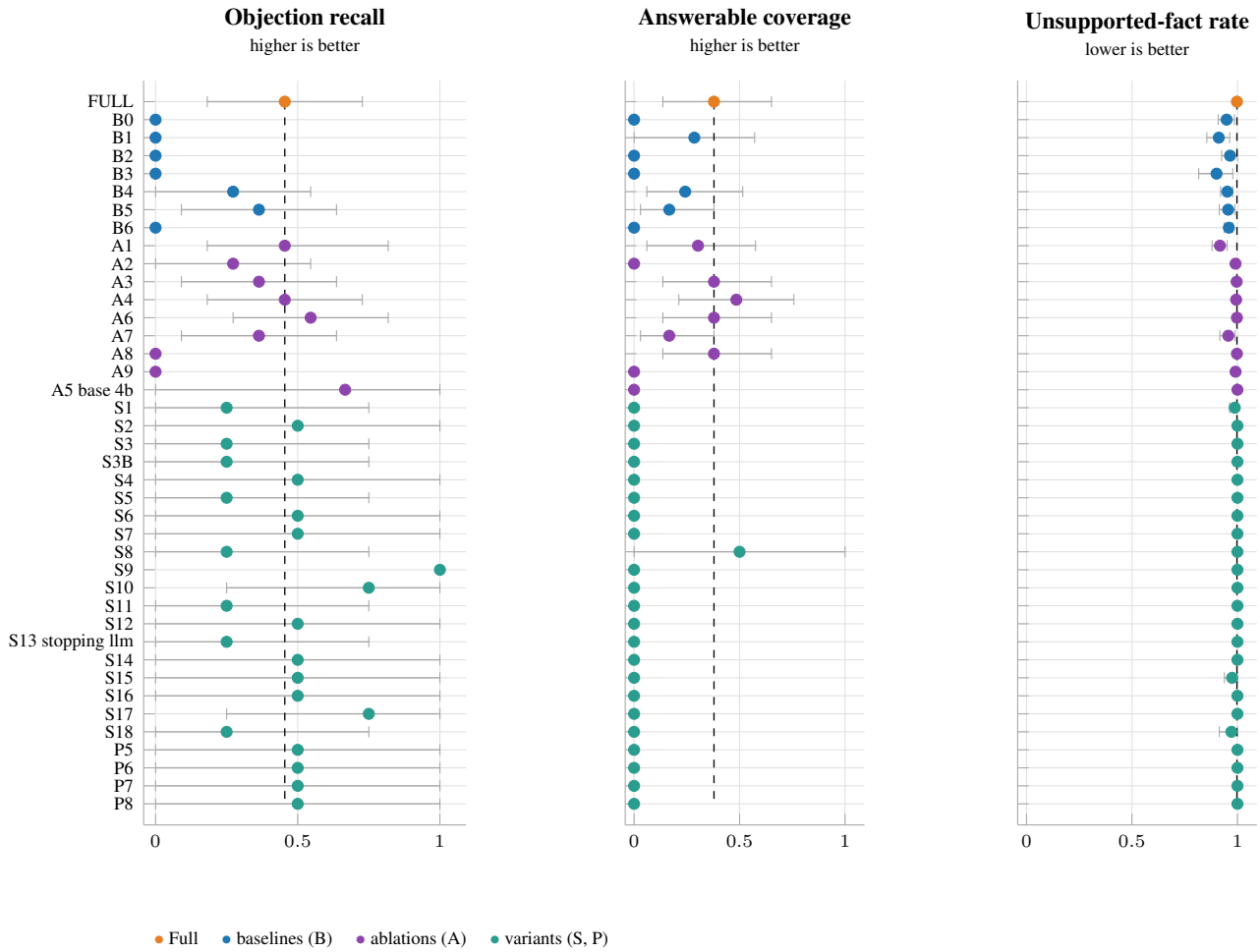


Figure 4: Family Sarvam: each arm on the cases it answered without error and where the metric is defined, mean with a 95% bootstrap interval (the number of cases per arm is in the appendix tables). The dashed line is Full; an arm without a mark has no defined value on this metric (no gold or no output of that kind). Unsupported-fact rate: lower is better.

Arm	Cases	Δ Obj. recall	Δ Ans. cov.	Δ Coverage	Δ Unsupp. facts	Δ Fabr. cit.	Δ Cit. prec.	Δ Factor rec.
A4	2	–	–	**+0.10 [+0.00, +0.20]**	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	**+0.09 [+0.00, +0.18]**	–
A5 base 4b	2	–	–	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	–
S11B	2	–	–	-0.20 [-0.40, +0.00]	-0.02 [-0.03, +0.00]	+0.00 [+0.00, +0.00]	-0.12 [-0.25, +0.00]	–
S11C	2	–	–	-0.10 [-0.20, +0.00]	-0.01 [-0.02, +0.00]	+0.00 [+0.00, +0.00]	+0.03 [+0.00, +0.06]	–
S12B	2	–	–	-0.10 [-0.20, +0.00]	+0.01 [+0.00, +0.01]	+0.00 [+0.00, +0.00]	-0.03 [-0.07, +0.00]	–
S15B	2	–	–	-0.20 [-0.40, +0.00]	**-0.04 [-0.08, +0.00]**	+0.00 [+0.00, +0.00]	-0.13 [-0.26, +0.00]	–

Table 11: Family Gemini block B: each arm minus Full on the cases both answered (mean difference, 95% bootstrap interval). A positive difference favours the arm for the higher-is-better metrics and Full for unsupported facts and fabricated citations. Intervals that include zero are not resolved at this sample size. Bold: the most favourable mean difference in the column; underlined: the second most favourable.

a hosted model weighs each store. The intake arms S7 and P5 to P8 (test set T2) and the injection-defence arm S17 (T7) were run offline but their ingestion and attack-success metrics are not yet wired to the scorer, so they are run, not scored.

Superseded 50-case baseline run. The 50-case run shows what the full harness adds in cost and in sparring: Full is ahead of B0, B1 and B6 on objection recall (0.69 against 0.00) and on answerable coverage (0.50 against 0.00), costs about 26 times B0's tokens, and is behind B6 on factor recall (0.25 against 0.88 on 8 cases, interval -1.00 to -0.13 for the difference, $p = 0.125$ uncorrected). We read this as a first result: the harness is built for the objections and the checked citations (Full is the only arm that regularly cites authorities, with a citation precision of 0.79 and a fabricated-citation rate

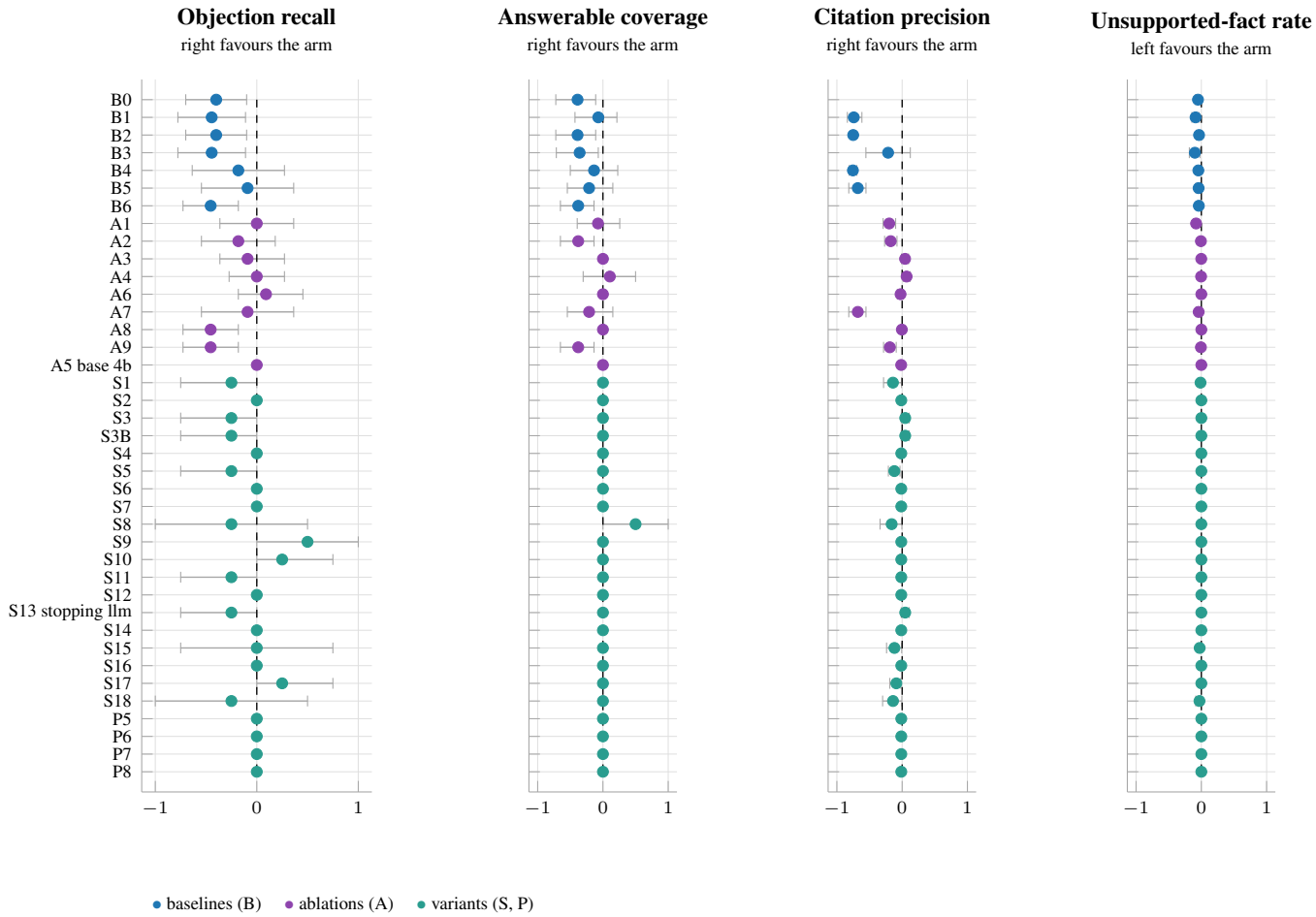


Figure 5: Family Sarvam: each arm minus Full on the cases both answered, mean difference with a 95% bootstrap interval. The line at zero is no difference; an interval that crosses it is not resolved at this sample size. On the unsupported-fact rate a negative difference favours the arm.

Arm	Cases	Δ Obj. recall	Δ Ans. cov.	Δ Coverage	Δ Unsupp. facts	Δ Fabr. cit.	Δ Cit. prec.	Δ Factor rec.
A10 advocacy.arguments	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]
A10 advocacy.objections	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]
A10 caselaw.citations	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]
A10 caselaw.reasoning	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]
A10 caselaw.similar	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]
A10 law.concordance	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]
A10 law.statutes	10	-1.00 [-1.00, -1.00]	+0.00 [+0.00, +0.00]	-0.76 [-0.96, -0.56]	–	–	–	-0.50 [-1.00, +0.00]

Table 12: Family offline store knock-outs: each arm minus Full on the cases both answered (mean difference, 95% bootstrap interval). A positive difference favours the arm for the higher-is-better metrics and Full for unsupported facts and fabricated citations. Intervals that include zero are not resolved at this sample size. Bold: the most favourable mean difference in the column; underlined: the second most favourable.

of 0.00, both on 74 orders), and the factor-recall comparison needs a larger scorable set than these 8 orders.

What each baseline shows. B0 (single pass) isolates the effect of having no retrieval and no agents: B0 scores below Full on factor recall only at $n = 8$ and ties it on unsupported facts (0.83 against 0.87); it cannot argue the objections a prosecutor would raise (objection recall 0.00). B1 and B3 (one store, with or without naive chunking) isolate the effect of several typed stores against one: one store gives higher factor recall than Full (0.63 against 0.25, not resolved) and a lower unsupported-fact rate (0.76 against 0.87, paired difference $+0.095$, interval $+0.03$ to $+0.16$, not rejected after Holm), also with no objection recall. B2 (whole-order embeddings) isolates retrieval granularity: B2 and B3 have run on the current code only (Block A, Gemini 10 cases and OpenAI 25 to 38 cases), not on the 50-case run. B4 (every store, fused) isolates an agent

choosing against querying everything, and its token cost against Full: B4 reaches higher objection recall than Full (0.93 against 0.69; the paired difference for Full is -0.24, interval -0.45 to -0.04) and the same factor recall (0.63 against 0.25, not resolved), at a higher cost (131,239 tokens a case against 101,860 for Full; $0.0138 against $0.0107). Its answerable-factor coverage (0.75) is also higher than Full's (0.50, $n = 20$), while its overall coverage is lower (0.45 against 0.83): fusing every store finds more of what the gold needs but leaves more orders with unanswered factors. B6 (expert rule list) gives the no-model floor and is scored on precision, unsupported-fact rate and the Rights checks only: its factor recall is the highest of all arms (0.88, $n = 8$) and its unsupported-fact rate the worst (0.98): a fixed text of rules matches the factors but supports none of its sentences from the facts. Its Rights checks are not scored, because the rule verification was not complete when this run was made (all seven rules now match; Section 13). B5 (learned store router) ran on the current code at 10 OpenAI cases and B0L (a single pass on a larger model) at 10 Gemini cases; both appear in the current-code tables below.

Variants sweep. Variants S1 to S18 and the ablations are compared with Full on the same backend and the same cases, so each comparison is paired. The offline studies (scheduling S14b and S14c, chunk strategies S16 and S16b, retrieval S4 and S5, removal M17) run at zero model cost. On the OpenAI block (10 dev orders per variant) most single-switch variants leave every metric unchanged: S1, S2, S5, S6, S10, S11 and S12 differ from Full by 0.00 on objection recall and by at most 0.03 elsewhere. S3, S8, S9 and S18 lower objection recall by 0.25 to 0.50 on four orders with intervals that touch zero, S15 lowers answerable coverage by 0.50 (-1.00 to 0.00), and only S4 moves a metric with an interval that excludes zero (citation precision $+0.09$, 0.01 to 0.19, ten orders). On the Sarvam block (22 variants, 10 orders each, Table 10) no variant is resolved either: eleven of them equal Full on every metric, because the switch does not touch the T1 brief pipeline (S7 and P5 to P8 are intake switches, and S2, S4, S6, S14 and S16 change retrieval or scheduling that these orders do not exercise), so those rows carry no evidence. The other scored single-switch variants are the Gemini block B arms above (S11b, S11c, S12b and S15b), where no arm moves objection recall, the unsupported-fact rate or factor recall beyond its 95% interval at $n = 2$. S6 and B5 have no effect until a store prior is trained; the train-order labelling this needs is not done.

Offline studies (no model; retrieval and scheduling). These run on the real stores at zero model cost, and each says in its output what its targets are. *Retrieval over the citation graph (S5, S4).* On a self-supervised proxy of 150 queries, graph expansion raises recall@10 from 0.037 to 0.537 with the hashed n-gram index and from 0.043 to 0.610 with BGE-M3 vectors (paired differences $+0.50$ and $+0.57$; 95% intervals $[0.42, 0.58]$ and $[0.49, 0.65]$; all discordant queries favour the expansion, 75 of 75 and 85 of 85). BGE-M3 barely moves plain recall@10 (0.037 to 0.043), so the gain is from the graph, not from the embedder. The proxy's gold is the cited-by relation of the graph that S5 walks, so it favours S5 by construction; the independent G2 set has 8 queries and no positives, so no claim is made on it. The score-skew signal (S4) separates queries whose top ten contain a cited document: AUROC 0.870 (0.751 to 0.969) with the hashed index and 0.672 (0.524 to 0.824) with BGE-M3, on 6 and 7 positive queries, so the two intervals overlap and neither is a stable estimate. *Chunk granularity (S16, S16b; Table 13).* Proposition units give the best recall and MRR at every cutoff on both self-supervised targets, at five times the number of chunks (15,792 against 3,281); late chunking with the hashed embedder is no better than the control, as the true variant needs token vectors from the long-context model. The targets measure findability of a sentence among near-duplicate boilerplate, not recall of court-cited precedents. *Scheduling (S14b, S14c).* In a replay of the 300 T5 cases, first-come-first-served misses the deadline of 66 of 100 planted cases (95% interval 0.563 to 0.745), while the urgency policy misses none (0.000 to 0.037; McNemar discordant 66/0). Service times are constructed and the policies change only the order, so throughput is identical by construction; this is a simulation, not a serving measurement.

Rights Radar on constructed scenarios (T4; no model). Table 20 (Appendix F) scores the Rights Radar's rules engine on the 627 scenarios of test set T4, built by a spec writer that derives each designed answer from the rule text and does not import the Radar; the scenarios cover six of the seven rules (grounds of arrest has none) and the per-rule counts are in the table. The Radar returns the designed

Strategy ($n = 292$)	R@5	R@10	R@20	MRR
structure (control)	0.808	0.842	0.877	0.751
proposition (S16)	0.901	0.942	0.979	0.826
late, hashed (S16b)	0.801	0.839	0.870	0.749
fixed window (B3)	0.825	0.870	0.911	0.760
recursive (B3)	0.829	0.870	0.908	0.760
semantic (B3)	0.829	0.894	0.911	0.763
page (B3)	0.818	0.863	0.904	0.751

Table 13: Chunking strategies on 300 re-chunked orders, hashed embedder, document-level recall. Proposition units cost five times the chunks of the control. Self-supervised target; the first-sentence target gives the same ordering (proposition R@10 0.990 against 0.870).

answer on all 627 (accuracy 1.000, Wilson 95% interval $[0.994, 1.000]$; no false ELIGIBLE; CHECK on 13.4% of scenarios, where the designed answer is also CHECK). This is a consistency check of the code against the rule text as read by the spec writer: the scenarios and the code share one reading of the rules, so it does not show that the readings are right in law, and the rule texts are checked separately, by script against the statute store (all seven rules match; Section 13). The two comparison rows are a naive offline stand-in reader (60 days or half the maximum term, no CHECK), not a language model: it reaches 0.488 and 17 false ELIGIBLE of 627, with and without the exact day count (the two rows are identical), and says nothing about any real model. No hosted or local model has been run on T4 yet, and the Radar was not compared with one.

Label-source sensitivity. The same outputs are scored against the agreed gold, one labeler alone and the keyword rules alone (Table 44). The ordering of the systems does not change: Full's factor recall stays below B0, B1 and B4 under the keyword rules (0.28 against 0.50, 0.61 and 0.67), under one labeler alone (0.33 against 0.44, 0.56 and 0.67) and under the agreed labels (0.25 against 0.50, 0.63 and 0.63), on 8 to 9 orders each, and every interval overlaps. The conclusion that factor recall is unresolved at this size does not depend on the label source.

Cost. Table 42 gives tokens, store calls, latency and USD per case. Full costs $0.0107 a case (101,860 tokens, 104 store calls, 280 seconds) against $0.0138 for B4 (131,239 tokens, a token ratio of 0.78, below the 1.0 to 1.6 the plan expected) and $0.0004 for B0 (3,862 tokens, about 26 times fewer).

9 Architecture search

Why Full loses. On the development runs Full does not beat every baseline (Section 8). Reading its outputs gives four causes. (1) The unsupported-fact rate checks every argument sentence against the facts of the case only; in Full's flagged sentences 55% state a legal principle from a cited authority, 24% say what the record does not state and 7% carry a pseudonym label, so only about 9% are real mismatches with the facts. (2) Munshi and Peshkar fill about 2.8 of the 13 dossier fields per case, and the rule-based stance picker never produces a custody-period, parity or trial-delay ground. (3) Abhiyojak's objections are checked against the material it was offered by an exact store label; on Sarvam Full kept 66 objections and discarded 230, and 263 of 320 dropped references named a paragraph that had in fact been offered. (4) Sentences that fail the check stay in the argument, marked, instead of becoming open issues.

Candidates. Each fix is an opt-in switch with the default byte-identical (Section 4): X1 fills dossier gaps with rules over the raw facts; X2 splits case-fact sentences (near-verbatim spans of the record) from legal-basis sentences; X3 adds one Abhiyojak call on factors with no ground; X4 removes sentences that still fail after repair, and "the record is silent" sentences, from the argument and lists them as open issues for counsel; X5 lets Vakeel draft from the raw facts while the dossier still drives the planner and Gate 1; X6 searches Abhiyojak's stores a second time keyed on the facts; X7 resolves an objection's reference to offered material by its id; X8 replaces a case-fact sentence that paraphrases the record by the record sentence.

Protocol. Sarvam-105B behind the pseudonymisation boundary. Search on the 30 dev orders t1dev-000..029 (single switches, then combinations); the 20 dev orders t1dev-030..049 are a holdout touched once, for the winner, Full and the baselines; the blind slice of the test set is run once from the freeze tag. Every arm of the final search table was rerun on one commit. A comparison is a win only when the paired-bootstrap interval at its Holm level excludes zero (10,000 resamples, seed 0; Holm at 0.05 over B0–B6 for each arm and metric). Quality metrics decide; cost is judged only against B4 and B5, the baselines of comparable capability; a metric a baseline cannot produce (objection precision when it raises no objection, the case-fact rate M6a

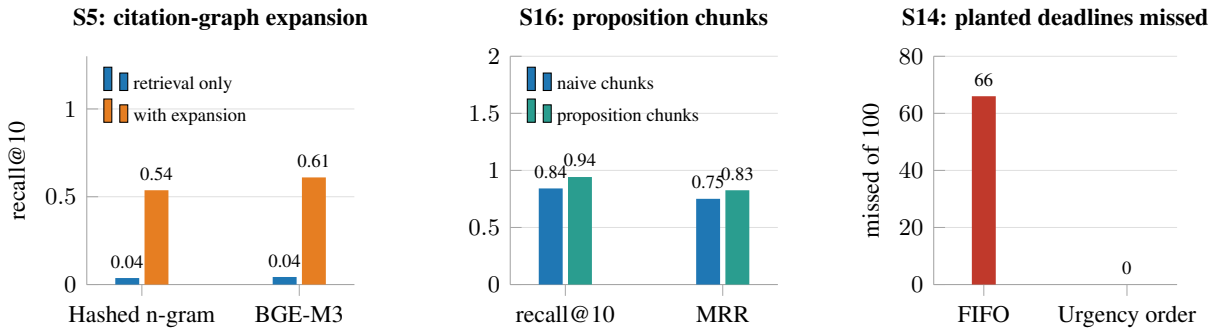


Figure 6: Offline studies, no language model. Left: expanding the citation graph raises recall@10 on a self-supervised proxy (150 queries; 75 queries gained and none lost on hashed vectors). Middle: proposition-sized chunks against naive chunks on the dropout target (292 orders). Right: a simulation of 100 batches with a planted deadline; urgency order misses none.

when it writes no sentence markers) is listed as undefined, not won.

Result. The winner, c1457, is Full with X1, X4, X5 and X7 (Table 7). On the 30 search orders it has 30 wins, 37 ties and 3 losses against B0–B6, where Full has 11, 49 and 10. Its objection recall rises from 0.36 to 1.00 (X7: the objections were there and were being discarded), answerable coverage from 0.38 to 0.61, coverage from 0.64 to 0.97, and the unsupported-fact rate falls from 0.99 to 0.90 (case-fact sentences alone: 0.84 to 0.47). It pays with citation precision (0.82 to 0.68, still a win over B4 and B5, which fabricate citations on 93% and 64% of orders) and 27% more tokens. X2 failed on Sarvam (most grounds could not meet the verbatim rule and came out empty) and X8 added a loss for three more wins, so neither is in the winner. On the holdout the winner repeats: 22 wins, 26 ties, 3 losses, the same three cells (Full: 10, 40 and 5).

The losses that remain are structural. (1) Unfinished-factor share against B4 and B5: the winner plans 130 factors on the search orders and 47 end with a required slot missing (trial stage, witness examination, antecedents, community roots). A search of the facts text (not the gold) finds none of these 62 slots stated in the record, so "missing" is the true state; B4 and B5 have no required slots and score near zero by construction. Filling the slots would be fabrication; not planning those factors would drop the prosecution concerns counsel must prepare for. (2) Confident-wrong rate against B4: the metric counts a sentence as faithful only when the facts of the case support it, so a sentence stating a legal principle counts as wrong unless the court's order happens to repeat it; B4 emits

Block	Arm	Metric	vs	Arm	Base	Outcome
search	c1457 (winner)	Tokens	B4	121065	118183	tie
search	c1457 (winner)	Tokens	B5	121065	76842	loss
search	c1457 (winner)	Store calls	B4	109	186	win
search	c1457 (winner)	Store calls	B5	109	87	loss
search	c1457 (winner)	Seconds	B4	534	176	loss
search	c1457 (winner)	Seconds	B5	534	138	loss
search	Full	Tokens	B4	95413	118183	win
search	Full	Tokens	B5	95413	76842	loss
search	Full	Store calls	B4	98	186	win
search	Full	Store calls	B5	98	87	loss
search	Full	Seconds	B4	271	176	loss
search	Full	Seconds	B5	271	138	loss
holdout	c1457 (winner)	Tokens	B4	108542	111186	tie
holdout	c1457 (winner)	Tokens	B5	108542	72603	loss
holdout	c1457 (winner)	Store calls	B4	99	185	win
holdout	c1457 (winner)	Store calls	B5	99	82	loss
holdout	c1457 (winner)	Seconds	B4	547	256	loss
holdout	c1457 (winner)	Seconds	B5	547	100	loss
holdout	Full	Tokens	B4	90914	111186	win
holdout	Full	Tokens	B5	90914	72603	loss
holdout	Full	Store calls	B4	95	185	win
holdout	Full	Store calls	B5	95	82	loss
holdout	Full	Seconds	B4	303	256	tie
holdout	Full	Seconds	B5	303	100	loss
blind	c1457 (winner)	Tokens	B4	122387	120614	tie
blind	c1457 (winner)	Tokens	B5	122387	79877	loss
blind	c1457 (winner)	Store calls	B4	110	188	win
blind	c1457 (winner)	Store calls	B5	110	89	loss
blind	c1457 (winner)	Seconds	B4	719	269	loss
blind	c1457 (winner)	Seconds	B5	719	160	loss
blind	Full	Tokens	B4	98814	120614	win
blind	Full	Tokens	B5	98814	79877	loss
blind	Full	Store calls	B4	103	188	win
blind	Full	Store calls	B5	103	89	loss
blind	Full	Seconds	B4	398	269	loss
blind	Full	Seconds	B5	398	160	loss

Table 14: Cost per order against B4 and B5, the baselines of comparable capability (lower is better). Against the one-call baselines B0–B3 and the template B6 cost is shown in the means tables, not counted.

7.5% of its grounds, a selected few sentences. (3) Coverage against B6, a template that scores 1.00 by construction (a loss over the pooled 50 orders and on the blind slice). Over the pooled 50 orders the winner also loses confident-wrong against B5 (B5's few sentences are drawn from the order's own grounds). None of these is chased, because closing them would change what the metric counts, not the brief.

Ablation	W/T/L	Winner loses on
minus X1 (dossier gaps from facts)	1/18/1	Unsupported facts, count
minus X4 (strict abstention)	4/16/0	none
minus X5 (draft from facts)	0/20/0	none
minus X7 (objections by id)	2/18/0	none
A3 no tools	0/20/0	none
A4 plain retrieval	1/19/0	none
A6 fixed router	0/19/1	Unfinished factors
A7 free planner	0/16/4	Unsupported facts (M6), Unsupported facts, count, Unfinished factors, Confident wrong
A1 no dossier (without X1, X5)	0/20/0	none
A2 no check-repair (without X4)	5/13/1	Unfinished factors
A8 no sparring (without X7)	2/17/0	none
A9 neither (without X4, X7)	6/11/1	Unfinished factors

Table 15: The winner c1457 against each ablation on the 30 search orders (wins, ties, losses of the winner; same statistics as Table 7).

Cost. Table 14: the winner needs fewer store calls than B4 (109 against 186 per order on the search orders, 110 against 188 on the blind slice, a win each time) and as many tokens (a tie); it is slower than both and uses more tokens and store calls than B5. Against the one-call baselines and the template its cost is necessarily higher and is shown, not counted.

Ablations of the winner. Table 15 compares the winner with the winner minus one of its switches and with A1–A4 and A6–A9 applied on top of it (A1, A2, A8 and A9 drop the winner switch they cannot combine with, which the switch validator refuses). Removing X4 costs four wins and gains none; removing X1 or X5 changes nothing that the 30 orders can resolve (X5 alone lowered the unsupported-fact rate in the single-switch round, which X4 and X1 now largely cover). Without check and repair (A2, A9) the winner gains five and six wins; without sparring (A8) two; no tools (A3), plain retrieval (A4) and no dossier (A1) change nothing the 30 orders resolve. Two rows go against the winner and neither is a better brief. Removing X1 wins the unsupported-sentence count because it writes fewer sentences (5.1 against 6.8 per order, with the same number supported). The free planner (A7) beats the winner on the unsupported-fact rate (0.50 against 0.90), the unsupported count, the confident-wrong rate and the unfinished-factor share, and against B0–B6 its only loss is coverage against the template. Three independent read-only checks of the saved outputs found the gains are measurement, not quality. The free planner records no plan or assessment, so the unfinished-factor share has nothing to count (88 contract gaps passed to counsel become 1). Its model rarely searches the precedent stores, so Vakeel restates the facts text under the factor headings: 18% of its grounds cite an authority against 90%, 19 of 30 briefs cite none, and it writes 29 fewer grounds and twice as many "counsel to draft it" notes. The like-for-like case-fact rate (M6a) is a tie. The unsupported-fact rate falls because law sentences, which the facts cannot support, give way to copied facts. These numbers are on dev, after selection, and were not taken to the blind slice.

Blind slice. The winner was registered with five hypotheses and taken, once, to the 80 sealed test orders; Section 7 reports the result (46 wins, 69 ties, 5 losses against B0–B6; four of five hypotheses hold).

10 Challenges and Mitigations

Table 16 lists what went wrong while building and running the system, how each problem was found, what we did about it, and what is left to do. The hardest were running out of free-tier quota and disk, a privacy boundary that refuses some orders for hosted models, and gold labels that cover only a few development orders.

11 Limitations and Risks

- **One model, one run, one slice.** The test result rests on 80 blind orders, one run per system, on one hosted model (Sarvam-105B). The development runs on OpenAI and Gemini used earlier code and 2 to 50 orders per arm, so they show direction, not a ranking across model families. c1457 was selected on 30 development orders and confirmed on a 20-order holdout before the blind slice; the blind slice was opened once.
- **RQ2 and fine-tuning not answered.** The ingestion arms on degraded, photographed and Hindi pages (S7, P5 to P8 on T2) ran only with offline stand-ins and were not scored, and the QLoRA pipeline for a 4B dossier extrac-

Challenge	What happened	Mitigation
Budget for hosted models	The plan needs about 4,500 Full case-runs; a Full case takes about 50 model calls. About 4.15 USD (OpenAI) was spent and the provider locked in code; Sarvam was locked, then opened on 1 November with an INR 5,000 cap, and the whole Sarvam block cost about INR 225.	Hosted runs moved to the Gemini free tier (500 requests a day per project per model), with a quota ledger and resumable shards; sample sizes are cut, never features; team keys are pooled.
Free-tier reliability	Gemma 4 31B failed on 86 of 94 calls in an hour (HTTP 500, 503); `gemini-3.5-flash-lite` rejected the default thinking setting with a bare 400.	Model chosen on a 3-case smoke (no failures on `gemini-3.5-flash-lite`); per-model thinking settings live in configuration; a case that hits the daily quota is not saved and reruns.
GPU serving	Qwen3-32B installed on the shared GPU nodes but never answered a prompt: the account limit on job starts blocked it.	Dropped for now; BGE-M3 vectors were built on the nodes (2 h 43 min) and applied locally.
Thin input facts	The facts part of many orders is short or only a caption (19 of 50 dev orders under 1,000 characters).	Answerable-factor rule; coverage columns; cases with no facts are reported, not dropped.
Privacy refusals on hosted calls	The boundary refused some dev orders for hosted models (8 of the first 25; 26 of 100 baseline case-runs), and the local fallback model was not running, so those cases saved an error and were scored as empty briefs.	The scorer counts errored cases explicitly; every paired comparison uses only the cases both arms answered; running a local fallback model is a phase-2 fix.
Sparse scorable gold	Only 8 of 50 dev orders have a gold factor that the facts make answerable, 20 have an agreed objection, and none has a resolved court citation.	Every cell reports its number of cases; a 24-order set chosen from gold labels alone was used for the current-code runs; recall of cited precedents is measured in the offline studies and on the test slice only.
Scoring time	MiniCheck runs on a laptop CPU: scoring one block of about 500 case-runs took over three hours.	Scoring runs unattended, one model family at a time; the larger blocks are scored for the final report.
Gold independence	The model labeler is the same model as the system in the OpenAI runs.	Rule-only gold reported beside the agreed gold; the phase-5 runs use a different model family from the labeler.
Checker circularity	The unsupported-fact metric first fell back to the system's own DeBERTa checker.	The scorer now uses MiniCheck and refuses the system's checker.
Privacy on the wire	Names were scrubbed but FIR numbers, police stations and dates went out.	Quasi-identifier aliasing and a per-case date shift; per-case consent for client text; free tiers take public text only.
Rights rules not verified	Rule texts were to be checked by hand against the bare acts, and the store held the wrong text for BNSS s.58.	A script checks each rule's cited sections against the statute store and sets the date only on a match; s.58 is now taken from the official Gazette. Seven of seven rules match, after the Companies Act s.212 was added to the store from the Ministry's text. Verification is bound to a hash of rule text and code; exports print "Not yet checked against the bare act" for any rule that is not verified.
Gated data	HLDC (Hindi district-court bail orders) and an Indic translation model are gated.	Native Hindi High Court orders as T2's Hindi set; access requested.
Disk	The laptop is at 97% disk use.	Caches cleared; large weights not in use are candidates for removal.

Table 16: Challenges met and how each was handled.

tor is built and unit-tested but no model was trained (no GPU time in the window). The tool-calling set T6 and the injection set T7 were not run on the hosted model.

- **Absolute quality is still low.** c1457 wins relative comparisons, but 0.89 of its argument sentences are not entailed by the facts alone (largely legal principles from authorities, which the facts cannot entail), 86% of the simulated prosecutor's objections stay unanswered, and the structured statutory position is empty in every brief (Section 7). The brief is a first draft for counsel, which is how the gates treat it.
- **Stand-in models.** The offline stand-in arms that are reported (the store knock-outs A10,

the intake arms S7 and P5 to P8, the injection arm S17) carry that label in the tables and measure plumbing, not a language model. The 4B and 14B models of the design have not been run on the GPU nodes for this report; serving Qwen3-32B on the shared nodes failed and was dropped.

- **Thin facts in T1.** The facts part is often only a caption. A single pass then answers "no facts supplied", while Full draws grounds from retrieved authorities not tied to the client's facts, so Full can win on recall through retrieved boilerplate.
- **Gold is not human.** Rules plus one non-Qwen labeler that must agree; κ is 0.60 for G1, and the agreed G1 gold leaves 81 of 250 orders without a factor. Only 37.3% of citations resolve, and no dev order has a resolved court citation. HLDC was unavailable, so the Hindi set is native High Court orders from one court and two years.
- **Rights Radar.** Seven of the seven rules match the statute store in an automatic check of each cited section's number, heading, sub-section and the conditions the rule tests (`scripts/check_rights_rules.py`); the check reads acts only, not the judgments a rule also relies on, and the store's Companies Act s.212 comes from a copy of the Ministry's text dated March 2016, so a later amendment to that section would not be seen. A rule's verification is bound to a hash of its text and code, so a later change clears it, and an unverified rule is printed in the export as "Not yet checked against the bare act". T4 is generated from the same rule table, so it measures conformance, not legal accuracy.
- **No authentication.** The gates are enforced in code, but there are no user accounts and the API token is off by default, so "approval only from counsel" is a property of the code path, not of an identity check.
- **Budgets.** Spent in total: about 4.15 USD on OpenAI (locked in code again) and about INR 990 on Sarvam by 13 November, under a cap of INR 5,000 that the ledger enforces (it refuses any call past it). The free Gemini tier allows 500 requests a day per project on the flash-lite models, about 10 Full cases a day (a Full case takes about 49 requests). The full plan (about 4,500 Full case-runs) does not fit, so sample sizes are cut, not features.
- **Known weaknesses.** Grounds are sometimes weak in substance (about ten open issues per case); the pseudonymiser still refuses one order of 250.

12 Threats to Validity

Internal. Small n and a single run per system limit what can be resolved; an effect smaller than the interval is reported as not resolved. The architecture search chose c1457 on 30 development orders, which favours it on those orders; the 20-order holdout and the sealed blind slice, opened once after the freeze tags and the registered hypotheses, are the guard. Thresholds fixed on 50 orders may be mis-set, so margins are rounded to whole points; the pass bar and NLI threshold were fixed before the freeze. The blind scoring code was changed once after a crash, before any score existed (deviation D10).

Construct. Gold from rules plus an LLM labeler, not humans, risks circularity. The model labeler is an OpenAI model, which was also a system backend in the dev runs (not on the blind slice, which runs Sarvam), so its reading of a factor and the drafter's may share biases; the rule-only gold is the declared sensitivity analysis for this reason, and on the blind slice it changes no hypothesis verdict (Section 7). A second circularity: the system reads the facts part of an order while gold is read from other parts of the same order, and the tagger that builds paragraph metadata is of the same family as the rule labeler. Controls are an 8-word-overlap check, agreed labels, label-source sensitivity, and B6 not reusing the tagger. Because T1 facts are often only a caption, retrieved boilerplate can inflate recall; the answerable-factor rule and the faithful/corroborated table are guards, not a cure. Factor recall against G1 measures what the defence argued, not what a good lawyer should argue; lenient precision against G1 $\cup$ G5 corrects this in part. The unsupported-fact checker (MiniCheck) is independent of the system's checker but was validated on other domains, so subtle legal entailment errors may pass.

External. One legal domain (bail), one source of orders, one court for the Hindi native set, and constructed data for T2 to T5, T7 and T8 limit transfer. Only 5.9% of orders split cleanly, so results

hold for orders that split cleanly. Qwen3 and the hosted labeler may have seen public orders; the release date bounds the cutoff but does not measure it. There is no user study, and no claim about lawyer time or outcomes.

13 Conclusion and Future Work

Answer to RQ1. On the same hosted model, an agent harness that plans retrieval per bail factor, checks every claim and spars with a simulated prosecutor writes better-grounded bail briefs than the simpler systems we compared it with, but only after its failures were diagnosed and fixed. The first full system did not beat every baseline on development orders. Reading its outputs traced the losses to four causes, and four opt-in switches fixed them. On 80 sealed test orders the resulting system, c1457, beats the baselines on 46 comparisons and loses 5. It cites real authorities where the comparable retrieval baselines fabricate them, states fewer unsupported facts than any baseline, answers more of the prosecution's objections than the first full system, and needs 41% fewer store calls than querying every store. One registered hypothesis failed, on a count that favours systems which write less. No brief left the system without counsel's approval. The lesson we draw is methodological: the harness only helped once each of its decisions was measured against a simpler alternative and its losses were read case by case.

What was built. Six agents with an event log; ten typed stores on four shelves; a page-level ingestion ladder for ten input kinds; evidence contracts for twelve bail factors and the orchestrator Sutradhar; claim checking and repair; prosecution sparring; truth maintenance with byte-identical redrafting; batch triage; the Rights Radar (all seven rules matched against the statute store by script); both approval gates and the pseudonymisation boundary; a web interface and API. For evaluation: test sets T1 to T9, gold G1 to G7 without manual annotation, baselines B0 to B6, 46 opt-in switches for literature variants (61 of 61 arms pass an offline end-to-end audit), a scoring pipeline with paired bootstrap and Holm correction, and spend ledgers that cap every paid call. The test suite has about 5,000 tests.

Future work. In order of expected value:

1. **Fill the statutory position.** Extract the charged sections into the dossier (they are in the order text) and look up bailability and maximum punishment in the statute store, so that the checker can reject objections and replies that misstate punishment (failure examples 3 and 4 of Section 7).
2. **Ground the prosecutor.** Check Abhiyojak's objections against the statute store and the facts before they reach counsel, and drop objections that do not apply (an NDPS bar in an IPC case); answer more of them, since 86% stay unanswered.
3. **Separate legal principles from case facts in the metric and the draft.** Sentences that state law should be checked against the cited paragraph and sentences that state facts against the record; the like-for-like case-fact rate already shows the gap (on the development search orders, 0.47 on case-fact sentences against 0.90 on all sentences).
4. **RQ2 and fine-tuning.** Train the 4B QLoRA dossier extractor (the pipeline and 110 tests exist), run ablation A5 against it, and score the ingestion arms on T2 with the real models, so that client documents can stay on the lawyer's machine.
5. **Tool selection and injection.** Run the tool-calling set T6 and the injection set T7 on the hosted model to measure tool selection and attack success, not only argument errors and refusals.
6. **More runs and a person in the loop.** Repeat the blind comparison on a second model family and with more runs per system, and have legal-aid lawyers use the interface on real files under consent, with authentication in front of the gates.

Code. All code, tests, documents, the report source, run manifests and the tags `prereg-v1` and `eval-freeze-v1` are in `https://github.com/vishakkashyapk30/lma-group-project`. Data, model weights and keys are not in it; `make setup`, `make data` and `make models` rebuild them (Appendix, Reproducibility).

Ethics and Privacy Statement

Client files stay on the user's machine. Text leaves it only through the pseudonymisation boundary, and only when the external-model mode and counsel's per-case consent allow it; external services are

off by default, paid providers are locked in code, and free-tier services, which may use prompts to improve their products, receive public court text only. On 250 T1 orders the party-name proxy found names in 80 orders before a fix to caption-name learning and 5 after; a second change now also aliases FIR, case and crime numbers, police stations, villages, addresses and ages and shifts dates per case, and the boundary refuses the 5 orders whose names OCR glued to other words. We do not claim zero leakage: the scrubber is rule-based, and Hindi coverage is thin. Keys live in environment variables, never in the repository, and spending is capped in code. The public corpus is court text already in the public domain; orders with the same FIR as a test order are excluded from the stores. The system makes no outcome predictions, shows no grant rates, and gives no advice to accused persons or the public; the landing page points them to free legal aid (NALSA helpline 15100). Export needs counsel's approval at Gate 2. Constructed test cases use synthetic names. No manual annotation was used.

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A Reproducibility

Code and data. The repository is the code of record; the pinned tag for the current dev runs is `dev-sweep-v3` (commit `7ad6ed4`; the tag `dev-sweep-v1`, commit `efd0ce4`, is the superseded 50-case run). Data (`interim`, stores, test sets, gold, and the raw sources: Open India Law shards, IPC2BNS-Verify, Kathbath) is not in the repository. It is a private dataset in the Hugging Face organisation `Vakeel-Sahayak`, `vakeel-sahayak-data`; access is granted by the owner. Helper models (BGE-M3, DeBERTa NLI, MiniCheck, Whisper small) are fetched by `make models`.

```
make setup
make data
make models
make smoke        # case-0007, offline
make eval-smoke
python -m vakeel_sahayak.eval.runners.driver \
  plan --sweep
python -m vakeel_sahayak.eval.scoring score \
  --runs data/eval/dev_runs --slice dev \
  --gold-source both --out data/eval/dev_scores
python -m vakeel_sahayak.eval.scoring report \
  --scores data/eval/dev_scores
```

Manifests. Each run writes a manifest with the system id and variant, prompt hash, config hash, model names and modes, test-set and gold hashes, shard count, and the code stamp (`git_sha`, `git_dirty`, `code_tree_hash`). The scoring loader refuses runs from different commits unless told otherwise, and a dirty stamp is not admissible for frozen runs. The hosted-model settings are `VS_MODELS=auto`, `VS_LLM_BACKEND=openai`, `VS_EXTERNAL=pseudonymised`, with keys read from an untracked `.env`; offline stand-ins are the default and need neither a GPU nor a key. Agreed-gold hashes: agreed-gold sha256 `5583690e3754dacd8f2ebd4e27987ae7cfa0c3df9eb2fd9e8bc9a879d86c05be`, shared by every run of the current code; code tag `dev-sweep-v3`, git sha `7ad6ed41ea39`, code tree hash `d85cdc64f123`; the 24-case file of scorable orders has cases hash `ec0611f23b3c7749`.

B Arms

Id	Idea (source)	Switch	Run so far
B0, B1, B4, B6, Full	baselines (Section 6)	driver system id	run on the earlier code (50 cases) and the current code; scored
B0L, B2, B3, B5	baselines	driver system id	run on the current code (B0L on Gemini only); scored on the cases available
B0S	baseline	driver system id	not run
S1	self-critique retrieval (Asai et al., 2024; Yan et al., 2024)	`retrieval_policy`	run (OpenAI, 10 cases)
S2	adaptive depth (Jeong et al., 2024)	`retrieval_policy`	run (OpenAI, 10 cases)
S3, S3b	sufficiency stopping (Joren et al., 2025)	`stopping`	run (OpenAI, 10 cases)
S4	score-skew weak signal	`weak_signal`	offline study
S5	citation-graph expansion (Gutiérrez et al., 2025)	`graph_expansion`	offline study
S6	learned store prior (Zhang et al., 2026)	`store_prior`	run, no effect without a trained prior
S7	hierarchy re-reading for Munshi	`munshi_gap_policy`	run offline (T2), not yet scored
S8	self-critique repair (Gou et al., 2024; Huang et al., 2024)	`repair_mode`	run (OpenAI, 10 cases)
S9, S10	sampling critic; no isolation (Choi et al., 2025; Wynn et al., 2025)	`critic_mode`, `abhiyojak_isolation`	run (OpenAI, 10 cases)
S11	three-role review, debate, vote (Kumar et al., 2025)	`critic_mode`, `draft_mode`	S11 run (OpenAI); S11b, S11c run (Gemini block B, 2 cases)
S12	topology: sequential, single agent (Cemri et al., 2025)	`topology`	S12 run (OpenAI); S12b run (Gemini block B, 2 cases)
S13	searched workflow (grid on dev)	none (grid over switches)	offline smoke
S14, S14b–d	plan cache, scheduling, prefix signal (Zheng et al., 2024)	`plan_cache`, `urgency_policy`	offline study
S15, S15b	output format (Tam et al., 2024)	`output_format`	S15 run (OpenAI); S15b run (Gemini block B, 2 cases)
S16, S16b	proposition units, late chunking (Chen et al., 2024)	`chunking`	offline study
S17	injection defences (Beurer-Kellner et al., 2025)	defence flags	run offline (T7), not yet scored
S18	authority stance (Wachsmuth et al., 2018)	`authority_stance`	run (OpenAI, 10 cases)
P5–P8	ingestion arms, image-only (Yu et al., 2025; Faysse et al., 2025)	`munshi_*`	run offline (T2), not yet scored
A1–A10	remove one part (dossier, repair, tools, filter, extractor, router, planner, sparring, a store)	`Switches` flags	A1 to A3, A6 to A9 run (OpenAI); A4, A5 run (Gemini block B); A10 run offline, scored

Table 17: Arms and their state. "Run" means the arm has produced outputs on the tag `dev-sweep-v3` on the stated backend and cases; "not run" means it is implemented and offline-smoked but has produced no outputs. Switch names follow `orchestration/switches.py`; S18 is `authority_stance=outcome` and S3b is `stopping=llm`.

C Metrics

Id	Metric	Definition and check	Set
M1–M3	factor recall, precision, F1	set comparison with G1 (and G5 for lenient precision) on answerable factors; argue-all F1 as floor	T1
M4	G5, G4, G6 recall	as M1 for court-discussed factors, objections, conditions	T1
M5	label-source sensitivity	M1 to M4 on rules-only, labeler, agreed; κ per factor	T1
M6	unsupported-fact rate	share of factual sentences MiniCheck or the NLI finds unsupported	T1
M7	factual error	wrong section, date, statute text, custody count against tools	T1
M9	fabricated citation	id not in index, or span not found verbatim	T1
M10, M11	recall@k, MRR, nDCG@10	court-cited precedents, with coverage columns	T1
M12	slot-fill rate	required slots filled; missing slots are open issues	T1
M13, M15	authority quality, good-law violations	metadata and graph lookup; known-treatment share	T1
M16, M17	citation precision, recall; removal	ALCE-style; claims unchanged when the paragraph is removed	T1
M18	stopping quality	over-search and stopped-too-early	T1
M19	weak-evidence AUROC	score skew predicts a top-10 gold hit	T1
M33	coverage-risk	abstention curve	T1
Ingestion	CER, boundary F1, type and field accuracy	against construction truth; escalation, GPU seconds	T2
Rights	accuracy, false-ELIGIBLE	designed answers by construction; Wilson interval	T4
Safety	attack success, unapproved exports	bounded by the rule of three	T7
Cost	tokens, calls, latency, USD	from manifests and ledgers; cost-quality frontier	all

Table 18: Metrics. Definitions are in the evaluation plan; none is an LLM-judge score.

D Glossary of Names Used in This Report

Full The complete system: all six agents, the ten stores, retrieval, verification and prosecution sparring.

B0 to B6 Baselines, simpler systems that Full must beat. B0: one model call with no retrieval. B1: retrieval from one store. B2: retrieval over whole-order embeddings. B3: retrieval over naively chunked text. B4: every store searched and fused, with no agent planning. B5: a learned router alone, with no contracts. B6: rules only, no model. B0L: B0 on a larger model.

A1 to A10 Ablations: Full with one part switched off, to show whether the part earns its place.

S1 to S18 Variants taken from the literature (Section 3). Each swaps one idea from a paper into the system as an opt-in switch. P5 to P8 are the same for intake.

Dev set The 50 development orders (T1 dev). Development numbers come from it; the test result comes from the 80 blind orders, opened once (Section 7).

Code tag A named snapshot of the code. `dev-sweep-v1` (27 October) is the code of the first 50-case baseline runs; it is superseded and reported only as such. `dev-sweep-v3` (29 October) is the current code; every other model run reported here comes from it. A result is comparable only with results from the same tag, and the scorer refuses to mix tags.

Block A batch of runs on one model and one set of cases: the OpenAI block (`gpt-6-luna`), Gemini block A (`gemini-3.5-flash-lite`) and Gemini block B (`gemini-3.1-flash-lite`). Arms are compared only with the Full of the same block (Table 19).

Agents Munshi (intake), Peshkar (dossier and Rights Radar), Sutradhar (orchestrator), Vakeel (drafter), Abhiyojak (prosecution sparring), Pairokar (batch desk).

Gates Gate 1 (dossier) and Gate 2 (brief): approvals that only counsel can give; export is impossible before Gate 2.

Metrics M1 to M35, listed in the metrics table of this appendix: M1 factor recall, M6 unsupported-fact rate, M10 recall of cited precedents, and so on.

E Models and Where They Ran

Block of runs	Provider	Model	Runs	Arms and size
Dev runs, code `dev-sweep-v1` (superseded)	OpenAI	`gpt-6-luna`	hosted	Full (2 runs), B0, B1, B4: 50 cases, scored; B6 is a rule baseline with no model
Block A (OpenAI)	OpenAI	`gpt-6-luna`	hosted	Full, B0 to B3, A1: 25 cases
Block A (Gemini)	Google, free tier	`gemini-3.5-flash-lite`; B0L on `gemini-3.6-flash`	hosted	Full, B0 to B3, B0L: 25 cases; A1: 10; scored
Block C (Sarvam)	Sarvam	`sarvam-105b`	hosted	Full, B0 to B6, A1 to A4, A6 to A9: 30 cases; 22 variants: 10; scored
Block B	Google, free tier	`gemini-3.1-flash-lite`	hosted	Full, A2, A8: 5 cases; run, not yet scored
Offline studies	none (no language model)	real stores, hashed n-gram and BGE-M3	local	S4 and S5 (150 queries), S14 (100 simulated batches), S16 (292 orders)
Gold labeler	OpenAI	`gpt-6-luna`	hosted	labels the dev and test orders (public court text only); a label counts only if the rule labeler agrees
Claim checker (system)	local	DeBERTa NLI	local	inside Vakeel
Claim checker (scoring)	local	MiniCheck-RoBERTa-Large	local	evaluation only
Retrieval	local	hashed n-gram; BGE-M3 (study only)	local	BGE-M3 vectors built on GPU nodes
Speech	local	Whisper small	local	T8

Table 19: Every block of runs, its provider and model, and whether it ran locally or on a hosted service. Only one model family serves as system in each block, so each comparison is paired within a block.

F Further Generated Tables

In the comparison tables of this appendix and of the results section, the best value in each column is **bold** and the second best is underlined (the direction follows the metric: lower for unsupported facts, fabricated

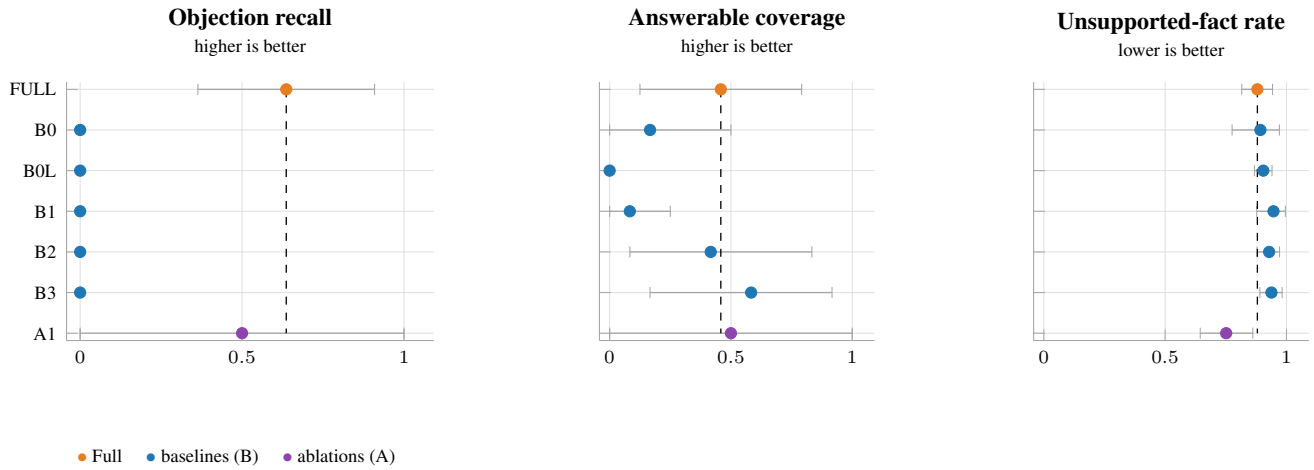


Figure 7: Family Gemini block A: each arm on the cases it answered without error and where the metric is defined, mean with a 95% bootstrap interval (the number of cases per arm is in the appendix tables). The dashed line is Full; an arm without a mark has no defined value on this metric (no gold or no output of that kind). Unsupported-fact rate: lower is better.

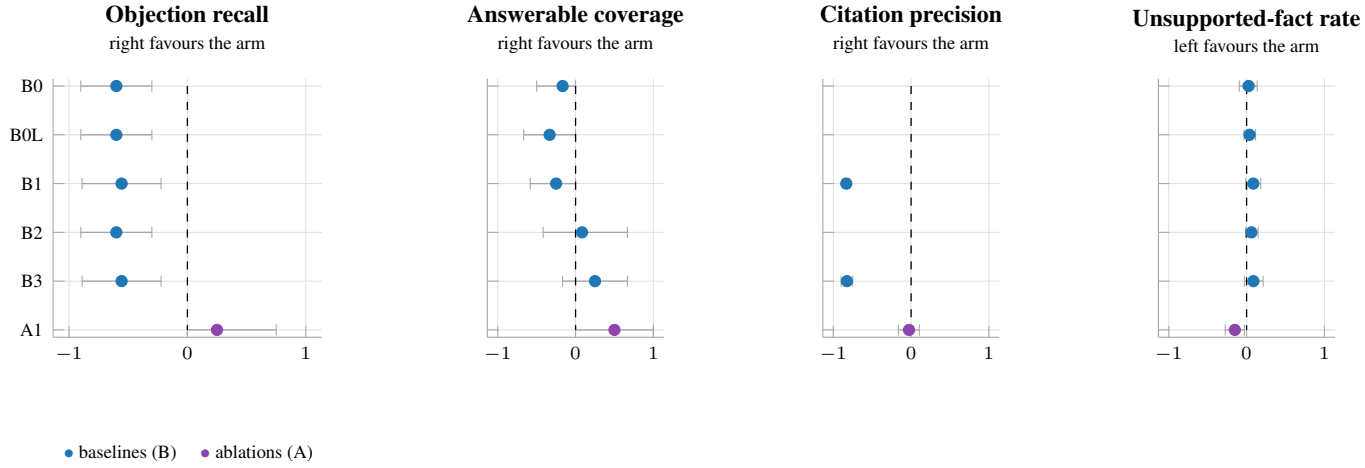


Figure 8: Family Gemini block A: each arm minus Full on the cases both answered, mean difference with a 95% bootstrap interval. The line at zero is no difference; an interval that crosses it is not resolved at this sample size. On the unsupported-fact rate a negative difference favours the arm.

citations, tokens, cost and risk; higher for the rest; in the paired-difference tables, the most favourable mean difference). Statistical tables (paired tests, Holm decisions) and label-source tables are not ranked.

Figures for the other model families and the cost plots

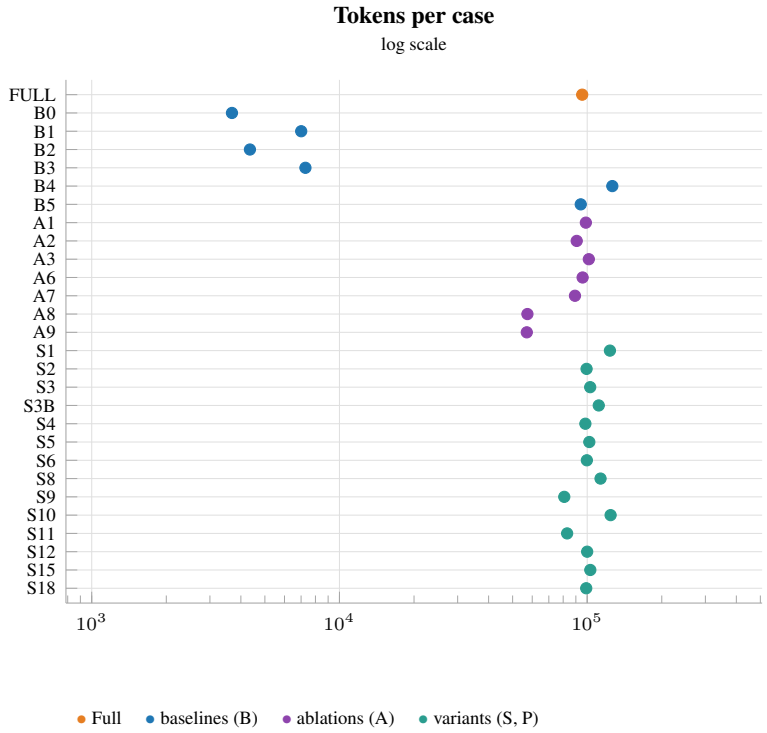


Figure 9: Family OpenAI: mean tokens per case (log scale), each arm on its own cases.

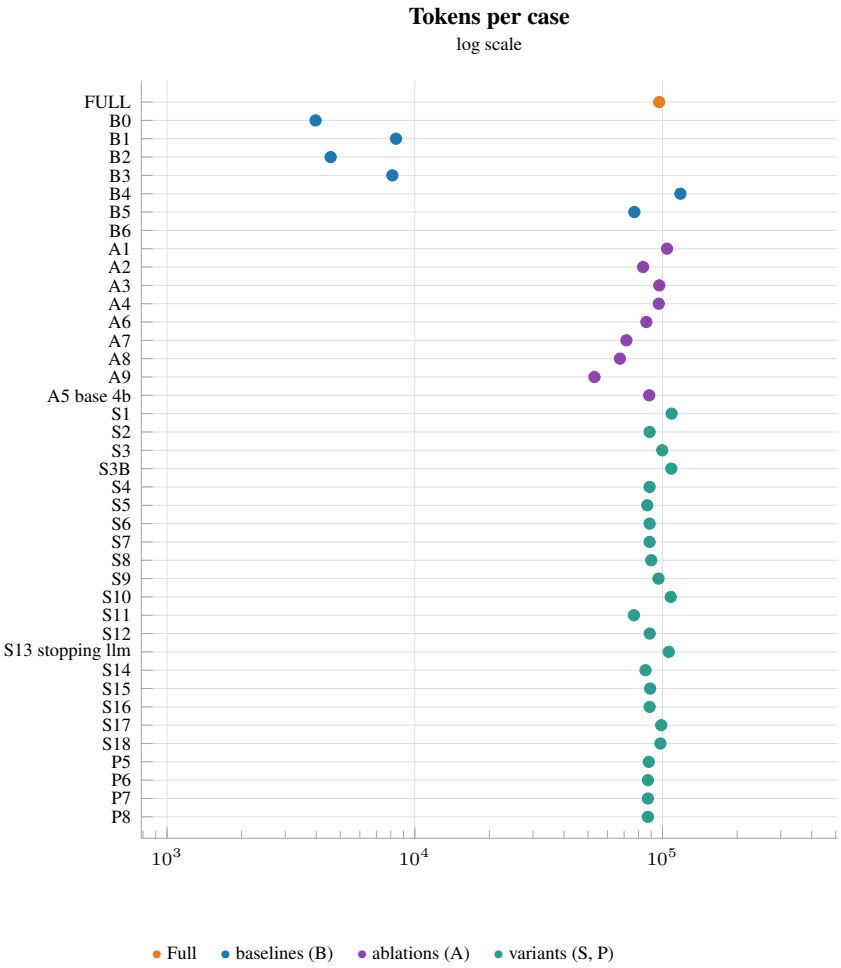


Figure 10: Family Sarvam: mean tokens per case (log scale), each arm on its own cases.

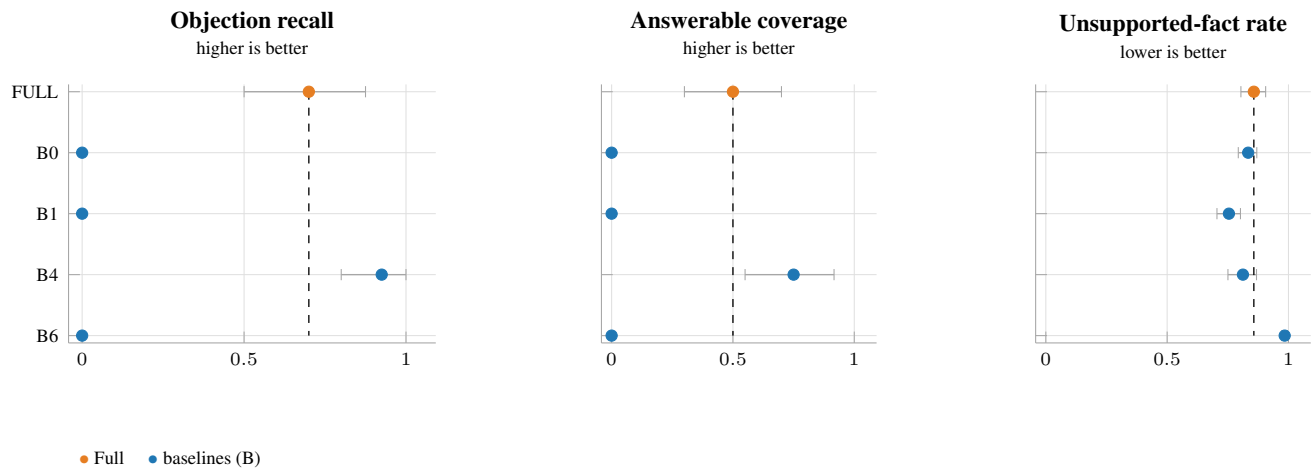


Figure 11: Family superseded 50-case run: each arm on the cases it answered without error and where the metric is defined, mean with a 95% bootstrap interval (the number of cases per arm is in the appendix tables). The dashed line is Full; an arm without a mark has no defined value on this metric (no gold or no output of that kind). Unsupported-fact rate: lower is better.

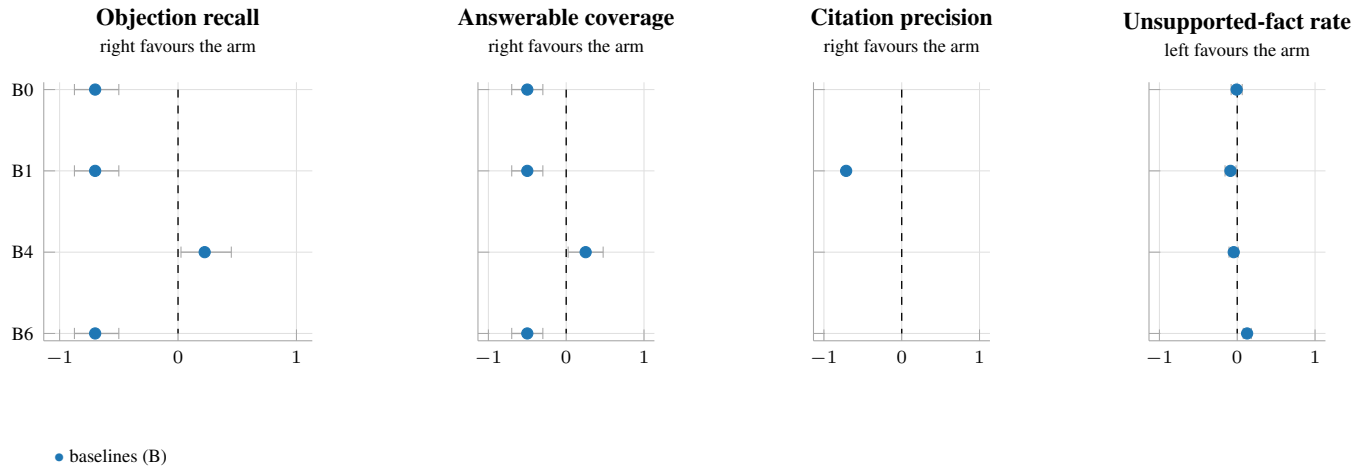


Figure 12: Family superseded 50-case run: each arm minus Full on the cases both answered, mean difference with a 95% bootstrap interval. The line at zero is no difference; an interval that crosses it is not resolved at this sample size. On the unsupported-fact rate a negative difference favours the arm.

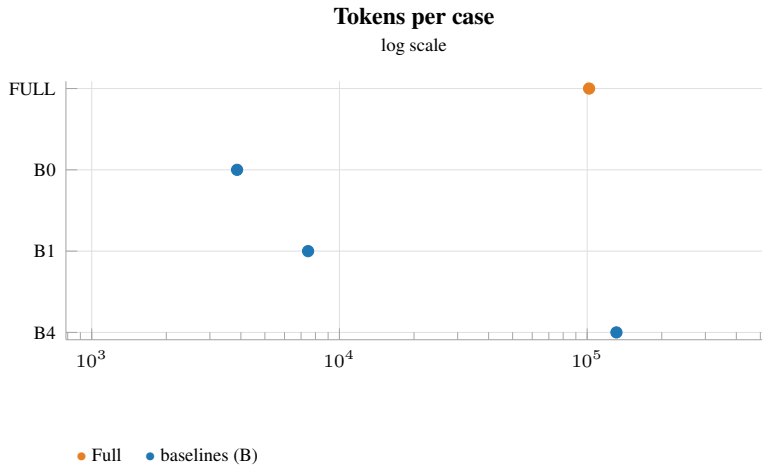


Figure 13: Family superseded 50-case run: mean tokens per case (log scale), each arm on its own cases.

Arm	n	Accuracy [95% Wilson]	Default bail (254)	Undertrial limit (232)	Notice of appearance (50)	Production 24 h (50)	Antil category (19)	Statutory bar (22)	False-ELIGIBLE	CHECK rate
Rights Radar (rules engine)	627	**1.000 [0.994, 1.000]**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**0 of 627**	0.134
Stand-in reader	627	0.488 [0.449, 0.527]	0.358	0.664	0.460	0.760	0.000	0.000	17 of 627	0.000
Stand-in reader + date_calc	627	0.488 [0.449, 0.527]	0.358	0.664	0.460	0.760	0.000	0.000	17 of 627	0.000

Table 20: Rights Radar on the constructed test set T4 ($n = 627$ scenarios; the number after each rule is its scenario count). The designed answer of each scenario is derived from the rule text by a spec writer that does not import the Radar, so the Radar's score is a consistency check of code against rule text, not a legal validation. The two stand-in rows are a deliberately naive offline reader, not a language model, and say nothing about any real model. T4 has scenarios for six of the seven rules; grounds of arrest is not scored. False-ELIGIBLE: predicted ELIGIBLE where the designed answer is not. Bold: best value in the column; underlined: second best (ties share a rank); the CHECK rate is not comparable.

Arm	B0	B1	B2	B3	B4	B5	Beats all
A1	2/6/0+3	1/5/0+12	2/6/0+3	2/4/0+5	3/7/0+3	2/3/1+16	no
A2	2/4/1+3	1/3/1+12	2/4/1+3	2/2/1+5	2/7/2+1	0/3/2+16	no
A3	2/6/0+3	1/5/0+12	2/6/0+3	2/4/0+5	3/8/1+1	1/3/1+17	no
A6	2/6/0+3	1/5/0+12	2/6/0+3	2/4/0+5	3/7/2+1	1/4/0+16	no
A7	2/5/1+3	1/4/1+5	2/5/1+3	2/3/1+5	2/8/2+1	0/5/0+7	no
A8	1/7/0+3	0/6/0+12	1/7/0+3	1/5/0+5	2/6/1+3	1/3/1+16	no
A9	1/5/1+3	0/4/1+12	1/5/1+3	1/3/1+5	1/6/3+1	0/3/2+15	no
FULL	2/6/0+3	2/4/0+12	2/6/0+3	2/4/0+5	3/8/1+1	1/3/1+17	no
S1	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S10	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S11	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S12	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S15	0/3/1+4	0/2/0+8	1/2/1+4	1/2/1+6	2/0/2+8	1/3/1+17	no
S18	0/2/2+4	0/2/0+8	1/1/2+4	1/1/2+6	2/0/2+7	1/3/1+16	no
S2	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S3	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S3B	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S4	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S5	0/4/0+4	0/2/0+8	1/3/0+4	1/2/1+6	2/2/0+8	1/3/1+17	no
S6	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no
S8	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/1/1+8	1/3/1+17	no
S9	0/4/0+4	0/2/0+8	1/3/0+4	1/3/0+6	2/2/0+8	1/3/1+17	no

Table 21: Family OpenAI: wins/ties/losses of each arm against each baseline over 25 metrics (paired bootstrap, 10,000 resamples, Holm at 0.05, one family per arm and metric over the baselines; a win only when the Holm-level interval excludes zero in the arm's favour). +n: comparisons with fewer than five paired cases (not a win). A dash: no metric defined for both. Beats all: every counted comparison is a win. Every comparison, with its interval, is in docs/results/wlt.

Arm	B0	B1	B2	B3	B4	B5	B6	Beats all
A1	1/5/0+2	1/5/0+11	1/3/0+10	1/5/0+11	7/8/1+6	4/11/2+5	1/4/1+4	no
A2	0/3/2+2	0/3/2+11	0/3/1+9	0/3/2+11	4/9/2+6	3/10/3+5	1/2/2+4	no
A3	1/3/2+2	1/3/2+11	1/3/0+10	1/3/2+11	5/8/3+6	4/10/3+5	1/2/3+4	no
A4	1/3/2+2	0/4/2+11	1/3/0+10	0/4/2+11	6/7/3+6	4/10/3+5	2/1/3+4	no
A5:BASE_4B	0/2/0+6	0/2/0+13	0/2/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
A6	2/2/2+2	2/2/2+11	2/2/0+10	2/2/2+11	6/8/2+5	4/11/2+5	2/1/3+4	no
A7	0/5/0+2	0/3/2+12	0/4/0+6	0/4/1+12	2/14/0+3	0/17/0+5	1/4/1+4	no
A8	1/3/2+2	1/3/2+11	1/3/0+10	1/3/2+11	5/7/3+6	4/8/4+5	1/2/3+4	no
A9	0/3/2+2	0/3/2+11	0/3/1+9	0/3/2+11	4/8/2+6	3/9/3+5	1/2/2+4	no
FULL	1/3/2+2	1/3/2+11	1/3/0+10	1/3/2+11	5/8/3+6	4/10/3+5	2/1/3+4	no
P5	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
P6	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
P7	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
P8	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S1	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/3/1+13	1/1/2+15	1/1/2+4	no
S10	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S11	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S12	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S13:STOPPING=LLM	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S14	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S15	0/2/0+6	0/2/0+13	1/1/0+6	0/2/0+6	5/3/1+13	1/1/2+15	0/3/1+4	no
S16	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S17	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/3/2+12	1/1/2+15	1/1/2+4	no
S18	0/2/0+6	0/2/0+12	0/2/0+6	0/2/0+6	4/6/0+12	0/4/0+15	0/2/1+5	no
S2	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S3	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S3B	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S4	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S5	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	6/3/1+12	1/1/2+15	0/2/2+4	no
S6	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S7	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S8	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/4/1+12	1/1/2+15	0/2/2+4	no
S9	0/2/0+6	1/1/0+13	1/1/0+6	0/2/0+6	5/5/0+12	1/2/1+15	0/2/2+4	no

Table 22: Family Sarvam: wins/ties/losses of each arm against each baseline over 25 metrics (paired bootstrap, 10,000 resamples, Holm at 0.05, one family per arm and metric over the baselines; a win only when the Holm-level interval excludes zero in the arm's favour). +n: comparisons with fewer than five paired cases (not a win). A dash: no metric defined for both. Beats all: every counted comparison is a win. Every comparison, with its interval, is in docs/results/wlt.

Arm	B0	B0L	B1	B2	B3	Beats all
A1	0/4/0+4	3/1/0+6	2/2/0+11	2/2/0+6	0/2/0+15	no
FULL	1/4/1+4	2/3/1+4	2/10/1+4	1/4/1+4	1/5/0+11	no

Table 23: Family Gemini block A: wins/ties/losses of each arm against each baseline over 25 metrics (paired bootstrap, 10,000 resamples, Holm at 0.05, one family per arm and metric over the baselines; a win only when the Holm-level interval excludes zero in the arm's favour). +n: comparisons with fewer than five paired cases (not a win). A dash: no metric defined for both. Beats all: every counted comparison is a win. Every comparison, with its interval, is in docs/results/wlt.

Arm	B0	B1	B4	B6	Beats all
FULL	3/7/0+1	3/7/0+8	2/7/3+1	5/4/1+1	no

Table 24: Family superseded 50-case run: wins/ties/losses of each arm against each baseline over 25 metrics (paired bootstrap, 10,000 resamples, Holm at 0.05, one family per arm and metric over the baselines; a win only when the Holm-level interval excludes zero in the arm's favour). +n: comparisons with fewer than five paired cases (not a win). A dash: no metric defined for both. Beats all: every counted comparison is a win. Every comparison, with its interval, is in docs/results/wlt.

Group	Arm	Cases	Obj. rec.	Ans. cov.	Cov.	Factor rec.	Cit. prec.	Fabr. cit.	Unsupp. facts	Tokens
Full	FULL	25	**0.64**	0.46	0.86	0.33	0.76	0.00	0.88	85128
Baselines	B0	20	0.00	0.17	1.00	0.00	–	–	0.89	**3032**
	B0L	20	0.00	0.00	1.00	0.50	–	–	0.90	3206
	B1	17	0.00	0.08	1.00	0.00	0.00	0.00	0.95	7136
	B2	20	0.00	0.42	1.00	0.50	–	–	0.93	3248
	B3	17	0.00	**0.58**	0.82	0.50	0.00	0.00	0.94	7494
Ablations	A1	10	0.50	0.50	0.90	0.50	**0.78**	0.00	**0.75**	96248

Table 25: Family Gemini block A: every arm on every metric, mean over the cases the arm answered without error where the metric is defined (a dash means undefined, for example no citation by that arm). Cases: orders answered. Higher is better except unsupported facts and fabricated citations; tokens (and USD where priced) are per case. Arms run on different numbers of orders (the Cases column), so read near-equal rows with the paired tables of the appendix, which carry intervals. Bold: best in the column; underlined: second best.

Arm	Cases	Δ Obj. recall	Δ Ans. cov.	Δ Coverage	Δ Unsupp. facts	Δ Fabr. cit.	Δ Cit. prec.	Δ Factor rec.
B0	30	-0.74 [-0.91, -0.50]	-0.33 [-0.63, -0.10]	+0.07 [-0.12, +0.23]	-0.01 [-0.08, +0.07]	–	–	+0.33 [-0.33, +0.83]
B1	23	-0.79 [-1.00, -0.57]	-0.42 [-0.75, -0.12]	+0.06 [-0.15, +0.25]	-0.07 [-0.16, +0.03]	+0.00	-1.00	**+0.50 [+0.00, +1.00]**
B2	30	-0.74 [-0.91, -0.50]	-0.28 [-0.60, +0.02]	**+0.10 [-0.07, +0.24]**	-0.05 [-0.15, +0.03]	–	–	+0.33 [-0.33, +0.83]
B3	23	-0.79 [-1.00, -0.57]	-0.42 [-0.75, -0.12]	**+0.10 [-0.08, +0.27]**	-0.04 [-0.11, +0.04]	–	–	**+0.50 [+0.00, +1.00]**
B4	30	+0.15 [-0.05, +0.35]	**+0.22 [+0.00, +0.47]**	-0.39 [-0.53, -0.25]	-0.07 [-0.13, -0.01]	–	–	+0.25 [+0.00, +0.62]
B5	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.63 [-0.80, -0.44]	-0.03 [-0.07, +0.00]	+0.00	+0.00	+0.00 [+0.00, +0.00]
A1	28	+0.14 [-0.03, +0.33]	-0.07 [-0.33, +0.23]	+0.09 [-0.05, +0.19]	**-0.10 [-0.18, -0.02]**	+0.00 [+0.00, +0.00]	-0.20 [-0.36, -0.03]	+0.12 [-0.25, +0.50]
A2	30	+0.03 [-0.07, +0.15]	-0.42 [-0.67, -0.17]	-0.82 [-0.89, -0.74]	+0.03 [+0.01, +0.06]	+0.00 [+0.00, +0.00]	-0.39 [-0.50, -0.29]	+0.00 [+0.00, +0.00]
A3	30	-0.03 [-0.17, +0.12]	+0.00 [+0.00, +0.00]	-0.01 [-0.02, +0.00]	+0.00 [-0.03, +0.04]	+0.00 [+0.00, +0.00]	-0.08 [-0.14, -0.02]	+0.00 [+0.00, +0.00]
A6	30	+0.07 [+0.00, +0.20]	+0.00 [+0.00, +0.00]	+0.02 [-0.03, +0.08]	+0.03 [-0.00, +0.07]	+0.00 [+0.00, +0.00]	+0.04 [-0.04, +0.12]	+0.00 [+0.00, +0.00]
A7	30	**+0.17 [+0.05, +0.35]**	-0.03 [-0.08, +0.00]	-0.47 [-0.58, -0.36]	-0.02 [-0.06, +0.03]	–	–	+0.00 [+0.00, +0.00]
A8	28	-0.75 [-0.94, -0.53]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
A9	30	-0.75 [-0.90, -0.57]	-0.42 [-0.67, -0.17]	-0.82 [-0.89, -0.74]	+0.03 [+0.01, +0.06]	+0.00 [+0.00, +0.00]	-0.39 [-0.50, -0.29]	+0.00 [+0.00, +0.00]
S1	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.02 [-0.09, +0.13]	-0.03 [-0.12, +0.08]	+0.00 [+0.00, +0.00]	+0.01 [-0.14, +0.17]	+0.00 [+0.00, +0.00]
S2	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S3	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S3B	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S4	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.02 [+0.00, +0.06]	+0.02 [+0.00, +0.06]	+0.00 [+0.00, +0.00]	+0.09 [+0.01, +0.19]	+0.00 [+0.00, +0.00]
S5	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.03 [+0.00, +0.10]	+0.00 [+0.00, +0.00]	+0.01 [-0.04, +0.05]	+0.00 [+0.00, +0.00]
S6	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S8	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.15, +0.12]	+0.02 [+0.00, +0.06]	+0.00 [+0.00, +0.00]	**+0.11 [-0.04, +0.24]**	+0.00 [+0.00, +0.00]
S9	10	-0.50 [-1.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S10	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S11	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S12	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
S15	10	+0.00 [+0.00, +0.00]	-0.50 [-1.00, +0.00]	+0.03 [-0.08, +0.15]	+0.05 [+0.00, +0.13]	+0.00 [+0.00, +0.00]	+0.00 [-0.12, +0.12]	+0.00 [+0.00, +0.00]
S18	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.07 [-0.03, +0.19]	+0.05 [+0.00, +0.13]	+0.00 [+0.00, +0.00]	+0.09 [-0.06, +0.26]	+0.00 [+0.00, +0.00]

Table 26: Family OpenAI: each arm minus Full on the cases both answered (mean difference, 95% bootstrap interval). A positive difference favours the arm for the higher-is-better metrics and Full for unsupported facts and fabricated citations. Intervals that include zero are not resolved at this sample size. Bold: the most favourable mean difference in the column; underlined: the second most favourable.

Arm	Cases	Δ Obj. recall	Δ Ans. cov.	Δ Coverage	Δ Unsupp. facts	Δ Fabr. cit.	Δ Cit. prec.	Δ Factor rec.
B0	25	-0.40 [-0.70, -0.10]	-0.39 [-0.72, -0.11]	-0.43 [-0.58, -0.25]	-0.05 [-0.09, -0.01]	–	–	-0.33 [-1.00, +0.00]
B1	20	-0.44 [-0.78, -0.11]	-0.07 [-0.43, +0.21]	-0.27 [-0.49, -0.05]	-0.09 [-0.15, -0.04]	+0.00 [+0.00, +0.00]	-0.74 [-0.84, -0.62]	-0.50 [-1.00, +0.00]
B2	25	-0.40 [-0.70, -0.10]	-0.39 [-0.72, -0.11]	-0.51 [-0.67, -0.33]	-0.04 [-0.07, +0.00]	+1.00	-0.75	-0.33 [-1.00, +0.00]
B3	20	-0.44 [-0.78, -0.11]	-0.36 [-0.71, -0.07]	-0.27 [-0.46, -0.07]	**-0.10 [-0.18, -0.02]**	+0.00 [+0.00, +0.00]	-0.22 [-0.56, +0.12]	+0.00 [-1.00, +1.00]
B4	30	-0.18 [-0.64, +0.27]	-0.14 [-0.50, +0.23]	-0.57 [-0.66, -0.47]	-0.05 [-0.07, -0.02]	+0.93 [+0.83, +1.00]	-0.75 [-0.81, -0.69]	+0.00 [+0.00, +0.00]
B5	30	-0.09 [-0.55, +0.36]	-0.21 [-0.55, +0.15]	-0.57 [-0.67, -0.46]	-0.04 [-0.08, -0.01]	+0.64 [+0.29, +0.93]	-0.68 [-0.82, -0.55]	+0.00 [+0.00, +0.00]
B6	30	-0.45 [-0.73, -0.18]	-0.38 [-0.65, -0.14]	**+0.36 [+0.28, +0.44]**	-0.04 [-0.06, -0.02]	–	–	**+0.25 [-0.50, +1.00]**
A1	30	+0.00 [-0.36, +0.36]	-0.08 [-0.39, +0.26]	+0.08 [-0.01, +0.17]	-0.08 [-0.12, -0.05]	+0.01 [+0.00, +0.02]	-0.20 [-0.29, -0.10]	-0.25 [-0.75, +0.00]
A2	30	-0.18 [-0.55, +0.18]	-0.38 [-0.65, -0.14]	-0.64 [-0.72, -0.56]	-0.01 [-0.02, +0.00]	+0.03 [+0.00, +0.08]	-0.18 [-0.27, -0.08]	+0.00 [+0.00, +0.00]
A3	30	-0.09 [-0.36, +0.27]	+0.00 [+0.00, +0.00]	+0.05 [-0.01, +0.12]	-0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.04 [-0.01, +0.10]	+0.00 [+0.00, +0.00]
A4	30	+0.00 [-0.27, +0.27]	+0.11 [-0.30, +0.50]	-0.11 [-0.18, -0.03]	-0.00 [-0.01, +0.00]	+0.00 [+0.00, +0.00]	**+0.07 [+0.02, +0.12]**	+0.00 [+0.00, +0.00]
A6	30	+0.09 [-0.18, +0.45]	+0.00 [+0.00, +0.00]	+0.05 [+0.00, +0.12]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.02 [-0.07, +0.02]	+0.00 [+0.00, +0.00]
A7	30	-0.09 [-0.55, +0.36]	-0.21 [-0.55, +0.15]	-0.58 [-0.67, -0.47]	-0.04 [-0.08, -0.01]	+0.64 [+0.29, +0.93]	-0.68 [-0.82, -0.55]	+0.00 [+0.00, +0.00]
A8	30	-0.45 [-0.73, -0.18]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.00 [-0.01, +0.00]	+0.00 [+0.00, +0.00]
A9	30	-0.45 [-0.73, -0.18]	-0.38 [-0.65, -0.14]	-0.64 [-0.72, -0.56]	-0.01 [-0.02, +0.00]	+0.03 [+0.00, +0.08]	-0.19 [-0.28, -0.09]	+0.00 [+0.00, +0.00]
A5 base 4b	9	+0.00 [+0.00, +0.00]	+0.00	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00
S1	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	-0.04 [-0.20, +0.12]	-0.01 [-0.04, +0.00]	+0.02 [+0.00, +0.05]	-0.14 [-0.29, -0.01]	+0.00 [+0.00, +0.00]
S2	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S3	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.07 [+0.00, +0.17]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.05 [-0.01, +0.11]	+0.00 [+0.00, +0.00]
S3B	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.07 [+0.00, +0.17]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.05 [-0.01, +0.11]	+0.00 [+0.00, +0.00]
S4	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S5	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.07 [+0.00, +0.17]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.12 [-0.21, -0.03]	+0.00 [+0.00, +0.00]
S6	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S7	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S8	10	-0.25 [-1.00, +0.50]	**+0.50 [+0.00, +1.00]**	+0.09 [-0.03, +0.22]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.16 [-0.34, -0.00]	+0.00 [+0.00, +0.00]
S9	10	**+0.50 [+0.00, +1.00]**	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S10	10	+0.25 [+0.00, +0.75]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S11	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S12	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S13 stopping llm	10	-0.25 [-0.75, +0.00]	+0.00 [+0.00, +0.00]	+0.07 [+0.00, +0.17]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.05 [-0.01, +0.11]	+0.00 [+0.00, +0.00]
S14	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S15	10	+0.00 [-0.75, +0.75]	+0.00 [+0.00, +0.00]	+0.03 [-0.13, +0.23]	-0.03 [-0.06, +0.00]	+0.00 [+0.00, +0.00]	-0.12 [-0.24, -0.01]	+0.00 [+0.00, +0.00]
S16	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
S17	10	+0.25 [+0.00, +0.75]	+0.00 [+0.00, +0.00]	+0.00 [-0.10, +0.10]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.09 [-0.19, -0.02]	+0.00 [+0.00, +0.00]
S18	10	-0.25 [-1.00, +0.50]	+0.00 [+0.00, +0.00]	-0.53 [-0.67, -0.28]	-0.03 [-0.09, +0.00]	+0.02 [+0.00, +0.05]	-0.14 [-0.30, -0.00]	+0.00 [+0.00, +0.00]
P5	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
P6	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
P7	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]
P8	10	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.01 [-0.04, +0.00]	+0.00 [+0.00, +0.00]

Table 27: Family Sarvam: each arm minus Full on the cases both answered (mean difference, 95% bootstrap interval). A positive difference favours the arm for the higher-is-better metrics and Full for unsupported facts and fabricated citations. Intervals that include zero are not resolved at this sample size. Bold: the most favourable mean difference in the column; underlined: the second most favourable.

Arm	Cases	Δ Obj. recall	Δ Ans. cov.	Δ Coverage	Δ Unsupp. facts	Δ Fabr. cit.	Δ Cit. prec.	Δ Factor rec.
B0	20	-0.60 [-0.90, -0.30]	-0.17 [-0.50, +0.00]	+0.14 [+0.05, +0.24]	+0.03 [-0.09, +0.14]	–	–	+0.00 [+0.00, +0.00]
B0L	20	-0.60 [-0.90, -0.30]	-0.33 [-0.67, +0.00]	+0.14 [+0.05, +0.24]	+0.04 [-0.03, +0.12]	–	–	+0.50 [+0.00, +1.00]
B1	17	-0.56 [-0.89, -0.22]	-0.25 [-0.58, +0.00]	**+0.17 [+0.06, +0.29]**	+0.09 [-0.01, +0.18]	**+0.00 [+0.00, +0.00]**	-0.83 [-0.87, -0.79]	+0.00 [+0.00, +0.00]
B2	20	-0.60 [-0.90, -0.30]	+0.08 [-0.42, +0.67]	+0.14 [+0.05, +0.24]	+0.06 [-0.01, +0.15]	–	–	+0.50 [+0.00, +1.00]
B3	17	-0.56 [-0.89, -0.22]	+0.25 [-0.17, +0.67]	-0.01 [-0.22, +0.19]	+0.09 [-0.03, +0.21]	**+0.00 [+0.00, +0.00]**	-0.82 [-0.90, -0.75]	+0.50 [+0.00, +1.00]
A1	10	**+0.25 [+0.00, +0.75]**	**+0.50 [+0.00, +1.00]**	-0.01 [-0.12, +0.10]	**-0.15 [-0.28, -0.03]**	**+0.00 [+0.00, +0.00]**	**-0.03 [-0.16, +0.11]**	+0.50 [+0.00, +1.00]

Table 28: Family Gemini block A: each arm minus Full on the cases both answered (mean difference, 95% bootstrap interval). A positive difference favours the arm for the higher-is-better metrics and Full for unsupported facts and fabricated citations. Intervals that include zero are not resolved at this sample size. Bold: the most favourable mean difference in the column; underlined: the second most favourable.

Arm	Cases	Δ Obj. recall	Δ Ans. cov.	Δ Coverage	Δ Unsupp. facts	Δ Fabr. cit.	Δ Cit. prec.	Δ Factor rec.
B0	50	-0.70 [-0.88, -0.50]	-0.50 [-0.70, -0.30]	+0.03 [-0.11, +0.15]	-0.01 [-0.08, +0.06]	–	–	+0.25 [-0.25, +0.75]
B1	50	-0.70 [-0.88, -0.50]	-0.50 [-0.70, -0.30]	+0.07 [-0.05, +0.18]	**-0.09 [-0.16, -0.02]**	**+0.00**	**-0.71**	+0.38 [+0.12, +0.75]
B4	50	**+0.23 [+0.03, +0.45]**	**+0.25 [+0.03, +0.47]**	-0.38 [-0.49, -0.28]	-0.05 [-0.11, +0.01]	–	–	+0.38 [+0.00, +0.75]
B6	50	-0.70 [-0.88, -0.50]	-0.50 [-0.70, -0.30]	**+0.17 [+0.11, +0.23]**	+0.13 [+0.08, +0.18]	–	–	**+0.62 [+0.12, +1.00]**

Table 29: Family superseded 50-case run: each arm minus Full on the cases both answered (mean difference, 95% bootstrap interval). A positive difference favours the arm for the higher-is-better metrics and Full for unsupported facts and fabricated citations. Intervals that include zero are not resolved at this sample size. Bold: the most favourable mean difference in the column; underlined: the second most favourable.

Arm	Cases	Obj. recall (arm / Full)	Ans. cov. (arm / Full)	Coverage (arm / Full)	Unsupp. facts (arm / Full)	Cit. prec. (arm / Full)	Factor rec. (arm / Full)
B0	30	0.00 / 0.74 (17)	0.00 / 0.33 (10)	0.83 / 0.77 (30)	0.87 / 0.87 (25)	–	0.50 / 0.17 (6)
B1	23	0.00 / 0.79 (14)	0.00 / 0.42 (8)	0.83 / 0.77 (23)	**0.82** / 0.88 (19)	0.00 / 1.00 (1)	**0.75** / 0.25 (4)
B2	30	0.00 / 0.74 (17)	0.05 / 0.33 (10)	**0.87** / 0.77 (30)	0.83 / 0.88 (26)	–	0.50 / 0.17 (6)
B3	23	0.00 / 0.79 (14)	0.00 / 0.42 (8)	**0.87** / 0.77 (23)	0.84 / 0.87 (20)	–	**0.75** / 0.25 (4)
B4	30	0.90 / 0.75 (20)	**0.64** / 0.42 (12)	0.42 / 0.82 (30)	**0.82** / 0.89 (30)	–	0.50 / 0.25 (8)
B5	10	**1.00** / 1.00 (4)	0.50 / 0.50 (2)	0.28 / 0.91 (10)	0.92 / 0.95 (10)	**0.50** / 0.50 (1)	0.50 / 0.50 (2)

Table 30: Family OpenAI: each baseline beside Full on the cases both answered without error. A cell is the arm's mean, then Full's mean on the same cases, over the cases where the metric is defined for both (the count in brackets). Unsupported facts: lower is better; the others: higher is better. The arm's value is bold for the best baseline in the column and underlined for the second best. A dash means the metric is undefined for the arm (for example, a baseline that cites no authority has no citation precision).

Arm	Cases	Obj. recall (arm / Full)	Ans. cov. (arm / Full)	Coverage (arm / Full)	Unsupp. facts (arm / Full)	Cit. prec. (arm / Full)	Factor rec. (arm / Full)
B0	25	0.00 / 0.40 (10)	0.00 / 0.39 (9)	0.24 / 0.67 (25)	0.95 / 1.00 (6)	–	0.00 / 0.33 (3)
B1	20	0.00 / 0.44 (9)	**0.29** / 0.36 (7)	0.40 / 0.67 (20)	0.91 / 1.00 (8)	0.00 / 0.74 (4)	0.00 / 0.50 (2)
B2	25	0.00 / 0.40 (10)	0.00 / 0.39 (9)	0.16 / 0.67 (25)	0.96 / 1.00 (4)	0.00 / 0.75 (1)	0.00 / 0.33 (3)
B3	20	0.00 / 0.44 (9)	0.00 / 0.36 (7)	0.40 / 0.67 (20)	**0.90** / 1.00 (8)	**0.50** / 0.72 (2)	0.50 / 0.50 (2)
B4	30	0.27 / 0.45 (11)	0.24 / 0.38 (11)	0.07 / 0.64 (30)	0.95 / 1.00 (30)	0.02 / 0.78 (30)	0.50 / 0.50 (4)
B5	30	**0.36** / 0.45 (11)	0.17 / 0.38 (11)	0.07 / 0.64 (30)	0.96 / 1.00 (30)	0.00 / 0.68 (7)	0.50 / 0.50 (4)
B6	30	0.00 / 0.45 (11)	0.00 / 0.38 (11)	**1.00** / 0.64 (30)	0.96 / 1.00 (30)	–	**0.75** / 0.50 (4)

Table 31: Family Sarvam: each baseline beside Full on the cases both answered without error. A cell is the arm's mean, then Full's mean on the same cases, over the cases where the metric is defined for both (the count in brackets). Unsupported facts: lower is better; the others: higher is better. The arm's value is bold for the best baseline in the column and underlined for the second best. A dash means the metric is undefined for the arm (for example, a baseline that cites no authority has no citation precision).

Arm	Cases	Obj. recall (arm / Full)	Ans. cov. (arm / Full)	Coverage (arm / Full)	Unsupp. facts (arm / Full)	Cit. prec. (arm / Full)	Factor rec. (arm / Full)
B0	20	0.00 / 0.60 (10)	0.17 / 0.33 (6)	**1.00** / 0.86 (20)	**0.89** / 0.87 (20)	–	0.00 / 0.00 (2)
B1	17	0.00 / 0.56 (9)	0.08 / 0.33 (6)	**1.00** / 0.83 (17)	0.95 / 0.86 (17)	**0.00** / 0.83 (5)	0.00 / 0.00 (2)
B2	20	0.00 / 0.60 (10)	0.42 / 0.33 (6)	**1.00** / 0.86 (20)	0.93 / 0.87 (20)	–	**0.50** / 0.00 (2)
B3	17	0.00 / 0.56 (9)	**0.58** / 0.33 (6)	0.82 / 0.83 (17)	0.94 / 0.85 (14)	**0.00** / 0.82 (2)	**0.50** / 0.00 (2)

Table 32: Family Gemini block A: each baseline beside Full on the cases both answered without error. A cell is the arm's mean, then Full's mean on the same cases, over the cases where the metric is defined for both (the count in brackets). Unsupported facts: lower is better; the others: higher is better. The arm's value is bold for the best baseline in the column and underlined for the second best. A dash means the metric is undefined for the arm (for example, a baseline that cites no authority has no citation precision).

Arm	Cases	Obj. recall (arm / Full)	Ans. cov. (arm / Full)	Coverage (arm / Full)	Unsupp. facts (arm / Full)	Cit. prec. (arm / Full)	Factor rec. (arm / Full)
B0	50	0.00 / 0.70 (20)	0.00 / 0.50 (20)	0.86 / 0.83 (50)	0.83 / 0.84 (43)	–	0.50 / 0.25 (8)
B1	50	0.00 / 0.70 (20)	0.00 / 0.50 (20)	0.90 / 0.83 (50)	**0.75** / 0.84 (45)	**0.00** / 0.71 (1)	0.62 / 0.25 (8)
B4	50	**0.93** / 0.70 (20)	**0.75** / 0.50 (20)	0.45 / 0.83 (50)	0.81 / 0.86 (50)	–	0.62 / 0.25 (8)
B6	50	0.00 / 0.70 (20)	0.00 / 0.50 (20)	**1.00** / 0.83 (50)	0.98 / 0.86 (50)	–	**0.88** / 0.25 (8)

Table 33: Family superseded 50-case run: each baseline beside Full on the cases both answered without error. A cell is the arm's mean, then Full's mean on the same cases, over the cases where the metric is defined for both (the count in brackets). Unsupported facts: lower is better; the others: higher is better. The arm's value is bold for the best baseline in the column and underlined for the second best. A dash means the metric is undefined for the arm (for example, a baseline that cites no authority has no citation precision).

Arm	Factor recall	Precision (lenient)	F1
FULL	0.25 [0.00, 0.62] (8)	0.19 [0.00, 0.44] (8)	0.21 [0.00, 0.50] (8)
B0	0.38 [0.12, 0.75] (8)	0.67 [0.25, 1.00] (4)	0.35 [0.10, 0.70] (8)
B1	0.38 [0.12, 0.75] (8)	0.67 [0.25, 1.00] (4)	0.35 [0.10, 0.70] (8)
B2	0.38 [0.12, 0.75] (8)	0.67 [0.25, 1.00] (4)	0.35 [0.10, 0.70] (8)
B3	0.38 [0.12, 0.75] (8)	0.67 [0.25, 1.00] (4)	0.35 [0.10, 0.70] (8)
B4	**0.50 [0.12, 0.88] (8)**	0.38 [0.08, 0.67] (8)	0.41 [0.12, 0.72] (8)
B5	**0.50 [0.00, 1.00] (2)**	**1.00 (1)**	**0.50 [0.00, 1.00] (2)**

Table 34: Family OpenAI: factor-level metrics, mean [95% bootstrap interval] over the orders where the metric is defined (their number in brackets). Each arm on its own orders. Bold: best in the column; underlined: second best.

Arm	Unsupported facts	Objection recall	Answerable coverage	Coverage	Tokens	USD per case
FULL	0.90 [0.86, 0.94] (38)	0.75 [0.57, 0.90] (20)	0.47 [0.24, 0.71] (15)	**0.79 [0.72, 0.86] (38)**	95466 [89311, 101838] (38)	0.0100 [0.0094, 0.0107] (38)
B0	0.87 [0.81, 0.92] (25)	0.00 [0.00, 0.00] (20)	0.00 [0.00, 0.00] (15)	0.66 [0.50, 0.82] (38)	**2907 [2398, 3363] (38)**	**0.0003 [0.0003, 0.0004] (38)**
B1	**0.82 [0.75, 0.88] (19)**	0.00 [0.00, 0.00] (20)	0.00 [0.00, 0.00] (15)	0.50 [0.34, 0.66] (38)	4242 [3094, 5310] (38)	0.0004 [0.0003, 0.0006] (38)
B2	0.83 [0.76, 0.88] (26)	0.00 [0.00, 0.00] (20)	0.03 [0.00, 0.10] (15)	0.68 [0.53, 0.82] (38)	3435 [2806, 4018] (38)	0.0004 [0.0003, 0.0004] (38)
B3	0.84 [0.77, 0.90] (20)	0.00 [0.00, 0.00] (20)	0.00 [0.00, 0.00] (15)	0.53 [0.37, 0.66] (38)	4412 [3191, 5563] (38)	0.0005 [0.0003, 0.0006] (38)
B4	**0.82 [0.74, 0.89] (30)**	0.90 [0.75, 1.00] (20)	**0.64 [0.39, 0.89] (12)**	0.42 [0.32, 0.53] (30)	126301 [120250, 132649] (30)	0.0133 [0.0126, 0.0139] (30)
B5	0.92 [0.84, 1.00] (10)	**1.00 [1.00, 1.00] (4)**	0.50 [0.00, 1.00] (2)	0.28 [0.14, 0.42] (10)	94180 [86213, 102432] (10)	0.0099 [0.0091, 0.0108] (10)

Table 35: Family OpenAI: unsupported-fact rate (lower is better), objection recall, answerable coverage, coverage, tokens and cost per case, mean [95% bootstrap interval] (n). Each arm on its own orders. Bold: best in the column; underlined: second best.

Arm	Factor recall	Precision (lenient)	F1
FULL	0.50 [0.00, 1.00] (4)	0.19 [0.00, 0.44] (8)	0.42 [0.00, 0.83] (4)
B0	0.00 [0.00, 0.00] (4)	–	0.00 [0.00, 0.00] (4)
B1	0.00 [0.00, 0.00] (4)	0.00 [0.00, 0.00] (2)	0.00 [0.00, 0.00] (4)
B2	0.00 [0.00, 0.00] (4)	0.00 (1)	0.00 [0.00, 0.00] (4)
B3	0.25 [0.00, 0.75] (4)	**0.50 [0.00, 1.00] (2)**	0.25 [0.00, 0.75] (4)
B4	0.50 [0.00, 1.00] (4)	0.19 [0.00, 0.44] (8)	0.42 [0.00, 0.83] (4)
B5	0.50 [0.00, 1.00] (4)	0.19 [0.00, 0.44] (8)	0.42 [0.00, 0.83] (4)
B6	**0.75 [0.25, 1.00] (4)**	0.47 [0.07, 0.87] (5)	**0.62 [0.25, 1.00] (4)**

Table 36: Family Sarvam: factor-level metrics, mean [95% bootstrap interval] over the orders where the metric is defined (their number in brackets). Each arm on its own orders. Bold: best in the column; underlined: second best.

Arm	Unsupported facts	Objection recall	Answerable coverage	Coverage	Tokens	USD per case
FULL	1.00 [0.99, 1.00] (30)	**0.45 [0.18, 0.73] (11)**	**0.38 [0.14, 0.65] (11)**	0.64 [0.57, 0.72] (30)	96945 [89413, 104304] (30)	–
B0	0.95 [0.91, 0.99] (6)	0.00 [0.00, 0.00] (11)	0.00 [0.00, 0.00] (11)	0.20 [0.07, 0.33] (30)	3315 [2711, 3835] (30)	–
B1	0.91 [0.85, 0.96] (8)	0.00 [0.00, 0.00] (11)	0.18 [0.00, 0.45] (11)	0.27 [0.13, 0.43] (30)	5599 [4084, 6968] (30)	–
B2	0.96 [0.93, 1.00] (4)	0.00 [0.00, 0.00] (11)	0.00 [0.00, 0.00] (11)	0.13 [0.03, 0.27] (30)	3812 [3065, 4529] (30)	–
B3	**0.90 [0.82, 0.98] (8)**	0.00 [0.00, 0.00] (11)	0.00 [0.00, 0.00] (11)	0.27 [0.13, 0.43] (30)	5412 [3928, 6755] (30)	–
B4	0.95 [0.92, 0.98] (30)	0.27 [0.00, 0.55] (11)	0.24 [0.06, 0.52] (11)	0.07 [0.04, 0.12] (30)	118094 [110012, 126671] (30)	–
B5	0.96 [0.91, 0.99] (30)	0.36 [0.09, 0.64] (11)	0.17 [0.03, 0.38] (11)	0.07 [0.03, 0.13] (30)	76978 [71343, 82871] (30)	–
B6	0.96 [0.94, 0.98] (30)	0.00 [0.00, 0.00] (11)	0.00 [0.00, 0.00] (11)	**1.00 [1.00, 1.00] (30)**	**0 [0, 0] (30)**	–

Table 37: Family Sarvam: unsupported-fact rate (lower is better), objection recall, answerable coverage, coverage, tokens and cost per case, mean [95% bootstrap interval] (n). Each arm on its own orders. Bold: best in the column; underlined: second best.

Arm	Factor recall	Precision (lenient)	F1
FULL	0.25 [0.00, 0.62] (8)	0.15 [0.00, 0.33] (12)	0.21 [0.00, 0.50] (8)
B0	0.50 [0.12, 0.88] (8)	0.41 [0.17, 0.69] (9)	0.43 [0.12, 0.74] (8)
B1	0.62 [0.25, 1.00] (8)	0.47 [0.20, 0.72] (10)	0.56 [0.25, 0.85] (8)
B4	0.62 [0.25, 0.88] (8)	0.27 [0.11, 0.46] (17)	0.52 [0.21, 0.81] (8)
B6	**0.88 [0.62, 1.00] (8)**	**0.56 [0.30, 0.80] (11)**	**0.77 [0.52, 0.96] (8)**

Table 38: Family superseded 50-case run: factor-level metrics, mean [95% bootstrap interval] over the orders where the metric is defined (their number in brackets). Each arm on its own orders. Bold: best in the column; underlined: second best.

Arm	Unsupported facts	Objection recall	Answerable coverage	Coverage	Tokens	USD per case
FULL	0.86 [0.80, 0.91] (50)	0.70 [0.50, 0.88] (20)	0.50 [0.30, 0.70] (20)	0.83 [0.77, 0.89] (50)	101770 [96599, 107360] (50)	0.0107 [0.0102, 0.0113] (50)
B0	0.83 [0.79, 0.87] (43)	0.00 [0.00, 0.00] (20)	0.00 [0.00, 0.00] (20)	0.86 [0.76, 0.96] (50)	3862 [3727, 3992] (50)	0.0004 [0.0004, 0.0004] (50)
B1	**0.75 [0.71, 0.80] (45)**	0.00 [0.00, 0.00] (20)	0.00 [0.00, 0.00] (20)	0.90 [0.82, 0.98] (50)	7472 [7188, 7766] (50)	0.0008 [0.0008, 0.0008] (50)
B4	0.81 [0.75, 0.87] (50)	**0.93 [0.80, 1.00] (20)**	**0.75 [0.55, 0.92] (20)**	0.45 [0.38, 0.52] (50)	131239 [127132, 135449] (50)	0.0138 [0.0134, 0.0142] (50)
B6	0.98 [0.97, 1.00] (50)	0.00 [0.00, 0.00] (20)	0.00 [0.00, 0.00] (20)	**1.00 [1.00, 1.00] (50)**	**0 [0, 0] (50)**	**0.0000 [0.0000, 0.0000] (50)**

Table 39: Family superseded 50-case run: unsupported-fact rate (lower is better), objection recall, answerable coverage, coverage, tokens and cost per case, mean [95% bootstrap interval] (n). Each arm on its own orders. Bold: best in the column; underlined: second best.

Table 40: Paired differences against Full. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed.

Comparison	Metric	n	Difference [95% CI]	p (test)	d_z / odds ratio	Family (Holm)
full minus b0	factor_f1	8	-0.225 [-0.625, +0.250]	0.6250 (wilcoxon)	-0.33	F-RQ1-guard: not rejected
full minus b0	factor_hits_pooled (McNemar, cluster bootstrap)	8	-0.250 [-0.750, +0.250]	0.6250 (mcnemar exact (items))	0.33	exploratory (uncorrected)
full minus b0	factor_precision_lenient	6	-0.306 [-0.639, -0.028]	0.2500 (wilcoxon)	-0.72	F-RQ1-guard: not rejected
full minus b0	factor_recall	8	-0.250 [-0.750, +0.250]	0.6250 (wilcoxon)	-0.35	F-RQ1-recall: not rejected
full minus b0	objection_recall	20	+0.688 [+0.500, +0.863]	0.0003 (wilcoxon)	1.58	exploratory (uncorrected)
full minus b0	pass_per_case (McNemar)	8	-0.250	0.6250 (mcnemar exact)	0.33	exploratory (uncorrected)
full minus b0	recall_at_10	0	too few cases			
full minus b0	store_calls	50	+103.970 [+100.890, +106.780]	0.0000 (wilcoxon)	9.66	exploratory (uncorrected)
full minus b0	tokens	50	+97998.450 [+93099.750, +102962.125]	0.0000 (wilcoxon)	5.39	exploratory (uncorrected)
full minus b0	unsupported_fact_rate	43	+0.017 [-0.052, +0.086]	0.6593 (wilcoxon)	0.07	F-RQ1-halluc: not rejected
full minus b1	factor_f1	8	-0.350 [-0.700, -0.100]	0.2500 (wilcoxon)	-0.72	F-RQ1-guard: not rejected
full minus b1	factor_hits_pooled (McNemar, cluster bootstrap)	8	-0.375 [-0.750, -0.125]	0.2500 (mcnemar exact (items))	0.00	exploratory (uncorrected)
full minus b1	factor_precision_lenient	7	-0.262 [-0.571, -0.024]	0.2500 (wilcoxon)	-0.64	F-RQ1-guard: not rejected
full minus b1	factor_recall	8	-0.375 [-0.750, -0.125]	0.2500 (wilcoxon)	-0.72	F-RQ1-recall: not rejected
full minus b1	objection_recall	20	+0.688 [+0.500, +0.863]	0.0003 (wilcoxon)	1.58	exploratory (uncorrected)
full minus b1	pass_per_case (McNemar)	8	-0.375	0.2500 (mcnemar exact)	0.00	exploratory (uncorrected)
full minus b1	recall_at_10	0	too few cases			
full minus b1	store_calls	50	+102.970 [+99.890, +105.780]	0.0000 (wilcoxon)	9.56	exploratory (uncorrected)
full minus b1	tokens	50	+94388.090 [+89546.867, +99290.061]	0.0000 (wilcoxon)	5.26	exploratory (uncorrected)
full minus b1	unsupported_fact_rate	45	+0.095 [+0.026, +0.162]	0.0166 (wilcoxon)	0.40	F-RQ1-halluc: not rejected
full minus b4	factor_f1	8	-0.312 [-0.646, -0.021]	0.2500 (wilcoxon)	-0.64	F-RQ1-guard: not rejected
full minus b4	factor_hits_pooled (McNemar, cluster bootstrap)	8	-0.375 [-0.750, -0.125]	0.2500 (mcnemar exact (items))	0.00	exploratory (uncorrected)
full minus b4	factor_precision_lenient	12	-0.069 [-0.194, +0.028]	0.5000 (wilcoxon)	-0.34	F-RQ1-guard: not rejected
full minus b4	factor_recall	8	-0.375 [-0.750, -0.125]	0.2500 (wilcoxon)	-0.72	F-RQ1-recall: not rejected
full minus b4	objection_recall	20	-0.237 [-0.450, -0.037]	0.0371 (wilcoxon)	-0.49	exploratory (uncorrected)
full minus b4	pass_per_case (McNemar)	8	-0.375	0.2500 (mcnemar exact)	0.00	exploratory (uncorrected)
full minus b4	recall_at_10	0	too few cases			
full minus b4	store_calls	50	-90.970 [-94.900, -86.080]	0.0000 (wilcoxon)	-5.67	exploratory (uncorrected)
full minus b4	tokens	50	-29379.190 [-33256.092, -25355.336]	0.0000 (wilcoxon)	-2.07	exploratory (uncorrected)
full minus b4	unsupported_fact_rate	50	+0.052 [-0.004, +0.114]	0.1549 (wilcoxon)	0.25	F-RQ1-halluc: not rejected
full minus b6	factor_f1	8	-0.562 [-0.958, -0.021]	0.1250 (wilcoxon)	-0.75	F-RQ1-guard: not rejected
full minus b6	factor_hits_pooled (McNemar, cluster bootstrap)	8	-0.625 [-1.000, -0.125]	0.1250 (mcnemar exact (items))	0.17	exploratory (uncorrected)
full minus b6	factor_precision_lenient	6	-0.222 [-0.583, +0.056]	0.5000 (wilcoxon)	-0.50	F-RQ1-guard: not rejected
full minus b6	factor_recall	8	-0.625 [-1.000, -0.125]	0.1250 (wilcoxon)	-0.84	exploratory (uncorrected)
full minus b6	objection_recall	20	+0.688 [+0.500, +0.863]	0.0003 (wilcoxon)	1.58	exploratory (uncorrected)
full minus b6	pass_per_case (McNemar)	8	-0.625	0.1250 (mcnemar exact)	0.17	exploratory (uncorrected)
full minus b6	recall_at_10	0	too few cases			
full minus b6	store_calls	50	+101.830 [+98.800, +104.580]	0.0000 (wilcoxon)	9.73	exploratory (uncorrected)
full minus b6	tokens	50	+101860.030 [+96914.999, +106871.670]	0.0000 (wilcoxon)	5.55	exploratory (uncorrected)
full minus b6	unsupported_fact_rate	50	-0.120 [-0.171, -0.072]	0.0000 (wilcoxon)	-0.66	exploratory (uncorrected)

Table 41: Holm-adjusted tests of the registered claims. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed.

Family	Claims	Registered tests (m)	Comparator	Metric	Difference	p	Holm decision
F-RQ1-recall	C1	6	b0	factor_recall	-0.250	0.6250	not rejected (sign against reference)
F-RQ1-recall	C1	6	b0l	factor_recall			not run
F-RQ1-recall	C1	6	b1	factor_recall	-0.375	0.2500	not rejected (sign against reference)
F-RQ1-recall	C1	6	b2	factor_recall			not run
F-RQ1-recall	C1	6	b3	factor_recall			not run
F-RQ1-recall	C1	6	b4	factor_recall	-0.375	0.2500	not rejected (sign against reference)
F-RQ1-halluc	C2	6	b0	unsupported_fact_rate	+0.017	0.6593	not rejected (sign against reference)
F-RQ1-halluc	C2	6	b0l	unsupported_fact_rate			not run
F-RQ1-halluc	C2	6	b1	unsupported_fact_rate	+0.095	0.0166	not rejected (sign against reference)
F-RQ1-halluc	C2	6	b2	unsupported_fact_rate			not run
F-RQ1-halluc	C2	6	b3	unsupported_fact_rate			not run
F-RQ1-halluc	C2	6	b4	unsupported_fact_rate	+0.052	0.1549	not rejected (sign against reference)
F-RQ1-guard	C3	8	b0	factor_precision_lenient	-0.306	0.2500	not rejected (sign against reference)
F-RQ1-guard	C3	8	b1	factor_precision_lenient	-0.262	0.2500	not rejected (sign against reference)
F-RQ1-guard	C3	8	b4	factor_precision_lenient	-0.069	0.5000	not rejected (sign against reference)
F-RQ1-guard	C3	8	b6	factor_precision_lenient	-0.222	0.5000	not rejected (sign against reference)
F-RQ1-guard	C3	8	b0	factor_f1	-0.225	0.6250	not rejected (sign against reference)
F-RQ1-guard	C3	8	b1	factor_f1	-0.350	0.2500	not rejected (sign against reference)
F-RQ1-guard	C3	8	b4	factor_f1	-0.312	0.2500	not rejected (sign against reference)
F-RQ1-guard	C3	8	b6	factor_f1	-0.562	0.1250	not rejected (sign against reference)
F-ablate-recall	C10-C12	6	a1	factor_recall			not run
F-ablate-recall	C10-C12	6	a3	factor_recall			not run
F-ablate-recall	C10-C12	6	a4	factor_recall			not run
F-ablate-recall	C10-C12	6	a6	factor_recall			not run
F-ablate-recall	C10-C12	6	a7	factor_recall			not run
F-ablate-recall	C10-C12	6	a1	unsupported_fact_rate			not run
F-sparring	C7, C8, C13, C28	6	s9	objection_recall			not run
F-sparring	C7, C8, C13, C28	6	s10	objection_recall			not run
F-sparring	C7, C8, C13, C28	6	a2	objection_recall			not run
F-sparring	C7, C8, C13, C28	6	a8	objection_recall			not run
F-sparring	C7, C8, C13, C28	6	a9	objection_recall			not run
F-sparring	C7, C8, C13, C28	6	s8	unsupported_fact_rate			not run
F-retrieval	C5, C14	4	a4	recall_at_10			not run
F-retrieval	C5, C14	4	s4	recall_at_10			not run
F-retrieval	C5, C14	4	s5	recall_at_10			not run
F-retrieval	C5, C14	4	s16	recall_at_10			not run

Table 42: Cost per case. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed. USD is 0 on the offline stand-ins and priced on OpenAI only; Ada (Qwen) and Sarvam runs have no USD price.

System	Tokens/case [mean, p90]	Store calls/case	Seconds/case	USD/case	USD per passed case (M31)	Backend
full	101860, 128420	104.0	279.6	0.0107	0.2676	openai
b0	3862, 4429	**0.0**	4.1	0.0004	0.0051	openai
b1	7472, 8956	1.0	5.0	0.0008	0.0079	openai
b4	131239, 150266	194.9	237.6	0.0138	0.1379	openai
b6	**0, 0**	2.1	**0.1**	**0.0000**	**0.0000**	openai

Table 43: Gold-label coverage per metric. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed.

System	Orders scored	G1 and answerable (M1)	G2 resolved (M10)	Mean G2 gold size	G4 agreed (objection)	G7 labelled	G2 retrieved from
full	50	8	0	n/a	20	0	brief_authorities, events
b0	50	8	0	n/a	20	0	brief_authorities
b1	50	8	0	n/a	20	0	brief_authorities
b4	50	8	0	n/a	20	0	brief_authorities, events
b6	50	8	0	n/a	20	0	brief_authorities

Table 44: Label-source sensitivity. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed. Same outputs, different label sources; mean [95% CI] with n.

System	Label source	M1 factor recall	M2 precision (strict)	M2 precision (lenient)	M3 F1	M4 G5 recall
full	keyword rules only	0.278 [0.000, 0.556] n=9	0.153 [0.000, 0.347] n=12	0.236 [0.042, 0.472] n=12	0.222 [0.000, 0.444] n=9	0.500 [0.000, 1.000] n=4
full	one labeler only (LLM, span-grounded)	0.333 [0.000, 0.667] n=9	0.208 [0.000, 0.458] n=12	0.236 [0.055, 0.472] n=12	0.296 [0.000, 0.593] n=9	0.250 [0.000, 0.750] n=4
full	agreed only (primary)	0.250 [0.000, 0.500] n=8	0.125 [0.000, 0.333] n=12	0.153 [0.000, 0.347] n=12	0.208 [0.000, 0.500] n=8	0.333 [0.000, 1.000] n=3
b0	keyword rules only	0.500 [0.167, 0.778] n=9	0.370 [0.111, 0.648] n=9	0.519 [0.241, 0.778] n=9	0.441 [0.167, 0.715] n=9	0.750 [0.250, 1.000] n=4
b0	one labeler only (LLM, span-grounded)	0.444 [0.111, 0.778] n=9	0.315 [0.056, 0.593] n=9	0.407 [0.148, 0.667] n=9	0.385 [0.111, 0.674] n=9	0.750 [0.250, 1.000] n=4
b0	agreed only (primary)	0.500 [0.125, 0.875] n=8	0.315 [0.056, 0.593] n=9	0.407 [0.148, 0.667] n=9	0.433 [0.125, 0.725] n=8	0.667 [0.000, 1.000] n=3
b1	keyword rules only	0.611 [0.333, 0.889] n=9	0.433 [0.183, 0.700] n=10	0.567 [0.316, 0.817] n=10	0.552 [0.274, 0.811] n=9	0.750 [0.250, 1.000] n=4
b1	one labeler only (LLM, span-grounded)	0.556 [0.222, 0.889] n=9	0.383 [0.133, 0.667] n=10	0.467 [0.217, 0.733] n=10	0.496 [0.200, 0.785] n=9	0.750 [0.250, 1.000] n=4
b1	agreed only (primary)	0.625 [0.250, 0.875] n=8	0.383 [0.133, 0.667] n=10	0.467 [0.217, 0.733] n=10	0.558 [0.225, 0.850] n=8	0.667 [0.000, 1.000] n=3
b4	keyword rules only	0.667 [0.333, 1.000] n=9	0.265 [0.098, 0.461] n=17	0.324 [0.157, 0.510] n=17	0.552 [0.278, 0.815] n=9	0.625 [0.250, 1.000] n=4
b4	one labeler only (LLM, span-grounded)	0.667 [0.333, 1.000] n=9	0.284 [0.108, 0.490] n=17	0.333 [0.147, 0.529] n=17	0.574 [0.296, 0.852] n=9	0.625 [0.250, 1.000] n=4
b4	agreed only (primary)	0.625 [0.250, 0.875] n=8	0.225 [0.068, 0.422] n=17	0.275 [0.118, 0.471] n=17	0.521 [0.208, 0.812] n=8	0.500 [0.000, 1.000] n=3
b6	keyword rules only	0.889 [0.667, 1.000] n=9	0.591 [0.348, 0.848] n=11	0.682 [0.439, 0.909] n=11	0.774 [0.548, 0.956] n=9	0.875 [0.625, 1.000] n=4
b6	one labeler only (LLM, span-grounded)	0.778 [0.444, 1.000] n=9	0.530 [0.273, 0.803] n=11	0.561 [0.318, 0.818] n=11	0.685 [0.407, 0.926] n=9	0.875 [0.625, 1.000] n=4
b6	agreed only (primary)	0.875 [0.625, 1.000] n=8	0.530 [0.273, 0.803] n=11	0.561 [0.318, 0.818] n=11	0.771 [0.521, 0.958] n=8	0.833 [0.500, 1.000] n=3

Table 45: Agreement of the rule and model labelers. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed.

Pair	Label	n first	n second	both	Cohen kappa	95% CI
G1 rule vs labeler	antecedents	9	11	8	0.75	[0.46, 0.94]
G1 rule vs labeler	charge_sheet_filed	8	8	4	0.40	[-0.00, 0.73]
G1 rule vs labeler	custody_period	15	17	13	0.72	[0.49, 0.90]
G1 rule vs labeler	flight_risk	3	2	2	0.79	[0.00, 1.00]
G1 rule vs labeler	gravity_and_punishment	6	4	2	0.34	[-0.06, 0.69]
G1 rule vs labeler	health_age_family	5	6	4	0.69	[0.19, 1.00]
G1 rule vs labeler	parity_with_co_accused	8	8	7	0.85	[0.62, 1.00]
G1 rule vs labeler	statutory_bar	1	1	1	1.00	[1.00, 1.00]
G1 rule vs labeler	trial_delay	2	5	2	0.55	[0.00, 1.00]
G1 rule vs labeler	weak_prima_facie	20	37	17	0.16	[-0.04, 0.36]
G1 rule vs labeler	witness_tampering	2	1	1	0.66	[0.00, 1.00]
G1 rule vs labeler	all labels pooled	79	100		0.62	n=50 orders
G4 rule vs labeler	antecedents	8	9	8	0.93	[0.73, 1.00]
G4 rule vs labeler	charge_sheet_filed	6	3	2	0.40	[-0.04, 0.79]
G4 rule vs labeler	custody_period	2	0	0	0.00	[0.00, 1.00]
G4 rule vs labeler	flight_risk	2	2	0	-0.04	[-0.07, 0.00]
G4 rule vs labeler	gravity_and_punishment	9	25	9	0.36	[0.17, 0.57]
G4 rule vs labeler	health_age_family	1	0	0	0.00	[0.00, 1.00]
G4 rule vs labeler	parity_with_co_accused	7	2	2	0.41	[0.00, 0.79]
G4 rule vs labeler	statutory_bar	3	2	2	0.79	[0.00, 1.00]
G4 rule vs labeler	trial_delay	1	3	0	-0.03	[-0.07, 0.00]
G4 rule vs labeler	undertrial_period	1	0	0	0.00	[0.00, 1.00]
G4 rule vs labeler	weak_prima_facie	0	5	0	0.00	[-0.00, 0.00]
G4 rule vs labeler	all labels pooled	40	51		0.46	n=50 orders
G5 rule vs labeler	antecedents	7	8	7	0.92	[0.70, 1.00]
G5 rule vs labeler	charge_sheet_filed	6	6	4	0.62	[0.21, 0.90]
G5 rule vs labeler	custody_period	8	8	4	0.40	[0.00, 0.73]
G5 rule vs labeler	flight_risk	9	3	2	0.27	[-0.04, 0.62]
G5 rule vs labeler	gravity_and_punishment	13	14	11	0.75	[0.52, 0.94]
G5 rule vs labeler	health_age_family	1	1	0	-0.02	[-0.04, 1.00]
G5 rule vs labeler	parity_with_co_accused	4	5	4	0.88	[0.48, 1.00]
G5 rule vs labeler	statutory_bar	4	3	3	0.85	[0.26, 1.00]
G5 rule vs labeler	trial_delay	3	5	2	0.46	[-0.05, 0.85]
G5 rule vs labeler	weak_prima_facie	1	5	0	-0.03	[-0.09, 0.00]
G5 rule vs labeler	witness_tampering	2	2	1	0.48	[-0.04, 1.00]
G5 rule vs labeler	all labels pooled	58	60		0.60	n=50 orders

Table 46: Citation quality. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed.

System	Fabricated citations (M9)	Court reasoning (M13)	Supreme Court (M13)	Dated before order (M13)	Good-law violations (M15)	Known treatment (M15)	Citation precision (M16)	Citation recall (M16)	Claims changed by removal (M17)
full	**0.000 [0.000, 0.000] n=38**	1.000 [1.000, 1.000] n=38	0.044 [0.010, 0.086] n=38	1.000 [1.000, 1.000] n=38	**0.000 [0.000, 0.000] n=38**	0.003 [0.000, 0.010] n=38	**0.790 [0.727, 0.849] n=38**	**0.778 [0.721, 0.833] n=38**	n/a (n=0)
b0	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)
b1	**0.000 [0.000, 0.000] n=1**	1.000 [1.000, 1.000] n=1	0.000 [0.000, 0.000] n=1	1.000 [1.000, 1.000] n=1	**0.000 [0.000, 0.000] n=1**	0.000 [0.000, 0.000] n=1	0.000 [0.000, 0.000] n=1	n/a (n=0)	n/a (n=0)
b4	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)
b6	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)	n/a (n=0)

Table 47: Abstention and the coverage-risk trade-off. Superseded first dev run (tag `dev-sweep-v1`), dev slice (50 orders; the n of each system is in the table), gold source agreed.

System	Can decline	Claims	Coverage at the cut (M33)	Risk at the cut (M33)	Coverage, all emitted	Risk, all emitted (M6)	AURC [95% CI]	Confident-wrong (M35)	Over-search (M18)	Stopped with a slot unfilled (M18)
full	yes	474	0.141	**0.000**	0.816	0.827	0.427 [0.338, 0.549]	0.781 [0.697, 0.857] n=50	**0.017 [0.007, 0.030] n=50**	0.546 [0.461, 0.635] n=50
b0	no	560	0.168	**0.000**	**1.000**	0.832	one point (cannot decline)	0.789 [0.746, 0.829] n=43	n/a (n=0)	n/a (n=0)
b1	no	450	**0.209**	**0.000**	**1.000**	0.791	one point (cannot decline)	0.717 [0.666, 0.767] n=45	n/a (n=0)	n/a (n=0)
b4	yes	617	0.186	**0.000**	0.455	**0.591**	**0.103 [0.064, 0.154]**	**0.581 [0.475, 0.682] n=45**	n/a (n=0)	n/a (n=0)
b6	no	395	0.018	**0.000**	**1.000**	0.982	one point (cannot decline)	0.940 [0.918, 0.960] n=50	n/a (n=0)	n/a (n=0)